Signalling Games in Ideatic Diffusion Theory

A Formal Theory of Meaning, Interpretation, and Strategic Communication

With 364 Machine-Verified Theorems in Lean 4

Halley Young

University of Pennsylvania

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"The limits of my language mean the limits of my world."

— Ludwig Wittgenstein, *Tractatus Logico-Philosophicus* (5.6)

"A sign is something which stands to somebody for something in some respect or capacity."

— Charles Sanders Peirce, *Collected Papers* (2.228)

"Culture is public because meaning is."

— Clifford Geertz, *The Interpretation of Cultures*

Preface

This book develops a formal theory of signalling games within the framework of Ideatic Diffusion Theory (IDT). Where classical game theory treats signals as tokens in an abstract communication channel, we treat them as *ideas*—elements of an ideatic space equipped with composition, depth, and resonance. The central insight is that *interpretation is composition*: when a receiver with background idea r encounters signal s, they do not simply "decode" it but rather *compose* it with their existing understanding, producing the interpretation $r \circ s$.

This compositional view of interpretation has profound consequences for game theory. Payoffs depend not on the signal itself but on what receivers *make of it*—how it composes with their existing repertoire of ideas. A signal that is profound to one receiver may be trivial to another, not because the signal differs but because their repertoires do. This is the formal kernel of hermeneutic theory: meaning is not in the text but in the encounter between text and reader.

The book is organized in four parts. Part I (Chapters 1 to 5) establishes the interpretation framework: repertoires, signalling problems, costly signalling, resonance activation, and misinterpretation. Part II (Chapters 6 to 10) develops equilibrium theory, persuasion bounds, information loss, echo chambers, and multi-receiver signalling. Part III (Chapters 11 to 15) studies populations: competition, matching, coalitions, auctions, and repertoire curation. Part IV (Chapters 16 to 20) addresses cooperation, bargaining, network diffusion, social choice, and offers a synthesis.

Every theorem in this book has been formally verified in Lean 4, building on the `IDT_Foundations` library. There are no `sorry` axioms. The 364 machine-checked theorems constitute, to our knowledge, the first comprehensive formal treatment of signalling games in an algebraic theory of ideas.

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Part I

The Interpretation Framework

Chapter 1

Hermeneutic Foundations: Repertoires, Interpretation, and the Hermeneutic Bound

> "The limits of my language mean the limits of my world."
>
> Ludwig Wittgenstein, *Tractatus Logico-Philosophicus* (5.6)

Every human mind is a repertoire: a stock of ideas accumulated through enculturation. When a signal arrives—a gesture, an utterance, a ritual act—the receiver does not passively absorb it. They *interpret* it by composing it with their existing ideas. This chapter builds the foundational structures on which the entire sequel rests.

In the language of Bourdieu [4], a repertoire is the cognitive component of *habitus*—the durable, transposable dispositions through which agents perceive and act in the social world. In the language of Saussure [27], each element of the repertoire is a signified, and interpretation is the operation that maps a signifier to a new signified by composing it with an existing one. In the language of Peirce [24], interpretation is the generation of an *interpretant* from the encounter of a sign with a prior sign.

The payoff in a signalling game is *never* about the signal itself—it is always about the interpretations the signal produces in other minds. This insight, already present in Gadamer's *Truth and Method* [13], receives its first fully formal treatment here.

Chapter overview. We introduce four core concepts: *repertoire* (§1.1), *interpretation* and the *hermeneutic bound* (§1.2), *silence* (§1.3), and *cultural consensus* (§1.4). We close with the *telephone game* (§1.5) and structural

invariants (§1.6). All sixteen theorems are machine-verified in Lean 4 with zero `sorry` axioms.

1.1 Core Definitions

We work within an ideatic space $(\mathcal{I}, \circ, \mathbf{0}, \mathrm{d}, \sim)$ as established in the IDT Foundations.

Definition 1.1 (Repertoire)**.** A *repertoire* is a finite list of ideas:

$$\mathcal{R} = [r_1, r_2, \ldots, r_n] \in \mathcal{I}^*.$$

In Bourdieu's terms, this is the *habitus* made formal—the totality of schemas through which an agent perceives and interprets the social world.

Definition 1.2 (Interpretation)**.** When a receiver with repertoire $\mathcal{R} = [r_1, \ldots, r_n]$ encounters signal s, they produce the list of interpretations

$$\mathrm{interpret}(\mathcal{R}, s) = [r_1 \circ s,\ r_2 \circ s,\ \ldots,\ r_n \circ s].$$

This is Gadamer's *Horizontverschmelzung* (fusion of horizons): each background idea merges with the incoming signal.

In Lean 4, the definition is:

```
def interpret (rep : Repertoire I) (signal : I) : List I :=
  rep.map (fun r => IdeaticSpace.compose r signal)
```

Listing 1.1: Interpretation in Lean 4

Definition 1.3 (Maximum Interpretation Depth)**.** The *maximum interpretation depth* of a list of ideas is

$$\mathrm{maxInterpDepth}([a_1, \ldots, a_n]) = \max_i \mathrm{d}(a_i).$$

Definition 1.4 (Depth Sum)**.** The *depth sum* of a list of ideas is $\mathrm{depthSum}([a_1, \ldots, a_n]) = \sum_i \mathrm{d}(a_i)$.

1.2 The Hermeneutic Bound

"You can only understand what you are prepared to understand." The following two theorems make this aphorism precise.

Theorem 1.5 (Total Hermeneutic Bound)**.** *For any repertoire $\mathcal{R}$ and signal s,*

$$\text{depthSum}(\text{interpret}(\mathcal{R}, s)) \;\leq\; \text{depthSum}(\mathcal{R}) + |\mathcal{R}| \cdot \text{d}(s).$$

Proof sketch. By induction on $|\mathcal{R}|$. The base case ($\mathcal{R} = []$) is trivial. For $\mathcal{R} = r :: \mathcal{R}'$, we use the depth-composition bound $\text{d}(r \circ s) \leq \text{d}(r) + \text{d}(s)$ and the inductive hypothesis on $\mathcal{R}'$. The full proof is verified in `IDT_Signal_Ch01.lean`, theorem `total_hermeneutic_bound`. □

Remark 1.6 (Mass Media)**.** This theorem explains why mass media (one signal, many receivers) has bounded total interpretive impact per receiver. A richer repertoire absorbs more, but the signal's contribution scales only linearly with audience size.

Theorem 1.7 (MaxDepth Interpretation Bound)**.** *For any repertoire $\mathcal{R}$ and signal s,*

$$\text{maxInterpDepth}(\text{interpret}(\mathcal{R}, s)) \;\leq\; \text{maxInterpDepth}(\mathcal{R}) + \text{d}(s).$$

Proof sketch. By induction on $|\mathcal{R}|$. At each step, we bound $\text{d}(r \circ s) \leq \text{d}(r) + \text{d}(s)$ and propagate through the maximum. See theorem `max_interp_depth_bound`. □

Remark 1.8. Even the most sophisticated listener adds only $\text{d}(s)$ to their best existing understanding. No single signal can vault a receiver beyond their current horizon by more than the signal's intrinsic complexity.

1.3 Silence and the Unsaid

Theorem 1.9 (Silence is Transparent)**.** *The void signal leaves the mind unchanged:*

$$\text{interpret}(\mathcal{R}, \mathbf{0}) = \mathcal{R}.$$

Proof sketch. By induction on $\mathcal{R}$. For each $r \in \mathcal{R}$, the void-right identity $r \circ \mathbf{0} = r$ yields the result immediately. See `silence_is_transparent`. □

Remark 1.10 (Ritual Silence)**.** Quaker meetings, Buddhist *noble silence*, the minute of silence for the dead—all exploit this principle. By saying nothing, the receiver's existing framework remains undisturbed. The void signal is the formal foundation of taboo: what must not be said leaves the mind in its current state.

Theorem 1.11 (The Blank Receiver)**.** *A receiver whose only idea is $\mathbf{0}$ interprets any signal as the signal itself:*

$$\text{interpret}([\mathbf{0}], s) = [s].$$

Proof sketch. Direct computation: $\mathbf{0} \circ s = s$ by the void-left identity. See `blank_receiver`. □

Remark 1.12 (The Objective Observer)**.** The impossible ideal of the "objective observer" requires total cultural amnesia: a repertoire consisting solely of $\mathbf{0}$. Only then does interpretation reduce to transparent reception.

Theorem 1.13 (Negative Capability / Keats's Theorem)**.** *If* $\mathbf{0} \in \mathcal{R}$*, then* $s \in \text{interpret}(\mathcal{R}, s)$.

Proof sketch. Since $\mathbf{0} \in \mathcal{R}$, the map sends $\mathbf{0} \mapsto \mathbf{0} \circ s = s$, so s appears in the output. See `negative_capability`. □

Remark 1.14. Keats described *negative capability* as the capacity for "being in uncertainties, Mysteries, doubts, without any irritable reaching after fact & reason" [18]. In our framework, a mind retaining a slot of pure receptivity ($\mathbf{0}$ in the repertoire) captures the signal in its original form alongside culturally mediated versions.

1.4 Cultural Consensus

The next three theorems establish that resonance in backgrounds implies resonance in interpretations—the structural foundation of cultural consensus.

Theorem 1.15 (Durkheimian Consensus)**.** *If* $r_1 \sim r_2$ *and both receive signal* s*, then* $r_1 \circ s \sim r_2 \circ s$.

Proof sketch. Direct application of the resonance-compose-right lemma from IDT Foundations. See `durkheimian_consensus`. □

Remark 1.16. Two members of the same culture (shared myths, shared categories) *always* produce resonant interpretations of the same event. Cultural consensus is a structural consequence of shared background—Durkheim's *conscience collective* [9].

Theorem 1.17 (Lévi-Strauss's Mythical Transformation)**.** *For any receiver* r*, if* $s_1 \sim s_2$ *then* $r \circ s_1 \sim r \circ s_2$.

Proof sketch. By resonance-compose-left. See `mythical_transformation`. □

Remark 1.18. Ritual repetition-with-variation works because the participant's invariant background ensures that structurally similar performances produce resonant experiences. Lévi-Strauss's mythical "transformations" are perceived as "the same story told differently" precisely because the listener's background resonates with both variants [19].

Theorem 1.19 (Intercultural Solidarity / Turner's Communitas)**.** *If $r_1 \sim r_2$ and $s_1 \sim s_2$, then*

$$r_1 \circ s_1 \;\sim\; r_2 \circ s_2.$$

Proof sketch. By the full resonance-compose lemma. See `turners_communitas`. □

Remark 1.20. Members of allied cultures witnessing similar rituals produce resonant interpretations. This is the formal basis of what Turner called *communitas* [35]: the spontaneous solidarity that arises when structurally similar participants undergo similar experiences.

1.5 The Telephone Game: Cultural Transmission Chains

Theorem 1.21 (Bartlett's Serial Reproduction)**.** *Two-stage interpretation through different minds equals single-stage interpretation through the composed mind:*

$$r_B \circ (r_A \circ s) \;=\; (r_B \circ r_A) \circ s.$$

Proof sketch. This is the associativity axiom of the ideatic space, applied to the triple (r_B, r_A, s). See `bartletts_serial_reproduction`. □

Remark 1.22. Bartlett's (1932) serial reproduction experiments [2] showed that stories change predictably through chains of retelling. This theorem shows *why*: the chain of interpreters is structurally equivalent to a single composite interpreter. Oral tradition scholars can study the "net effect" of a transmission chain without tracking each individual retelling.

Theorem 1.23 (Telephone Depth Bound)**.** *The depth after two-stage interpretation satisfies*

$$\mathrm{d}(r_B \circ (r_A \circ s)) \;\leq\; \mathrm{d}(r_B) + \mathrm{d}(r_A) + \mathrm{d}(s).$$

Proof sketch. Rewrite using associativity, then apply depth-compose-le twice. See `telephone_depth`. □

Theorem 1.24 (Resonant Transmission Lines)**.** *If $r_{A_1} \sim r_{A_2}$ and $r_{B_1} \sim r_{B_2}$, then*

$$r_{B_1} \circ (r_{A_1} \circ s) \;\sim\; r_{B_2} \circ (r_{A_2} \circ s).$$

Proof sketch. Apply associativity to both sides, then use resonance-compose twice. See `resonant_transmission_lines`. □

Remark 1.25. Independent lines of ritual transmission (e.g., different griots in West Africa, different monastic orders transmitting the same texts) produce resonant outcomes if corresponding specialists resonate. Ritual orthodoxy can be maintained across geographically separated communities without central authority.

1.6 Structural Invariants of Interpretation

Theorem 1.26 (Void is Unpersuasive)**.** $\mathrm{d}(r \circ \mathbf{0}) = \mathrm{d}(r)$. *Silence contributes zero marginal depth.*

Proof sketch. By the depth-compose-void identity. See `void_is_unpersuasive`. □

Remark 1.27 (Game-theoretic significance)**.** The null message (babbling) contributes zero information. This is *why* babbling equilibria always exist: silence costs nothing and changes nothing.

Theorem 1.28 (Interpretation Conserves Count)**.** $|\,\mathrm{interpret}(\mathcal{R}, s)| = |\mathcal{R}|$.

Proof sketch. By the length-preserving property of `List.map`. See `interpretation_conserves` □

Theorem 1.29 (Durkheim's Division of Labor)**.** $\mathrm{interpret}(\mathcal{R}_1 \mathbin{++} \mathcal{R}_2, s) = \mathrm{interpret}(\mathcal{R}_1, s) \mathbin{++} \mathrm{interpret}(\mathcal{R}_2, s)$.

Proof sketch. By `List.map_append`. See `durkheims_division_of_labor`. □

Remark 1.30. A polymath's interpretation equals the union of specialist interpretations. Interdisciplinary understanding is *not* synergistic—it is the disjoint union of specialist understandings. This formalizes Durkheim's cognitive division of labor [9].

Theorem 1.31 (Silence Preserves DepthSum)**.** $\mathrm{depthSum}(\mathrm{interpret}(\mathcal{R}, \mathbf{0})) = \mathrm{depthSum}(\mathcal{R})$.

Proof sketch. Follows immediately from Theorem 1.9. See `silence_preserves_depthSum`. □

Theorem 1.32 (Three-Stage Telephone)**.**

$$r_C \circ (r_B \circ (r_A \circ s)) \;=\; ((r_C \circ r_B) \circ r_A) \circ s.$$

Proof sketch. Three applications of associativity. See `three_stage_telephone`. □

Remark 1.33. Arbitrarily long chains of cultural transmission collapse into a single composite interpreter. This is why we can meaningfully speak of "the Western tradition" as a single interpretive lens, despite it being the product of thousands of individual acts of interpretation.

Chapter Summary

This chapter established the foundational apparatus of the signalling-game framework within IDT. We introduced *repertoires* as formal models of culturally constituted minds, and *interpretation* as compositional fusion of horizons. The *Total Hermeneutic Bound* (Theorem 1.5) shows that no signal can inject more complexity into a mind than the signal's depth times the repertoire's size. *Silence is Transparent* (Theorem 1.9) provides the formal foundation for babbling equilibria. The *Durkheimian Consensus* (Theorem 1.15) and *Turner's Communitas* (Theorem 1.19) establish that resonant backgrounds guarantee resonant interpretations, providing the formal basis of cultural consensus. The *Telephone Game* theorems (Theorems 1.21–1.24) show that sequential interpretation chains collapse into single composite interpreters via associativity.

All sixteen theorems are verified in `FormalAnthropology/IDT_Signal_Ch01.lean`.

Chapter 2

The Signalling Problem

> "Words are also actions, and actions are a kind of words."
>
> Ralph Waldo Emerson

In every human society, actors choose signals strategically: a politician picks rhetoric, a suitor offers gifts, a priest designs liturgy. The sender's fundamental problem is that they cannot control how receivers will *interpret* the signal—each receiver composes the signal with their own repertoire.

This chapter formalises the core game-theoretic structure that governs strategic communication. Following Spence [32] and Crawford & Sobel [6], we model the signalling problem as a game in which a sender chooses a signal, a receiver interprets it through their repertoire, and both receive payoffs from the resulting interpretations. The crucial departure from classical game theory is that payoffs depend not on the signal itself but on what receivers *make of it*—how it composes with their existing repertoire of ideas.

Chapter overview. We define the *signalling game* (§2.1), establish the *babbling equilibrium* (§2.2), analyse *audience structure* (§2.3), prove *depth bounds* on outcomes (§2.4), study *resonance in game outcomes* (§2.5), and explore *strategic properties* (§2.6). All eighteen theorems are machine-verified.

2.1 The Signalling Game

Definition 2.1 (Signalling Game)**.** A *signalling game* $G = (\mathcal{R}, \pi_S, \pi_R)$ consists of:

- A receiver repertoire $\mathcal{R} \in \mathcal{I}^*$;
- A sender payoff function $\pi_S : \mathcal{I}^* \to \mathbb{N}$;

- A receiver payoff function $\pi_R : \mathcal{I}^* \to \mathbb{N}$.

The payoff depends on the *interpretation* (the list of composed ideas), not the signal directly.

```
structure SignallingGame (I : Type*) [IdeaticSpace I]
    where
  repertoire : Repertoire I
  sender_payoff : List I -> Nat
  receiver_payoff : List I -> Nat
```

Listing 2.1: SignallingGame in Lean 4

Definition 2.2 (Outcome)**.** The *outcome* of sending signal s in game G is

$$\text{outcome}(G, s) = \text{interpret}(G.\mathcal{R}, s).$$

Definition 2.3 (Optimal Signal)**.** A signal s is *optimal* for the sender if no alternative yields strictly higher payoff:

$$\forall s'.\ \pi_S(\text{outcome}(G, s')) \leq \pi_S(\text{outcome}(G, s)).$$

Definition 2.4 (Constant Sender Payoff)**.** A game has *constant sender payoff* if $\pi_S(\mathbf{x}) = \pi_S(\mathbf{y})$ for all interpretation lists $\mathbf{x}, \mathbf{y}$.

2.2 Babbling Equilibrium

Theorem 2.5 (Babbling Outcome)**.** *Sending void produces the original repertoire unchanged:*

$$\text{outcome}(G, \mathbf{0}) = G.\mathcal{R}.$$

Proof sketch. Unfold the definitions and apply Theorem 1.9 from Chapter 1. See `babbling_outcome`. □

Remark 2.6. The "babbling equilibrium" of Crawford & Sobel [6] exists in every signalling game because silence is transparent—the receiver's mind is left exactly as it was.

Theorem 2.7 (Constant Payoff Babbling Optimality)**.** *If sender payoff is constant, every signal is optimal—including void.*

Proof sketch. If π_S is constant, $\pi_S(\text{outcome}(G, s')) = \pi_S(\text{outcome}(G, s))$ for all s, s', so the optimality condition holds trivially. See `constant_payoff_optimal`. □

Remark 2.8. When the sender genuinely does not care what the receiver thinks, information transmission collapses entirely. This is the cheapest of cheap talk.

Theorem 2.9 (Outcome Length Invariance)**.** *Every signal produces the same number of interpretations:*

$$|\,\mathrm{outcome}(G, s)| = |G.\mathcal{R}|.$$

Proof sketch. By Theorem 1.28. See `outcome_length`. □

Remark 2.10. You cannot make someone think *more* thoughts by choosing a cleverer signal. The number of schemas activated is a fixed property of the receiver, not the sender.

Theorem 2.11 (Singleton Repertoire Outcome)**.** *A receiver with a single idea r produces $[r \circ s]$.*

Proof sketch. Direct computation. See `singleton_outcome`. □

Remark 2.12. The fanatic (single-idea mind) has a perfectly predictable interpretation of any signal. Totalitarian receivers are, paradoxically, the easiest to communicate with.

2.3 Audience Structure

Theorem 2.13 (Merged Audience)**.** *Signalling to a combined audience equals the concatenation of separate outcomes:*

$$\mathrm{outcome}(\mathcal{R}_1 \mathbin{+\!\!+} \mathcal{R}_2, s) = \mathrm{outcome}(\mathcal{R}_1, s) \mathbin{+\!\!+} \mathrm{outcome}(\mathcal{R}_2, s).$$

Proof sketch. By `List.map_append`. See `merged_audience_outcome`. □

Remark 2.14. A political speech to a diverse audience (workers ∪ elites) produces interpretations that are the disjoint union of what each subgroup would have produced separately. There is no "synergistic" interpretation effect from mere co-presence.

Theorem 2.15 (Empty Audience)**.** *Signalling to no one produces nothing:* $\mathrm{outcome}([], s) = []$.

Proof sketch. By definition. See `empty_audience`. □

Remark 2.16. If nobody is there to interpret, there are no interpretations. A signal without a receiver is not a communicative act—formalising Wittgenstein's point that meaning requires use [38].

Theorem 2.17 (Blank Audience Transparency)**.** *A receiver whose repertoire is all void interprets any signal as n copies of the signal itself:*

$$\text{outcome}(\text{replicate}(n, \mathbf{0}), s) = \text{replicate}(n, s).$$

Proof sketch. By induction on n, using $\mathbf{0} \circ s = s$ at each step. See `blank_audience`. □

2.4 Depth Bounds on Outcomes

Theorem 2.18 (Outcome Depth Bound)**.**

$$\text{depthSum}(\text{outcome}(G, s)) \leq \text{depthSum}(G.\mathcal{R}) + |G.\mathcal{R}| \cdot \text{d}(s).$$

Proof sketch. Direct application of the Total Hermeneutic Bound (Theorem 1.5). See `outcome_depth_bound`. □

Remark 2.19. The sender's capacity to inject complexity into the receiver's mind is linearly bounded. This is why propaganda targets breadth (many receivers) rather than depth (one deep signal).

Theorem 2.20 (Void Signal Zero Cost)**.** *The void signal adds zero total depth to the outcome:*

$$\text{depthSum}(\text{outcome}(G, \mathbf{0})) = \text{depthSum}(G.\mathcal{R}).$$

Proof sketch. By Theorem 1.31. See `void_signal_zero_cost`. □

2.5 Resonance in Game Outcomes

Theorem 2.21 (Resonant Repertoire Consensus)**.** *If all elements in the repertoire resonate with a fixed idea a, then all interpretations of any signal also resonate with $a \circ s$.*

Proof sketch. For each $r \in \mathcal{R}$ with $r \sim a$, by Theorem 1.15 we get $r \circ s \sim a \circ s$. See `resonant_repertoire_consensus`. □

Remark 2.22. A culturally homogeneous audience (everyone resonates with the same archetype) produces interpretations that all resonate with that archetype's interpretation. Cultural consensus is self-reinforcing.

Theorem 2.23 (Resonant Signals Produce Resonant Outcomes)**.** *If $s_1 \sim s_2$, then for each position i, the interpretations $r_i \circ s_1$ and $r_i \circ s_2$ resonate.*

Proof sketch. By induction on the repertoire, applying resonance-compose-left at each step. See `resonant_signals_resonant_outcomes`. □

2.6 Strategic Properties

Theorem 2.24 (Self-Signalling). *If sender_idea* $\in \mathcal{R}$*, then sender_idea* $\circ$ $s \in \mathrm{outcome}(G, s)$.

Proof sketch. The sender's idea is mapped to its composition with s by the interpretation function. See `self_signalling`. □

Remark 2.25. An insider—someone whose perspective is represented in the audience— can always produce resonant interpretations by sending signals resonant with their own position.

Theorem 2.26 (Composition Signal).

$$\mathrm{outcome}(\mathcal{R}, s_1 \circ s_2) = \mathrm{outcome}(\mathrm{interpret}(\mathcal{R}, s_1), s_2).$$

Proof sketch. By associativity of composition: for each r, $r \circ (s_1 \circ s_2) = (r \circ s_1) \circ s_2$. See `composition_signal`. □

Remark 2.27. A complex ritual (compound signal) has the same effect as first "priming" the audience with s_1 and then delivering s_2. Ritual structure exploits associativity.

Theorem 2.28 (Void Prepend). $\mathrm{outcome}(\mathcal{R}, \mathbf{0} \circ s) = \mathrm{outcome}(\mathcal{R}, s)$.

Proof sketch. By void-right: $r \circ (\mathbf{0} \circ s) = r \circ s$ after applying associativity and the void-right identity. See `void_prepend_signal`. □

Theorem 2.29 (Sequential Signal).

$$\mathrm{interpret}(\mathrm{interpret}(\mathcal{R}, s_1), s_2) = \mathrm{interpret}(\mathcal{R}, s_1 \circ s_2).$$

Proof sketch. By functoriality of map and associativity. See `sequential_signal`. □

Remark 2.30. Multi-stage rituals (initiation sequences, liturgical calendars) produce the same interpretive effect as a single complex ritual. Ceremonial structure is algebraically decomposable.

Theorem 2.31 (Void Captures Signal). *If* $\mathbf{0} \in G.\mathcal{R}$*, then* $s \in \mathrm{outcome}(G, s)$.

Proof sketch. By Theorem 1.13. See `void_captures_signal`. □

Remark 2.32. An "open-minded" receiver (one who retains $\mathbf{0}$ = pure receptivity) always captures the sender's intended signal.

Theorem 2.33 (MaxDepth Outcome Bound).

$$\mathrm{maxInterpDepth}(\mathrm{outcome}(G, s)) \leq \mathrm{maxInterpDepth}(G.\mathcal{R}) + \mathrm{d}(s).$$

Proof sketch. By Theorem 1.7. See `max_outcome_depth`. □

Remark 2.34. No matter how profound a signal, the deepest interpretation anyone can produce is bounded by their existing depth plus the signal's depth. The receiver's capacity limits understanding, not the sender's intent—a formalisation of "you can't teach beyond the student's readiness."

Theorem 2.35 (Babbling Payoff)**.**

$$\pi_S(\text{outcome}(G, \mathbf{0})) = \pi_S(G.\mathcal{R}).$$

Proof sketch. By Theorem 2.5. See `babbling_payoff`. □

Remark 2.36. The babbling payoff equals the "status quo" payoff. A signal is worth sending only if it yields strictly more than silence.

Chapter Summary

This chapter formalised the signalling problem within IDT. The central game-theoretic structure—a signalling game with repertoire-dependent payoffs—was introduced, and the *babbling equilibrium* (Theorem 2.5) was established: silence always leaves the receiver's mind unchanged. We showed that outcomes have invariant length (Theorem 6.10), bounded depth (Theorem 6.14), and that resonant signals produce resonant outcomes (Theorem 6.16). The composition signal theorem (Theorem 2.26) reveals that compound signals decompose via associativity, providing the algebraic foundation for analysing multi-stage rituals.

All eighteen theorems are verified in `FormalAnthropology/IDT_Signal_Ch02.lean`.

Chapter 3

Costly Depth and Ritual Sacrifice

> "A signal that is cheap to produce is a signal that is easy to fake."
>
> Amotz Zahavi, *The Handicap Principle*

Zahavi's (1975) handicap principle [39] and Sosis's (2003) costly signalling theory of ritual [31] propose that honest communication requires *costly* signals. The depth of an idea—its structural complexity—is a natural cost metric: deeper signals require more cognitive resources to produce and maintain.

In classical signalling theory (Spence [32]), the cost of education serves as a separating device between high-ability and low-ability workers. In our framework, the cost of a signal is its *depth*: the structural complexity that must be generated and maintained. A simple slogan has low cost; a multi-layered ritual performance has high cost.

The key anthropological insight is that rituals involving genuine sacrifice—scarification, potlatch, fasting—are *honest depth signals*: they cannot be cheaply faked because the depth (complexity) is intrinsic to the act itself.

Chapter overview. We identify depth with signal cost (§3.1), study iterated signals and ritual repetition (§3.2), formalise the single-crossing property (§3.3), bound interpretation cost (§3.4), and analyse the interaction between resonance and cost (§3.5).

3.1 Depth as Signal Cost

Definition 3.1 (Signal Cost)**.** The *cost* of signal s is its depth: $\mathrm{signalCost}(s) = \mathrm{d}(s)$. In Zahavi's handicap principle, this is the metabolic, social, or cognitive cost of producing the signal.

Theorem 3.2 (Void is Costless)**.** $\mathrm{signalCost}(\mathbf{0}) = 0$.

Proof sketch. By $\mathrm{d}(\mathbf{0}) = 0$. See `void_costless`. □

Remark 3.3. Silence is always free. Every signalling system must contend with the zero-cost outside option of saying nothing.

Theorem 3.4 (Cost Subadditivity)**.**

$$\mathrm{signalCost}(s_1 \circ s_2) \leq \mathrm{signalCost}(s_1) + \mathrm{signalCost}(s_2).$$

Proof sketch. By the depth-compose bound. See `cost_subadditive`. □

Remark 3.5. A two-part ritual (fasting + scarification) costs at most as much as the parts separately. Compound signals can be *cheaper* than their parts (subadditivity), but never more expensive. This formalises the "economies of ritual combination" observed in initiation ceremonies that bundle multiple ordeals.

Theorem 3.6 (Void Right-Compose Cost)**.** $\mathrm{signalCost}(s \circ \mathbf{0}) = \mathrm{signalCost}(s)$.

Proof sketch. By $\mathrm{d}(s \circ \mathbf{0}) = \mathrm{d}(s)$. See `void_compose_cost`. □

Theorem 3.7 (Void Left-Compose Cost)**.** $\mathrm{signalCost}(\mathbf{0} \circ s) = \mathrm{signalCost}(s)$.

Proof sketch. By $\mathrm{d}(\mathbf{0} \circ s) = \mathrm{d}(s)$. See `void_left_compose_cost`. □

Remark 3.8. You cannot fake costly signals by "adding silence." Prefixing or appending a dramatic pause to a sacrifice does not change its cost.

Theorem 3.9 (Trivial Composition is Costless)**.** *If* $\mathrm{signalCost}(s_1) = 0$ *and* $\mathrm{signalCost}(s_2) = 0$*, then* $\mathrm{signalCost}(s_1 \circ s_2) = 0$.

Proof sketch. By the depth-zero-closed property of ideatic spaces. See `trivial_compose_costl` □

Remark 3.10. Banality composed with banality remains banal. You cannot bootstrap costly signalling from costless components—depth must come from somewhere.

3.2 Iterated Signals and Ritual Repetition

Theorem 3.11 (Iteration Cost Bound)**.** *For n-fold repetition of signal s:*

$$\mathrm{signalCost}(\mathrm{composeN}(s, n)) \leq n \cdot \mathrm{signalCost}(s).$$

Proof sketch. By induction on n, using subadditivity at each step. See `iteration_cost_bound`. □

Remark 3.12. Ritual repetition (daily prayer, weekly sabbath, annual pilgrimage) has linearly bounded cumulative cost. A devotee who prays 5 times daily pays at most 5× the cost of a single prayer. This linear scaling is why religious systems can demand iterative commitment without unbounded cost.

Theorem 3.13 (Zero-Cost Iteration)**.** $\mathrm{signalCost}(\mathrm{composeN}(\mathbf{0}, n)) = 0$ *for all* n.

Proof sketch. $\mathrm{composeN}(\mathbf{0}, n) = \mathbf{0}$ by the void-composeN identity, so $\mathrm{d} = 0$. See `zero_cost_iteration`. □

Remark 3.14. Empty ritual (void composed with void, repeated endlessly) never generates genuine cost. "Going through the motions" with no internal depth is formally costless.

Theorem 3.15 (Single Repetition Cost)**.** $\mathrm{signalCost}(\mathrm{composeN}(s, 1)) \leq \mathrm{signalCost}(s)$.

Proof sketch. Specialise Theorem 3.11 with $n = 1$. See `single_rep_cost`. □

3.3 The Single-Crossing Property

Definition 3.16 (Costly Signal Model)**.** A *costly signal model* consists of two cost functions:

- $c_H : \mathcal{I} \to \mathbb{N}$ (high-quality type), with $c_H(s) \leq \mathrm{d}(s)$;
- $c_L : \mathcal{I} \to \mathbb{N}$ (low-quality type), with $\mathrm{d}(s) \leq c_L(s)$.

The high type can produce depth efficiently; the low type cannot.

```
structure CostlySignalModel (I : Type*) [IdeaticSpace
    I] where
  highTypeCost : I -> Nat
  lowTypeCost : I -> Nat
  high_efficient : forall s, highTypeCost s <= depth s
  low_costly : forall s, depth s <= lowTypeCost s
```

Listing 3.1: CostlySignalModel in Lean 4

Theorem 3.17 (High Type Cost Advantage)**.** *For any costly signal model and any signal* s*:* $c_H(s) \leq c_L(s)$.

Proof sketch. Chain the two constraints: $c_H(s) \leq \mathrm{d}(s) \leq c_L(s)$. See `high_type_advantag` □

Remark 3.18. In Sosis's theory, genuine believers find costly rituals less onerous than hypocrites. The devout faster suffers less than the faker because their internal "type" (depth of belief) makes the signal cheaper to produce. This is the formal foundation of ritual honesty.

Theorem 3.19 (Void is Free for All Types)**.** $c_H(\mathbf{0}) \leq 0$ *(and hence* $c_H(\mathbf{0}) = 0$*).*

Proof sketch. $c_H(\mathbf{0}) \leq \mathrm{d}(\mathbf{0}) = 0$. See `void_free_all_types`. □

Remark 3.20. The babbling option is costless for all types. Separation in costly signalling must come from *non-void* signals—only signals with genuine depth can separate types.

3.4 Interpretation Cost and Audience Effects

Theorem 3.21 (Total Interpretation Cost Bound)**.**

$$\mathrm{depthSum}(\mathrm{interpret}(\mathcal{R}, s)) \leq \mathrm{depthSum}(\mathcal{R}) + |\mathcal{R}| \cdot \mathrm{signalCost}(s).$$

Proof sketch. By the Total Hermeneutic Bound (Theorem 1.5), noting that $\mathrm{signalCost}(s) = \mathrm{d}(s)$. See `total_interpretation_cost`. □

Remark 3.22. The "impact" of a costly signal on an audience is bounded by the signal's cost times the audience size plus the audience's existing complexity. A potlatch (costly gift-giving) impresses proportionally to its cost × audience.

Theorem 3.23 (Costless Signal Preserves Total Depth)**.** *If* $\mathrm{signalCost}(s) = 0$*, then* $\mathrm{depthSum}(\mathrm{interpret}(\mathcal{R}, s)) \leq \mathrm{depthSum}(\mathcal{R})$.

Proof sketch. Set $\mathrm{d}(s) = 0$ in the hermeneutic bound to eliminate the linear term. See `costless_preserves_depth`. □

Theorem 3.24 (Atomic Signal Bounded Cost)**.** *If* s *is atomic, then* $\mathrm{signalCost}(s) \leq 1$.

Proof sketch. By $\mathrm{d}(s) \leq 1$ for atomic ideas. See `atomic_signal_cost`. □

Theorem 3.25 (Compound Atomic Cost)**.** *If* s_1, s_2 *are both atomic, then* $\mathrm{signalCost}(s_1 \circ s_2) \leq 2$.

Proof sketch. By $\mathrm{d}(s_1 \circ s_2) \leq \mathrm{d}(s_1) + \mathrm{d}(s_2) \leq 2$. See `compound_atomic_cost`. □

Remark 3.26. The simplest possible compound ritual (two atomic acts) has strictly bounded cost—complex enough to potentially separate types but simple enough to be tractable.

3.5 Resonance and Cost Interaction

Theorem 3.27 (Resonant Signals Have Comparable Outcomes)**.** *If $s_1 \sim s_2$, then for each $r \in \mathcal{R}$, $r \circ s_1 \sim r \circ s_2$.*

Proof sketch. By resonance-compose-left applied to each repertoire element. See `resonant_signals_comparable`. □

Remark 3.28. Two rituals that "resonate" produce similar effects on any audience, even if one costs more than the other. This is why cheap imitations of costly rituals can sometimes "work"—if they resonate with the original, the audience's interpretations resonate.

Theorem 3.29 (Self-Resonant Interpretation)**.** *For any repertoire $\mathcal{R}$ and signal s,* $\mathrm{interpret}(\mathcal{R}, s)$ *is pairwise self-resonant.*

Proof sketch. Apply Theorem 3.27 with $s_1 = s_2 = s$, using reflexivity of resonance. See `self_resonant_interpretation`. □

Theorem 3.30 (Bounded Repertoire, Bounded Outcome)**.** *If $\mathrm{d}(r) \leq d$ for all $r \in \mathcal{R}$, then each interpretation $x \in \mathrm{interpret}(\mathcal{R}, s)$ satisfies $\mathrm{d}(x) \leq d + \mathrm{signalCost}(s)$.*

Proof sketch. For each $r \in \mathcal{R}$, $\mathrm{d}(r \circ s) \leq \mathrm{d}(r) + \mathrm{d}(s) \leq d + \mathrm{d}(s)$. See `bounded_repertoire_bounded_outcome`. □

Remark 3.31. Receivers with bounded sophistication cannot produce arbitrarily deep interpretations. Even the most costly signal only adds its own cost to each receiver's maximum capacity.

Theorem 3.32 (Interpretation Filtration)**.** *If $\mathcal{R} \subseteq \mathcal{F}_n$ (the depth-n filtration), then* $\mathrm{interpret}(\mathcal{R}, s) \subseteq \mathcal{F}_{n+\mathrm{signalCost}(s)}$.

Proof sketch. Each $r \in \mathcal{R}$ has $\mathrm{d}(r) \leq n$, so $\mathrm{d}(r \circ s) \leq n + \mathrm{d}(s)$. See `interpretation_filtration`. □

Remark 3.33. Costly signals push interpretations up the depth hierarchy by at most their cost. A depth-3 ritual received by depth-5 minds produces at most depth-8 interpretations. This is the formal "budget" of cultural transformation through signalling.

Chapter Summary

This chapter connected IDT depth to signal cost theory. The identification $\mathrm{signalCost}(s) = \mathrm{d}(s)$ allowed us to prove that void is costless (Theorem 3.2), cost is subadditive (Theorem 3.4), and ritual repetition has linearly bounded cumulative cost (Theorem 3.11). The *single-crossing property* (Theorem 3.17) establishes that high-quality types pay less for depth, providing the formal foundation of Zahavi's handicap principle and Sosis's costly signalling theory of ritual. Resonant signals produce comparable outcomes regardless of cost difference (Theorem 3.27), and the *interpretation filtration* theorem (Theorem 3.32) shows that costly signals push interpretations up the depth hierarchy by exactly their cost.

All eighteen theorems are verified in `FormalAnthropology/IDT_Signal_Ch03.lean`.

Chapter 4

Resonance Activation and Cultural Priming

> "The question of relevance is the question of what is activated."
>
> Dan Sperber, *Explaining Culture*

When a signal arrives at a mind, it does not activate the entire repertoire uniformly. It selectively *activates* those ideas with which it resonates—Sperber's (1996) relevance theory [33] formalised. A Catholic seeing a cross activates christological schemas. A Hindu seeing the same cross might activate geometric symmetry schemas. The signal is identical; the activation pattern differs because different repertoire elements resonate with it.

This chapter builds the formal theory of activation: signals do not just produce interpretations—they *select which interpretations matter* through resonance. We connect this to Bourdieu's *habitus* [4]: the culturally acquired dispositions that determine what is "relevant" are precisely the resonance structure of the repertoire.

Chapter overview. We define activation (§4.1), study its interaction with interpretation (§4.2), analyse sequential (double) activation (§4.3), establish depth bounds (§4.4), and trace resonance chains (§4.5).

4.1 Activation

Definition 4.1 (Activation)**.** An idea r is *activated* by signal s if they resonate: $\text{activated}(r, s) \iff r \sim s$. Activation is the formal counterpart of Sperber's "relevance": a signal is relevant to a schema if and only if it resonates with it.

Definition 4.2 (Activated Subset)**.** The *activated subset* of repertoire $\mathcal{R}$ under selection predicate σ is $\mathrm{activatedSubset}(\mathcal{R}, \sigma) = \mathrm{filter}(\sigma, \mathcal{R})$.

Definition 4.3 (Activated Interpretations)**.** The *activated interpretations* compose only the selected elements with the signal:

$$\mathrm{activatedInterpretations}(\mathcal{R}, s, \sigma) = \mathrm{map}(\lambda r.\ r \circ s,\ \mathrm{filter}(\sigma, \mathcal{R})).$$

This models selective attention: only relevant schemas participate in interpretation.

Theorem 4.4 (Self-Activation)**.** *Every idea activates itself:* $\mathrm{activated}(r, r)$ *for all* r.

Proof sketch. By reflexivity of resonance. See `self_activation`. □

Remark 4.5. The formal basis of self-recognition. Familiar symbols are immediately meaningful because they resonate with what is already there.

Theorem 4.6 (Activation Symmetry)**.** *If* $\mathrm{activated}(r, s)$*, then* $\mathrm{activated}(s, r)$.

Proof sketch. By symmetry of resonance. See `activation_symm`. □

Remark 4.7. Relevance is reciprocal. If a Christian schema is activated by a cross, then the cross-schema would be activated by a Christian symbol. Cultural exchange is inherently bidirectional.

Theorem 4.8 (Void Self-Activates)**.** $\mathrm{activated}(\mathbf{0}, \mathbf{0})$.

Proof sketch. By reflexivity: $\mathbf{0} \sim \mathbf{0}$. See `void_self_activates`. □

Theorem 4.9 (Composition Preserves Activation (Right))**.** *If* $\mathrm{activated}(r, s)$*, then* $\mathrm{activated}(r \circ c,\ s \circ c)$.

Proof sketch. By resonance-compose-right. See `activation_compose_right`. □

Theorem 4.10 (Composition Preserves Activation (Left))**.** *If* $\mathrm{activated}(r, s)$*, then* $\mathrm{activated}(c \circ r,\ c \circ s)$.

Proof sketch. By resonance-compose-left. See `activation_compose_left`. □

Remark 4.11. A ritual frame (opening prayer, spatial arrangement) that precedes two resonant symbols does not destroy their mutual activation. Context preserves relevance.

4.2 Activation and Interpretation Interaction

Theorem 4.12 (Activated Interpretations Are a Sublist)**.**

$$|\,\mathrm{activatedInterpretations}(\mathcal{R}, s, \sigma)| \leq |\,\mathrm{interpret}(\mathcal{R}, s)|.$$

Proof sketch. Filtering can only reduce list length; mapping preserves it. See `activated_interp_sublist`. □

Remark 4.13. Attention is a filter, not a generator. Selective attention can only *reduce* the set of interpretations, never add new ones.

Theorem 4.14 (Universal Activation = Full Interpretation)**.** *If* $\sigma(r) = \texttt{true}$ *for all* r*, then* $\mathrm{activatedInterpretations}(\mathcal{R}, s, \sigma) = \mathrm{interpret}(\mathcal{R}, s)$*.*

Proof sketch. Filtering with the constant-true predicate is the identity. See `universal_activation`. □

Remark 4.15. A mind with zero attentional filters (the "open mind" ideal) interprets everything. Pure receptivity means every schema participates.

Theorem 4.16 (Null Activation = Empty Interpretation)**.** *If* $\sigma(r) = \texttt{false}$ *for all* r*, then* $\mathrm{activatedInterpretations}(\mathcal{R}, s, \sigma) = []$*.*

Proof sketch. Filtering with the constant-false predicate yields the empty list. See `null_activation`. □

Remark 4.17. Total cognitive shutdown—when nothing in your repertoire is deemed relevant—produces zero interpretations. This models culture shock or the "deer in headlights" response to incomprehensible stimuli.

4.3 Double Activation: Sequential Signals

Theorem 4.18 (Sequential Interpretation Associativity)**.**

$$\mathrm{interpret}(\mathrm{interpret}(\mathcal{R}, s_1), s_2) = \mathrm{interpret}(\mathcal{R}, s_1 \circ s_2).$$

Proof sketch. By functoriality of map and associativity of composition. See `sequential_activation`. □

Remark 4.19. Cultural priming works because sequential exposure to signals is equivalent to a single compound signal. Ritual sequences (opening chant → sacrifice → closing prayer) produce the same interpretations as a single complex ritual.

Theorem 4.20 (Resonant Sequential Identity)**.** *For any background r and signals s_1, s_2:*

$$r \circ (s_1 \circ s_2) = (r \circ s_1) \circ s_2.$$

Proof sketch. By associativity. See `resonant_sequential_identity`. □

Theorem 4.21 (Void Priming is Identity)**.**

$$\text{interpret}(\text{interpret}(\mathcal{R}, \mathbf{0}), s) = \text{interpret}(\mathcal{R}, s).$$

Proof sketch. $\text{interpret}(\mathcal{R}, \mathbf{0}) = \mathcal{R}$ by silence transparency, so the second interpretation acts on the original repertoire. See `void_priming`. □

Remark 4.22. A ritual preceded by silence produces the same interpretations as the ritual alone. The "pregnant pause" before a sermon adds drama but not information.

Theorem 4.23 (Post-Void is Identity)**.**

$$\text{interpret}(\text{interpret}(\mathcal{R}, s), \mathbf{0}) = \text{interpret}(\mathcal{R}, s).$$

Proof sketch. By silence transparency applied to $\text{interpret}(\mathcal{R}, s)$. See `post_void`. □

4.4 Activation Depth Bounds

Theorem 4.24 (Activated Interpretation Depth Bound)**.**

$$\text{maxInterpDepth}(\text{activatedInterpretations}(\mathcal{R}, s, \sigma)) \leq \text{maxInterpDepth}(\mathcal{R}) + \text{d}(s).$$

Proof sketch. By induction on $|\mathcal{R}|$. Filtering can only select elements whose depth is bounded by $\text{maxInterpDepth}(\mathcal{R})$; composition adds at most $\text{d}(s)$. See `activated_depth_bound`. □

Remark 4.25. Selective attention does not change the depth bound. Relevance-filtering is a horizontal cut, not a vertical one.

Theorem 4.26 (Activation-Filtered Count Bound)**.**

$$|\,\text{activatedInterpretations}(\mathcal{R}, s, \sigma)| \leq |\mathcal{R}|.$$

Proof sketch. By the length bound on filtered lists. See `activated_count_bound`. □

Remark 4.27. You cannot hallucinate more schemas than you actually have, no matter how stimulating the signal.

4.5 Resonance Chains and Activation Propagation

Theorem 4.28 (Resonant Backgrounds Produce Resonant Activations)**.** *If* $r_1 \sim r_2$*, then* $r_1 \circ s \ \sim \ r_2 \circ s$*.*

Proof sketch. By resonance-compose-right. See `resonant_backgrounds_resonant_acti`

□

Remark 4.29. Culturally similar people who are primed by the same stimulus produce resonant responses. This is Durkheim's collective effervescence: a crowd sharing cultural schemas, exposed to the same ritual stimulus, produces a harmonious collective response [10].

Theorem 4.30 (Activation Under Composition)**.** *If* $\mathrm{activated}(r, s_1)$ *and* $\mathrm{activated}(s_1, s_2)$*, then* $\mathrm{activated}(r \circ s_1,\ s_1 \circ s_2)$*.*

Proof sketch. By the full resonance-compose lemma. See `activation_composition_chai`

□

Remark 4.31. Resonance "propagates" through compositional chains. Multi-layered rituals create cascading activations through the audience's schemas.

Theorem 4.32 (Void Is Universally Self-Activatable)**.** $\mathrm{activated}(\mathbf{0}, \mathbf{0})$*.*

Proof sketch. By reflexivity of resonance. See `void_universally_self_activatable`.

□

Theorem 4.33 (Double Signal Depth Bound)**.**

$$\mathrm{d}(r_2 \circ (r_1 \circ s)) \leq \mathrm{d}(r_2) + \mathrm{d}(r_1) + \mathrm{d}(s).$$

Proof sketch. Apply depth-compose-le twice. See `double_signal_depth_bound`.

□

Remark 4.34. Cultural complexity does not compound super-linearly through chains of interpretation. A three-layer cultural transmission has bounded total complexity.

Chapter Summary

This chapter formalised the theory of *resonance activation*: signals selectively activate repertoire elements through resonance, producing filtered interpretations. Self-activation (Theorem 4.4) and activation symmetry (Theorem 4.6) establish the basic structure. Universal activation recovers full interpretation (Theorem 4.14), while null activation produces cognitive shutdown (Theorem 4.16). Sequential signals compose associatively (Theorem 4.18), and void

priming is the identity (Theorem 4.21). The activated depth bound (Theorem 4.24) shows that selective attention does not change the depth ceiling. Resonance propagates through activation chains (Theorem 4.30), providing the formal basis for understanding how multi-layered rituals create cascading effects through culturally constituted minds.

All eighteen theorems are verified in `FormalAnthropology/IDT_Signal_Ch04.lean`.

Chapter 5

Misinterpretation and Cultural Distance

> "Understanding is always more than merely re-creating someone else's meaning."
>
> Hans-Georg Gadamer, *Truth and Method*

The sender intends the receiver to compose signal s with background r^* (the "shared understanding"). But the receiver actually composes with r_{actual}—their real, possibly different, cultural background. The gap between $r^* \circ s$ and $r_{\text{actual}} \circ s$ is the formal measure of *misinterpretation*.

This chapter develops the formal theory of interpretive divergence. Connecting to the Sapir–Whorf hypothesis [37], which holds that linguistic background shapes interpretation, and to Gadamer's hermeneutic circle [13], which insists that understanding requires pre-understanding, we show that cultural distance between backgrounds governs interpretive distance between outcomes.

Chapter overview. We define the interpretation gap (§5.1), analyse void backgrounds (§5.2), develop iterated re-interpretation (§5.3), study resonance under iteration (§5.4), and formalise cultural distance (§5.5).

5.1 The Interpretation Gap

Definition 5.1 (Intended Interpretation)**.** The sender's *intended interpretation* is $\text{intendedInterp}(r_{\text{intended}}, s) = r_{\text{intended}} \circ s$.

Definition 5.2 (Actual Interpretation)**.** The receiver's *actual interpretation* is $\text{actualInterp}(r_{\text{actual}}, s) = r_{\text{actual}} \circ s$.

Definition 5.3 (Well-Interpreted)**.** An interpretation is *well-interpreted* if the actual and intended interpretations resonate:

$$\text{wellInterpreted}(r_{\text{int}}, r_{\text{act}}, s) \iff r_{\text{int}} \circ s \sim r_{\text{act}} \circ s.$$

This is weaker than equality but stronger than "no relation."

Theorem 5.4 (Resonant Backgrounds Guarantee Well-Interpretation)**.** *If* $r_{\text{int}} \sim r_{\text{act}}$*, then the signal* s *is well-interpreted.*

Proof sketch. By resonance-compose-right applied to the resonant backgrounds. See `resonant_backgrounds_well_interpreted`. □

Remark 5.5. Members of the same culture (resonant backgrounds) always produce well-interpreted outcomes for any signal. Gadamer's "fusion of horizons" is guaranteed when horizons resonate [13].

Theorem 5.6 (Self-Interpretation Always Succeeds)**.** $\text{wellInterpreted}(r, r, s)$ *for all* r, s*.*

Proof sketch. By reflexivity of resonance on the backgrounds. See `self_interpretation`. □

Remark 5.7. You always understand yourself. Soliloquy is the degenerate case of communication.

Theorem 5.8 (Well-Interpretation is Symmetric)**.** *If* $\text{wellInterpreted}(r_1, r_2, s)$*, then* $\text{wellInterpreted}(r_2, r_1, s)$*.*

Proof sketch. By symmetry of resonance. See `well_interpretation_symm`. □

Remark 5.9. Mutual intelligibility is symmetric. If I can understand your ritual, you can understand mine. Cultural exchange is inherently bidirectional—a formalisation of the anthropological observation that first contact is always reciprocal.

5.2 The Void Background

Theorem 5.10 (Void Intended Transparency)**.** $\text{intendedInterp}(\mathbf{0}, s) = s$*.*

Proof sketch. By $\mathbf{0} \circ s = s$*.* See `void_intended_transparency`. □

Remark 5.11. A sender who assumes zero shared context intends the signal to be received "as is." Any deviation from the raw signal is the receiver's cultural contribution. This is the formal model of the "objective journalist" fantasy.

Theorem 5.12 (Void Actual Transparency)**.** $\text{actualInterp}(\mathbf{0}, s) = s$.

Proof sketch. By $\mathbf{0} \circ s = s$. See `void_actual_transparency`. □

Theorem 5.13 (Void-Void Perfect Interpretation)**.** $\text{wellInterpreted}(\mathbf{0}, \mathbf{0}, s)$ *for all* s.

Proof sketch. Both interpret s as s itself, so the interpretations are identical (and hence resonant). See `void_void_perfect`. □

Remark 5.14. In the (impossible) limit of zero culture, there is zero misinterpretation. Misunderstanding is entirely a product of cultural difference.

Theorem 5.15 (The Hermeneutic Circle)**.** *For any* r, s*:* $\text{actualInterp}(r, s) \sim \text{actualInterp}(r, s)$.

Proof sketch. By reflexivity of resonance. See `hermeneutic_circle`. □

Remark 5.16. Understanding always loops back to self-resonance. You cannot escape your own interpretive framework—even when you try to "step outside," the result still resonates with your starting point. This is the formal content of Gadamer's hermeneutic circle [13].

5.3 Iterated Re-Interpretation

Definition 5.17 (Iterated Interpretation)**.** The n-fold re-interpretation of signal s through background r is:

$$\text{iterateInterp}(r, s, n) = \begin{cases} s & \text{if } n = 0, \\ r \circ \text{iterateInterp}(r, s, n-1) & \text{if } n > 0. \end{cases}$$

This models rumination, echo chambers, and iterative cultural processing.

```
def iterateInterp (r s : I) : Nat -> I
  | 0 => s
  | n + 1 => IdeaticSpace.compose r (iterateInterp r s
      n)
```

Listing 5.1: Iterated interpretation in Lean 4

Theorem 5.18 (Zero Iteration = Raw Signal)**.** $\text{iterateInterp}(r, s, 0) = s$.

Proof sketch. By definition. See `iterate_zero`. □

Theorem 5.19 (Single Iteration = Basic Interpretation)**.** $\text{iterateInterp}(r, s, 1) = r \circ s$.

Proof sketch. By definition. See `iterate_one`. □

Theorem 5.20 (Void Iteration Identity)**.** $\mathrm{iterateInterp}(\mathbf{0}, s, n) = s$ *for all* n.

Proof sketch. By induction on n, using $\mathbf{0} \circ x = x$ at the inductive step. See `void_iterate`. □

Remark 5.21. An "empty mind" processing a signal repeatedly never changes it. Without cultural content, there is no interpretive drift. Echo chambers require content to echo.

Theorem 5.22 (Iterated Interpretation Depth Bound)**.**

$$\mathrm{d}(\mathrm{iterateInterp}(r, s, n)) \leq n \cdot \mathrm{d}(r) + \mathrm{d}(s).$$

Proof sketch. By induction on n. At step $n + 1$, the depth-compose bound gives $\mathrm{d}(r \circ x) \leq \mathrm{d}(r) + \mathrm{d}(x)$; applying the inductive hypothesis yields the result. See `iterate_depth_bound`. □

Remark 5.23. Echo chambers have linearly bounded complexity. Even infinite rumination through a fixed cultural lens can only add linear depth. Obsessive rumination is unproductive: each pass adds at most $\mathrm{d}(r)$ more complexity.

Theorem 5.24 (Void Signal Single Iteration)**.** $\mathrm{iterateInterp}(r, \mathbf{0}, 1) = r$.

Proof sketch. $r \circ \mathbf{0} = r$ by void-right. See `void_signal_iterate_one`. □

Remark 5.25. Processing "nothing" through a cultural lens yields the lens itself. An empty news feed processed through conspiracy-theory schemas yields the conspiracy schema unchanged.

5.4 Resonance Under Iteration

Theorem 5.26 (Resonant Iteration Step)**.** *If* $r_1 \sim r_2$ *and* $x_1 \sim x_2$*, then* $r_1 \circ x_1 \sim r_2 \circ x_2$.

Proof sketch. By the full resonance-compose lemma. See `resonant_iteration_step`. □

Theorem 5.27 (Full Resonant Iteration)**.** *If* $r_1 \sim r_2$*, then for all* n*:*

$$\mathrm{iterateInterp}(r_1, s, n) \sim \mathrm{iterateInterp}(r_2, s, n).$$

Proof sketch. By induction on n. Base case: $s \sim s$ by reflexivity. Inductive step: apply Theorem 5.26 with the inductive hypothesis. See `resonant_iteration`. □

Remark 5.28. Cultural similarity guarantees interpretive similarity at every stage of processing. This is why "deep" cultural understanding between allied groups is possible—resonance propagates through all layers of interpretation.

Theorem 5.29 (Self-Iteration Self-Resonance)**.** $\mathrm{iterateInterp}(r, s, n) \sim \mathrm{iterateInterp}(r, s,$ *for all* r, s, n.

Proof sketch. By reflexivity. See `self_iteration_resonance`. □

Remark 5.30. The echo chamber's output is always "consistent" in the sense of self-resonance—it just may not resonate with *anything else*. This is the formal trap of intellectual isolation.

5.5 Cultural Distance and Interpretation

Theorem 5.31 (Composition Preserves Well-Interpretation)**.** *If* s_1 *and* s_2 *are both well-interpreted under the same background pair* (r_1, r_2)*, then*

$$(r_1 \circ s_1) \circ (r_1 \circ s_2) \ \sim \ (r_2 \circ s_1) \circ (r_2 \circ s_2).$$

Proof sketch. By the full resonance-compose lemma applied to the two well-interpretation hypotheses. See `compound_well_interpreted`. □

Remark 5.32. If two cultures understand each other's simple signals, they understand compound signals too. Mutual intelligibility at the atomic level guarantees mutual intelligibility at the molecular level. This is why pidgins can bootstrap into creoles.

Theorem 5.33 (Interpretation Depth Gap)**.**

$$\mathrm{d}(\mathrm{actualInterp}(r_{\mathrm{actual}}, s)) \leq \mathrm{d}(r_{\mathrm{actual}}) + \mathrm{d}(s).$$

Proof sketch. By depth-compose-le. See `interpretation_depth_gap`. □

Remark 5.34. Misinterpretation does not create unbounded complexity. Even the most "wrong" interpretation is bounded by the receiver's actual depth plus the signal's depth.

Theorem 5.35 (Iteration Composition / Ritual Accretion)**.**

$$\mathrm{iterateInterp}(r, s, n + m) = \mathrm{iterateInterp}(r,\ \mathrm{iterateInterp}(r, s, m),\ n).$$

Proof sketch. By induction on n. See `iteration_composition`. □

Remark 5.36. A lifetime of cultural processing ($n + m$ iterations) decomposes into childhood processing (m iterations) followed by adult processing (n iterations). Cultural development is *sequentially decomposable*: we can study "what did early formation contribute?" and "what did later experience add?" as separable phases. This is the formal basis of Bourdieu's distinction between primary and secondary habitus formation [4].

Chapter Summary

This chapter developed the formal theory of misinterpretation within IDT. The *interpretation gap* between intended and actual interpretation is governed by the resonance between sender and receiver backgrounds (Theorem 5.4). Well-interpretation is symmetric (Theorem 5.8) and guaranteed for identical backgrounds (Theorem 5.6). *Iterated re-interpretation* has linearly bounded depth (Theorem 5.22), and *resonant iteration* (Theorem 5.27) shows that culturally similar backgrounds produce resonant interpretations at every stage. The *iteration composition* theorem (Theorem 5.35) provides the formal basis for decomposing cultural development into sequential phases.

This concludes Part I. We have established the interpretive framework: repertoires, signalling games, costly signals, resonance activation, and misinterpretation. Part II develops the equilibrium theory.

All eighteen theorems are verified in `FormalAnthropology/IDT_Signal_Ch05.lean`.

Part II

Equilibrium and Dynamics

Chapter 6

Interpretive Equilibrium

> "A convention is a regularity in behavior, sustained by an interest in coordination and an expectation that others will do their part."
>
> David Lewis, *Convention* (1969)

6.1 Introduction: Why Cultures Don't Change

Every anthropologist eventually confronts the same puzzle: if cultures are "arbitrary" in Saussure's sense—if there is no deep reason for driving on the left rather than the right, for shaking hands rather than bowing, for calling a dog *dog* rather than *chien*—then why do cultures persist at all? Why doesn't every generation re-randomize its conventions?

The game-theoretic answer, originating with Thomas Schelling's *Strategy of Conflict* (1960) and formalized by David Lewis in *Convention* (1969), is deceptively simple: cultural forms persist because *no individual can profitably deviate*. They are Nash equilibria of an implicit coordination game. A greeting ritual need not be optimal in any absolute sense; it need only be *locally stable*—better than any unilateral alternative.

This chapter transplants that insight into the framework of Ideatic Diffusion Theory. We define an *interpretation game*—a triple of a repertoire (the receiver's cultural background), a sender payoff, and a receiver payoff—and show that the concept of equilibrium has rich structural consequences when the action space is an ideatic space $\mathcal{I}$ equipped with composition $\circ$, void $\mathbf{0}$, depth d, and resonance $\sim$.

The central results of the chapter fall into three groups. First, *existence*: void (silence) is always an equilibrium when it weakly dominates, and

constant-payoff games admit universal equilibria (§6.3). Second, *structure*: outcomes preserve repertoire length, depth is linearly bounded, and void preserves depth exactly (§6.4–§6.5). Third, *robustness*: equilibria survive payoff agreement, monotone transforms, and void extension (§6.6–§6.8).

Throughout, we provide anthropological commentary and Lean 4 code listings that verify every theorem with zero `sorry` axioms.

6.2 Core Definitions

Definition 6.1 (Repertoire)**.** A *repertoire* is a finite list of ideas in $\mathcal{I}$:

$$\mathrm{Rep} \;=\; \mathrm{List}\,\mathcal{I}.$$

It represents the receiver's stock of cultural concepts, categories, and schemas—what they bring to any interpretive encounter.[1]

Definition 6.2 (Interpretation)**.** Given a repertoire $R = [r_1, \ldots, r_n]$ and a signal $s \in \mathcal{I}$, the *interpretation* of s through R is

$$\mathrm{interpret}(R, s) \;=\; [r_1 \circ s,\; \ldots,\; r_n \circ s].$$

Each background idea is composed with the incoming signal, producing a list of new ideas—one per schema the receiver already holds.

```
def interpret (rep : Repertoire I) (signal : I) : List
    I :=
  rep.map (fun r => IdeaticSpace.compose r signal)
```

Listing 6.1: Interpretation in Lean 4.

Definition 6.3 (Interpretation Game)**.** An *interpretation game* $G = (R, u_S, u_R)$ consists of

1. a repertoire R (the receiver's cultural background),
2. a sender payoff $u_S : \mathrm{List}\,\mathcal{I} \to \mathbb{N}$,
3. a receiver payoff $u_R : \mathrm{List}\,\mathcal{I} \to \mathbb{N}$.

Payoffs depend on the *interpretation* (the list of composed ideas), not on the signal directly. This captures the anthropological insight that meaning is always mediated by the receiver's culture.

[1] In Bourdieu's terms, the repertoire is the agent's *habitus* rendered as a finite enumeration of conceptual dispositions.

```
structure InterpGame (I : Type*) [IdeaticSpace I]
    where
  repertoire : Repertoire I
  sender_payoff : List I -> ℕ
  receiver_payoff : List I -> ℕ
```

Listing 6.2: The interpretation game structure.

Definition 6.4 (Outcome)**.** The *outcome* of sending signal s in game G is the interpretation produced by composing the repertoire with s:

$$\text{outcome}(G, s) \;=\; \text{interpret}(G.\text{Rep},\; s).$$

Definition 6.5 (Equilibrium)**.** A signal s is an *equilibrium* of game G if no alternative signal s' yields a strictly higher sender payoff:

$$\text{IsEquilibrium}(G, s) \iff \forall\, s'.\; u_S\big(\text{outcome}(G, s')\big) \;\leq\; u_S\big(\text{outcome}(G, s)\big).$$

This is the Nash equilibrium condition for the sender, given the receiver's fixed repertoire.[2]

```
def IsEquilibrium (G : InterpGame I) (s : I) : Prop :=
  forall (s' : I), G.sender_payoff (outcome G s') <=
    G.sender_payoff (outcome G s)
```

Listing 6.3: Equilibrium in Lean 4.

6.3 Babbling Equilibrium and Constant-Payoff Games

The deepest result of Lewis's *Convention* is often overlooked: in *every* signalling game, there exists a "babbling" equilibrium in which signals carry no information. In IDT, this corresponds to the void signal.

Theorem 6.6 (Outcome of Void)**.** $\text{outcome}(G, \mathbf{0}) = G.\text{Rep}$. *Silence leaves meaning unchanged: the interpretation of void through any repertoire is the repertoire itself.*

Proof sketch. By induction on the repertoire list. The base case is trivial. For the inductive step, $r \circ \mathbf{0} = r$ by the right-identity axiom of $\mathcal{I}$, so the head is preserved and the tail follows from the inductive hypothesis. □

[2] We suppress the receiver's strategic choice in this formalization, treating the repertoire as exogenous. Chapter 10 relaxes this.

This result is the linchpin of everything that follows. It says that silence is *transparent*: the receiver's world is unchanged. In anthropological terms, a ritual that communicates nothing leaves the participants exactly where they were.

Theorem 6.7 (Babbling Equilibrium / Schelling's Silence (6.1))**.** *If no signal produces a strictly better payoff than silence, then* $\mathbf{0}$ *is an equilibrium:*

$$\big(\forall\, s.\ u_S(\text{outcome}(G, s)) \leq u_S(\text{outcome}(G, \mathbf{0}))\big) \implies \text{IsEquilibrium}(G, \mathbf{0}).$$

Proof sketch. Immediate from the definition of $\mathsf{IsEquilibrium}$: the hypothesis *is* the conclusion. □

Lewis (1969) showed that babbling equilibria exist in every signalling game. In IDT, the proof is one line—because the algebraic structure of $\mathcal{I}$ (specifically, the existence of $\mathbf{0}$) guarantees a universal fallback. This is the formal foundation of Schelling's focal-point theory: when nothing you say can improve on silence, silence is optimal.

Theorem 6.8 (Constant-Payoff Equilibrium (6.2))**.** *If every signal yields the same payoff* k*, then every signal is an equilibrium:*

$$\big(\forall\, s.\ u_S(\text{outcome}(G, s)) = k\big) \implies \forall\, s.\ \text{IsEquilibrium}(G, s).$$

Proof sketch. For any candidate deviator s', rewrite both payoffs to k and apply $k \leq k$. □

Anthropologically, constant-payoff games formalize *radical cultural relativism*: if all rituals are equally valued, none faces selection pressure, and every cultural form persists indefinitely. The theorem explains why highly egalitarian societies display persistent ritual variation—there is no payoff gradient to drive convergence.

Theorem 6.9 (Equilibrium is Reflexive (6.3))**.** *Every signal is at least as good as itself:* $u_S(\text{outcome}(G, s)) \leq u_S(\text{outcome}(G, s))$.

This is the trivial direction of the Nash definition—every strategy weakly dominates itself. Though mathematically obvious, it serves as the base case for inductive proofs of equilibrium preservation in later chapters.

6.4 Outcome Structural Properties

Outcomes have a rigid structure: signals transform but never create or destroy ideas.

Theorem 6.10 (Outcome Length Invariance (6.4)). $|\text{outcome}(G, s)| = |G.\text{Rep}|$ *for all signals* s.

Proof sketch. outcome is a `List.map`, which preserves length. □

Cultural contact does not change how many schemas a society holds—it transforms each one. The "size" of a culture's conceptual inventory is invariant under external signalling. This is a non-trivial constraint: it rules out models in which signals can spontaneously generate new categories in the receiver's mind.[3]

Theorem 6.11 (Outcome Length Independence (6.5)). *Two different signals produce interpretation lists of the same length:* $|\text{outcome}(G, s_1)| = |\text{outcome}(G, s_2)|$.

An immediate corollary of Theorem 6.10: both sides equal $|G.\text{Rep}|$. Game-theoretically, the "action space" of interpretations has constant dimensionality across all possible signals.

Theorem 6.12 (Empty Repertoire Trivial Outcome (6.6)). *A receiver with no ideas produces no interpretations:* $\text{outcome}(\langle [], \ldots \rangle, s) = []$.

A mind with zero cultural content cannot interpret anything. There is no "raw perception"—interpretation requires prior ideas. This is the formal counterpart of Kant's dictum: "Intuitions without concepts are blind."

Theorem 6.13 (Singleton Repertoire Outcome (6.7)). *A single-idea mind produces exactly one interpretation:* $\text{outcome}(\langle [r], \ldots \rangle, s) = [r \circ s]$.

Monocultural societies have exactly one interpretation of every event. The singleton case is the simplest non-trivial example and will serve as a running test case in Chapters 7–10.

6.5 Depth Bounds on Outcomes

The depth function d measures interpretive complexity. How does it behave under the `outcome` operation?

Theorem 6.14 (Outcome Depth Bound (6.8)).

$$\sum_i \mathrm{d}\big(\text{outcome}(G, s)_i\big) \;\leq\; \sum_i \mathrm{d}(r_i) \;+\; |R| \cdot \mathrm{d}(s).$$

The total depth of all interpretations is bounded by the repertoire's total depth plus the repertoire size times the signal depth.

[3] Category *creation* requires mechanisms beyond simple composition—see the discussion of "conceptual blending" in Chapter 8.

Proof sketch. Induction on the repertoire. At each step, the depth of $r \circ s$ is at most $\mathrm{d}(r) + \mathrm{d}(s)$ by the subadditivity of composition depth. Summing over $|R|$ elements yields the bound. □

Game-theoretically, the sender's "persuasive reach" is linearly bounded by signal complexity times audience size. This result foreshadows the persuasion bounds of Chapter 7: even before introducing formal limits on marginal depth, we see that total impact is $O(|R| \cdot \mathrm{d}(s))$.

Theorem 6.15 (Void Outcome Preserves Depth (6.9))**.** $\mathrm{depthSum}(\mathrm{outcome}(G, \mathbf{0})) = \mathrm{depthSum}(G.\mathrm{Rep})$.

Silence preserves total interpretive complexity exactly—not just approximately, but with zero loss. The "cost" of silence is exactly zero in every dimension.

```
theorem outcome_depth_bound
    (rep : Repertoire I) (s : I) (sp rp : List I -> ℕ)
        :
    depthSum (outcome (<rep, sp, rp> : InterpGame I) s
        ) <=
    depthSum rep + rep.length * IdeaticSpace.depth s
```

Listing 6.4: Outcome depth bound in Lean 4.

6.6 Resonance and Equilibrium

Resonance ($\sim$) captures structural similarity between ideas. How does it interact with equilibrium?

Theorem 6.16 (Resonant Signals Produce Resonant Outcomes (6.10))**.** *If $s_1 \sim s_2$, then for each background idea r in the repertoire, $r \circ s_1 \sim r \circ s_2$.*

Structurally similar rituals—Christmas and Hanukkah, weddings across cultures—produce structurally similar experiences for each participant. This is why comparative anthropology is possible at all: resonant signals yield resonant interpretations, ensuring that cross-cultural comparison is not an act of pure projection.

Theorem 6.17 (Self-Resonance of Outcomes (6.11))**.** *Every outcome resonates with itself at each position: $r \circ s \sim r \circ s$.*

A trivial consequence of the reflexivity of resonance, but conceptually important: a single signal always produces self-consistent interpretations. No signal can produce an internally contradictory outcome.

Theorem 6.18 (Equilibrium Under Payoff Agreement (6.12)). *If two games G_1, G_2 have the same repertoire and their sender payoffs agree on all outcomes, then an equilibrium for G_2 is an equilibrium for G_1:*

$$\big(G_1.\mathrm{Rep} = G_2.\mathrm{Rep}\big) \wedge \big(\forall\, s.\ u_{S,1}(\mathrm{outcome}(G_1, s)) = u_{S,2}(\mathrm{outcome}(G_2, s))\big) \implies \big(\mathrm{IsEqu}$$

Proof sketch. Substitute the payoff agreement into the equilibrium condition for G_2 and apply it to G_1. □

If a cultural form is stable in one payoff environment, it remains stable in any environment that produces the same relative rankings. This is the formal version of the sociological observation that institutions survive "rebranding" as long as the underlying incentive structure is preserved.

6.7 Composition and Sequential Signalling

Cultural communication is rarely a single signal. Rituals have multiple acts; conversations have turns. This section studies how equilibrium interacts with sequential composition.

Theorem 6.19 (Sequential Signal Composition (6.13)). $(r \circ s_1) \circ s_2 = r \circ (s_1 \circ s_2)$.

A ritual with two acts can be understood as a single compound act—but only from the perspective of someone who has already absorbed the first act into their background. This is the associativity axiom of $\mathcal{I}$, applied to the signalling context.

Theorem 6.20 (Equilibrium Stability Under Void Extension (6.14)). $\mathrm{outcome}(G,\ s \circ \mathbf{0}) = \mathrm{outcome}(G, s)$.

Proof sketch. For each r in the repertoire, $r \circ (s \circ \mathbf{0}) = r \circ s$ by the right-identity axiom. Hence the map is pointwise equal. □

Padding a message with silence doesn't change the game. This is the formal version of "cheap talk is irrelevant when it carries no information." In anthropological terms, the elaborate silences that frame ritual speech (the pause before a toast, the breath before a prayer) are formally inert with respect to the equilibrium structure.

Theorem 6.21 (Dual Repertoire Combination (6.15)). $\mathrm{outcome}(\langle R_1 {+\!\!+} R_2, \ldots\rangle, s) = \mathrm{outcome}(\langle R_1, \ldots\rangle, s) {+\!\!+}\ \mathrm{outcome}(\langle R_2, \ldots\rangle, s)$.

A bilingual mind's interpretation is the union of each language's interpretation. There is no mysterious "third space" of bilingual meaning—just the disjoint combination.[4]

```
theorem dual_repertoire_outcome
    (rep_1 rep_2 : Repertoire I) (s : I) (sp rp : List
        I -> ℕ) :
    outcome (<rep_1 ++ rep_2, sp, rp> : InterpGame I)
        s =
    outcome (<rep_1, sp, rp> : InterpGame I) s ++
    outcome (<rep_2, sp, rp> : InterpGame I) s
```

Listing 6.5: Dual repertoire combination in Lean 4.

6.8 Advanced Equilibrium Properties

We close with three theorems on the robustness of equilibria under transformations of the payoff structure.

Theorem 6.22 (Maximum Payoff Equilibrium (6.16))**.** *If a signal s achieves the maximum possible sender payoff M, then s is an equilibrium:*

$$u_S(\text{outcome}(G, s)) = M \wedge \forall s'.\ u_S(\text{outcome}(G, s')) \leq M \implies \text{IsEquilibrium}(G, s).$$

The most sacred ritual—the one that maximizes collective effervescence (Durkheim)—is always in equilibrium because nothing can improve on the maximum. This is tautological in isolation, but gains force in combination with depth bounds: since persuasion is bounded (Chapter 7), the maximum payoff is itself bounded.

Theorem 6.23 (Equilibrium Preserved Under Monotone Transform (6.17))**.** *If $f : \mathbb{N} \to \mathbb{N}$ is monotone and s is an equilibrium under payoff u_S, then s is an equilibrium under $f \circ u_S$:*

$$(a \leq b \implies f(a) \leq f(b)) \wedge \text{IsEquilibrium}(G, s) \implies \forall s'.\ f(u_S(\text{outcome}(G, s'))) \leq f(u_S(\text{out}$$

Cultural stability is robust under monotone revaluation. If a society decides that all payoffs should be doubled (inflation), squared (increasing returns), or logarithmically compressed (diminishing marginal utility), the same cultural forms remain stable. The theorem identifies the essential feature: it is the *ordering* of payoffs, not their magnitudes, that determines equilibrium.

[4] This is a strong claim that will be qualified in Chapter 9, where iterated self-interpretation can create genuinely new structures through repeated composition.

Theorem 6.24 (Void Equilibrium in Zero-Payoff Games (6.18))**.** *In a game where every signal yields zero payoff, void is an equilibrium:* $\mathrm{IsEquilibrium}(\langle R, \lambda_ . 0, \lambda_ . 0$

Proof sketch. Every signal gives payoff 0, so every signal is an equilibrium (Theorem 6.8 with $k = 0$). In particular, $\mathbf{0}$ is. □

In a society where communication carries no reward (Durkheim's *anomie*), silence is trivially optimal. This is the formal model of communicative breakdown: when nothing matters, nothing is said.

```
theorem void_equilibrium_zero_game (rep : Repertoire I
    ) :
     IsEquilibrium
       (<rep, fun _ => 0, fun _ => 0> : InterpGame I)
       (IdeaticSpace.void : I)
```

Listing 6.6: Void equilibrium in zero-payoff games.

6.9 Conclusion

This chapter established the game-theoretic foundations of interpretive stability within IDT. We showed that the algebraic structure of ideatic spaces—composition, void, depth, resonance—yields a rich theory of equilibrium without any additional axioms.

Three themes will recur in the chapters ahead. First, *void as universal fallback*: silence is always available and always stable (Theorem 6.7). Second, *linear depth bounds*: the impact of any signal is bounded by its own complexity times the audience size (Theorem 6.14). Third, *resonance preservation*: structurally similar inputs yield structurally similar outputs (Theorem 6.16).

Chapter 7 turns from equilibrium existence to the *limits* of persuasion: how much can a signal change a receiver's depth, and why propaganda is subadditive.

Chapter 7

Persuasion Bounds and Propaganda

> "The ideas of the ruling class are in every epoch the ruling ideas: i.e., the class which is the ruling material force of society is at the same time its ruling intellectual force."
>
> Karl Marx, *The German Ideology* (1846)

7.1 Introduction: The Algebra of "Hearts and Minds"

Every propagandist, preacher, and politician confronts the same constraint, whether they know it or not: *you cannot add more complexity to someone's worldview than the complexity of your message allows.* A bumper sticker cannot produce a Ph.D.-level understanding; a three-word slogan cannot install a twenty-variable causal model. Conversely, a complex treatise cannot simplify a mind below its current depth—complexity is added, not subtracted, by composition.

Antonio Gramsci grasped this intuitively when he argued that hegemony requires not just force but *cultural production*—the sustained manufacture of complex ideological structures capable of reshaping the depth of popular consciousness.[1] Michel Foucault extended the insight: power operates not by loud commands but by quiet restructuring of the categories through which

[1] Gramsci, *Prison Notebooks*, Q. 12 (1932). The concept of "cultural hegemony" names exactly the problem that this chapter formalizes: how much ideological depth can a ruling class actually install in a population?

subjects perceive the world—what he called *power/knowledge*.[2]

This chapter provides the mathematical backbone for these intuitions. We define *marginal depth*—how much a signal adds to a receiver's interpretive complexity—and prove that it is bounded above by the signal's own depth (the *fundamental persuasion bound*). We then show that chaining messages is *subadditive*: the total persuasive capacity of a compound message is at most the sum of its parts. Repetition helps, but only linearly—never exponentially. Even totalitarian propaganda, with unlimited repetition, faces a linear ceiling.

These results have sharp implications for anthropological theory. They explain why "hearts and minds" campaigns fail not because of insufficient repetition, but because of the fundamental *subadditivity* of ideatic composition. You cannot shout your way to depth.

7.2 Core Definitions

Definition 7.1 (Marginal Depth)**.** The *marginal depth* of signal s for receiver r is the increase in depth caused by composition:

$$\text{marginalDepth}(r, s) \;=\; \mathrm{d}(r \circ s) - \mathrm{d}(r),$$

where $-$ denotes truncated (natural number) subtraction.

The marginal depth of a sermon to a parishioner is how much *more* complex their worldview becomes after hearing it, relative to what they had before. It is inherently receiver-dependent: the same lecture may add five "layers" to a novice and zero to an expert.

Definition 7.2 (Persuasion Bound)**.** The *persuasion bound* of a signal s is its own depth:

$$\text{persuasionBound}(s) \;=\; \mathrm{d}(s).$$

This is an *absolute* ceiling, independent of the receiver.

```
def marginalDepth (r s : I) : ℕ :=
  IdeaticSpace.depth (IdeaticSpace.compose r s) -
    IdeaticSpace.depth r

def persuasionBound (s : I) : ℕ := IdeaticSpace.depth
    s
```

Listing 7.1: Marginal depth and persuasion bound in Lean 4.

[2]Foucault, *Discipline and Punish* (1975); *The History of Sexuality*, vol. 1 (1976).

7.3 The Fundamental Persuasion Bound

Theorem 7.3 (Marginal Depth Bound (7.1) — The Fundamental Theorem of Propaganda)**.** *For all receivers r and signals s:*

$$\text{marginalDepth}(r, s) \;\leq\; \text{persuasionBound}(s) \;=\; \mathrm{d}(s).$$

Proof sketch. By subadditivity of composition depth: $\mathrm{d}(r \circ s) \leq \mathrm{d}(r) + \mathrm{d}(s)$. Subtracting $\mathrm{d}(r)$ from both sides gives $\text{marginalDepth}(r, s) \leq \mathrm{d}(s)$. □

No matter how receptive the audience, a message cannot add more complexity than it intrinsically contains. A bumper sticker cannot produce a Ph.D.-level understanding. This is the **fundamental theorem of propaganda theory**: persuasive capacity is bounded by signal depth.

The result vindicates Gramsci's insistence that effective hegemony requires *organic intellectuals*—agents capable of producing signals of sufficient depth to restructure popular consciousness. Simple slogans (low depth) have low persuasion bounds. Only sustained, complex cultural production can deeply transform a worldview.

Theorem 7.4 (Void Has Zero Marginal Depth (7.2))**.** $\text{marginalDepth}(r, \mathbf{0}) = 0$ *for all r.*

Proof sketch. $r \circ \mathbf{0} = r$ by the right-identity axiom, so $\mathrm{d}(r \circ \mathbf{0}) - \mathrm{d}(r) = 0$. □

Silence cannot persuade. Quaker silence, Buddhist *noble silence*, and the minute of silence for the dead change nothing about the listener's depth. They are powerful *socially*—as markers of solidarity, grief, or contemplation—but informationally null.[3]

Theorem 7.5 (Void Has Zero Persuasion Bound (7.3))**.** $\text{persuasionBound}(\mathbf{0}) = 0$.

Independent of the receiver, the null message has zero persuasive capacity. This is a property of the signal itself, not of any particular audience.

Theorem 7.6 (Persuasion Bound Nonnegativity (7.4))**.** $\text{persuasionBound}(s) \geq 0$ *for all s.*

Trivially true for natural numbers, but conceptually important: there are no "anti-signals" that reduce the maximum possible effect below zero. Every signal has non-negative persuasive potential.

[3] This creates a sharp distinction between the *social* and *informational* functions of silence, a distinction that Foucault's analysis of "incitement to discourse" (*History of Sexuality*, vol. 1) explores from the opposite direction.

7.4 Deeper Signals Are More Persuasive

Theorem 7.7 (Monotonicity of Persuasion Bound (7.5)). *If* $\mathrm{d}(s_1) \leq \mathrm{d}(s_2)$, *then* $\mathrm{persuasionBound}(s_1) \leq \mathrm{persuasionBound}(s_2)$.

A complex ritual has *more potential* to transform a participant's worldview than a simple one. This doesn't mean it *will*—marginal depth depends on the receiver—but it *can*. Complexity is a *necessary* (not sufficient) condition for deep persuasion.

This result gives formal content to Foucault's concept of *power/knowledge*: the capacity to restructure categories of thought (persuasion) scales with the complexity of the discursive apparatus (signal depth). Simple commands have low persuasion bounds; complex disciplinary regimes have high ones.

Theorem 7.8 (Self-Composition Persuasion Bound (7.6)). $\mathrm{persuasionBound}(s \circ s) \leq 2 \cdot \mathrm{persuasionBound}(s)$.

Proof sketch. $\mathrm{d}(s \circ s) \leq \mathrm{d}(s) + \mathrm{d}(s) = 2\,\mathrm{d}(s)$ by subadditivity. □

Repeating a message doubles the *maximum possible* effect, but the *actual* effect depends on the receiver. Propaganda works by repetition, but with diminishing returns.

Theorem 7.9 (ComposeN Persuasion Bound (7.7)). $\mathrm{persuasionBound}(sn) \leq n \cdot \mathrm{persuasionBound}(s)$.

Even with unlimited repetition, persuasive capacity grows at most linearly—never exponentially. This is why totalitarian propaganda, despite its volume, has bounded effectiveness. The 20th century demonstrated this repeatedly: neither Nazi Germany nor the Soviet Union achieved total ideological control despite saturation-level messaging.[4]

```
theorem composeN_persuasion_bound (s : I) (n : ℕ) :
    persuasionBound (composeN s n) <=
      n * persuasionBound s
```

Listing 7.2: ComposeN persuasion bound in Lean 4.

7.5 Subadditivity: Chaining Doesn't Multiply

Theorem 7.10 (Persuasion Subadditivity (7.8)).

$$\mathrm{persuasionBound}(s_1 \circ s_2) \leq \mathrm{persuasionBound}(s_1) + \mathrm{persuasionBound}(s_2).$$

[4] Hannah Arendt's *Origins of Totalitarianism* (1951) documents both the ambition and the failure of totalitarian propaganda to achieve total depth penetration.

Proof sketch. Direct from the subadditivity of depth under composition: $\mathrm{d}(s_1 \circ s_2) \leq \mathrm{d}(s_1) + \mathrm{d}(s_2)$. □

A two-part message cannot be more persuasive than the sum of its parts. This rules out "synergistic propaganda"—the idea that combining messages could produce super-additive persuasion. It is the formal foundation of bounded rationality in cultural transmission.

The subadditivity theorem has immediate policy implications. If a state seeks to increase persuasive capacity, it must increase the *depth* of its messages, not merely their *number*. Quantity is strictly dominated by quality as a persuasion strategy.

Theorem 7.11 (Three-Message Persuasion Bound (7.9))**.**

$$\text{persuasionBound}\big((s_1 \circ s_2) \circ s_3\big) \;\leq\; \text{persuasionBound}(s_1) + \text{persuasionBound}(s_2) + \text{persuasi}$$

Multi-act rituals (liturgy with readings, sermon, and communion) have bounded total impact $\leq$ the sum of individual act complexities. The extension to n messages follows by induction.

Theorem 7.12 (Marginal Depth of Void Receiver (7.10))**.** $\text{marginalDepth}(\mathbf{0}, s) = \mathrm{d}(s)$.

Proof sketch. $\mathbf{0} \circ s = s$ by left-identity, and $\mathrm{d}(\mathbf{0}) = 0$, so $\mathrm{d}(s) - 0 = \mathrm{d}(s)$. □

The "blank slate" receiver (*tabula rasa*) absorbs every signal fully. This is the formal version of Locke's epistemology—and its anthropological critique: no real human is a blank slate, so no real human absorbs a signal fully. The void receiver is a theoretical limit, never achieved in practice.

7.6 Persuasion Through Chains

What happens when a signal passes through an intermediary? A teacher receives a text, processes it through their own background, and transmits the result to a student. How much depth can this chain deliver?

Theorem 7.13 (Chain Persuasion Bound (7.11))**.** $\text{marginalDepth}(r, r_1 \circ s) \leq \mathrm{d}(r_1) + \mathrm{d}(s)$.

A teacher cannot transmit more depth than their own understanding plus the source material's complexity. Great teaching requires both deep knowledge *and* rich source texts. This formalizes what every graduate student knows intuitively: a mediocre advisor with a brilliant text, or a brilliant advisor with a mediocre text, both produce bounded results.

Theorem 7.14 (Marginal Depth of Self-Composition (7.12)). $\mathrm{marginalDepth}(r, r) \leq \mathrm{d}(r)$.

Reflecting on your own ideas (self-composition) cannot add more than your existing complexity. Introspection has bounded returns. This result anticipates Chapter 9's analysis of echo chambers: if self-reflection is bounded, then *iterated* self-reflection is bounded too.

Theorem 7.15 (Marginal Depth Triangle Inequality (7.13)). $\mathrm{marginalDepth}(r, s_1 \circ s_2) \leq \mathrm{d}(s_1) + \mathrm{d}(s_2)$.

The marginal impact of a compound message is bounded by the sum of its parts' complexities. No synergy, no multiplication—just addition. This is the triangle inequality for marginal depth, and it gives the persuasion space a quasi-metric structure.

7.7 Propaganda Effectiveness Bounds

We now turn to the specific problem of propaganda: what happens when the same message is repeated many times?

Theorem 7.16 (Bounded Propaganda Effectiveness (7.14)). *After composing receiver r with n repetitions of signal s:*

$$\mathrm{d}\big(r \circ sn\big) \;\leq\; \mathrm{d}(r) + n \cdot \mathrm{d}(s).$$

Proof sketch. Combine the subadditivity of composition depth with the bound $\mathrm{d}(sn) \leq n \cdot \mathrm{d}(s)$ from the composeN depth theorem. □

Propaganda saturation—hearing the same message n times—increases depth at most linearly. No amount of repetition can produce exponential radicalization. This is why "deprogramming" is possible: the damage is bounded, and bounded damage can be reversed.[5]

Theorem 7.17 (Resonant Propaganda (7.15)). *If $s_1 \sim s_2$, then $r \circ s_1 \sim r \circ s_2$ for all r.*

Competing narratives that "resonate" with the same cultural themes produce resonant (structurally similar) effects on receivers. When Fox News and MSNBC draw on the same patriotic imagery, they produce resonant—not identical—reactions in their audiences. The content differs; the *structure* of the interpretive effect is preserved.

[5] The bounded-propaganda theorem provides a formal counterargument to the popular "brainwashing" hypothesis. See Lifton, *Thought Reform and the Psychology of Totalism* (1961), for the empirical evidence that even extreme indoctrination has bounded effects.

Theorem 7.18 (Double Propagandist Bound (7.16))**.** $\text{marginalDepth}(r, s_1 \circ s_2) \leq \text{persuasionBound}(s_1) + \text{persuasionBound}(s_2)$.

Allied propaganda campaigns—joint military-religious messaging, corporate-governmental public relations—cannot exceed the sum of their individual complexities. Coalition propaganda doesn't synergize; it merely adds.

Theorem 7.19 (Atomic Signal Minimal Persuasion (7.17))**.** *If s is atomic, then* $\text{persuasionBound}(s) \leq 1$.

Single words, simple gestures, basic symbols—all have persuasion bound at most 1. To deeply transform a worldview, you need *structured* complexity, not atomic simplicity. A thumbs-up cannot substitute for a treatise.[6]

Theorem 7.20 (Iterated Propaganda Marginal Decay (7.18))**.** $\text{marginalDepth}(r, sn) \leq n \cdot \text{persuasionBound}(s)$.

Proof sketch. By the composition depth bound and the composeN depth bound: $\mathrm{d}(r \circ sn) \leq \mathrm{d}(r) + n \cdot \mathrm{d}(s)$, so $\text{marginalDepth} \leq n \cdot \mathrm{d}(s)$. □

The k-th repetition of propaganda is no more effective than the first—the bound is uniform. Each repetition has the same ceiling on marginal depth, but the *actual* marginal depth can only decrease in realistic models. This formalizes the "diminishing returns" of propaganda.

```
theorem iterated_propaganda_bound (r s : I) (n : ℕ) :
    marginalDepth r (composeN s n) <=
      n * persuasionBound s
```

Listing 7.3: Iterated propaganda bound in Lean 4.

7.8 Conclusion

This chapter established the fundamental limits of persuasion within IDT. The central result—the marginal depth bound (Theorem 7.3)—states that no signal can add more complexity than it contains. The subadditivity theorem (7.10) rules out synergistic propaganda, and the bounded propaganda theorem (7.16) caps the effect of repetition at linear growth.

Together, these results provide formal foundations for Gramsci's theory of cultural hegemony (persuasion requires complex cultural production), Foucault's power/knowledge (persuasive capacity scales with discursive depth),

[6] This result formalizes the linguistic insight that single lexemes have bounded semantic impact. The power of language lies in *compositional* complexity, not lexical richness.

and the empirical observation that totalitarian propaganda has bounded effectiveness.

Chapter 8 turns from the *sender's* constraints to the *receiver's*: what happens when the reception process itself is lossy?

Chapter 8

Lossy Reception and Oral Tradition

> "The story becomes at each reproduction more concise, more schematized ... Proper names drop out. Vivid details vanish. What remains is a general form."
>
> Frederic Bartlett, *Remembering* (1932)

8.1 Introduction: The Thermodynamics of Storytelling

In 1932, Frederic Bartlett conducted a series of serial reproduction experiments that became foundational for cognitive anthropology. He asked a subject to read a Native American folk tale ("The War of the Ghosts"), then reproduce it from memory; the next subject read only the reproduction, and so on through a chain of ten or more retellings. The results were dramatic: details dropped out, unfamiliar concepts were rationalized into familiar categories, and the story simplified with each retelling until it converged on a stable, culturally assimilated "skeleton."[1]

Oral tradition—the transmission of myths, genealogies, ritual instructions, and folk knowledge from person to person without writing—is the primordial form of cultural transmission. Before the invention of writing (c. 3200 BCE in Sumer), *all* culture was oral. Even today, the vast majority

[1] Bartlett, F. C., *Remembering: A Study in Experimental and Social Psychology* (Cambridge, 1932), ch. 5. The experiment has been replicated hundreds of times with consistent results; see Mesoudi & Whiten (2008) for a meta-analysis.

of human cultural knowledge is transmitted orally: recipes, gossip, jokes, parenting advice, religious instruction.

This chapter formalizes the defining feature of oral transmission: *lossiness*. A receiver who processes a signal through an oral channel—through memory, paraphrase, reinterpretation—can never produce output more complex than the input. Depth can only decrease or remain constant; it cannot increase. This is the "second law of cultural thermodynamics."[2]

We define three levels of receiver: a generic *Receiver* (preserving void and distributing over composition), a *LossyReceiver* (depth non-increasing), and a *StrictlyLossy* receiver (depth strictly decreasing for complex ideas). We prove that lossy reception composes, iterates, and ultimately converges: only shallow ideas survive the "telephone game."

8.2 Core Structures

Definition 8.1 (Receiver)**.** A *receiver* is an endomorphism $\varphi : \mathcal{I} \to \mathcal{I}$ satisfying:

1. *Void preservation:* $\varphi(\mathbf{0}) = \mathbf{0}$.
2. *Distributivity:* $\varphi(a \circ b) = \varphi(a) \circ \varphi(b)$.

A receiver is a "cultural filter"—a formalized version of how a particular mind processes incoming signals. The two axioms are natural: silence in yields silence out, and hearing a compound message is the same as hearing the parts and composing the results.

```
structure Receiver (I : Type*) [IdeaticSpace I] where
  process : I -> I
  process_void :
    process IdeaticSpace.void = IdeaticSpace.void
  process_compose : forall (a b : I),
    process (IdeaticSpace.compose a b) =
    IdeaticSpace.compose (process a) (process b)
```

Listing 8.1: The Receiver structure in Lean 4.

Definition 8.2 (Lossy Receiver)**.** A *lossy receiver* is a receiver φ such that

$$\mathrm{d}(\varphi(a)) \;\leq\; \mathrm{d}(a) \quad \text{for all } a \in \mathcal{I}.$$

Processing never increases depth.

[2]The analogy to thermodynamic entropy is imperfect but illuminating. In thermodynamics, the second law states that entropy never decreases in an isolated system. Here, the analog is that *complexity* (depth) never increases through lossy reception. The "entropy" of cultural transmission is the *loss* of depth.

This is the defining property of oral transmission: you can never come out of a retelling with *more* complexity than you went in with. Written transmission is different (you can re-read, annotate, cross-reference), but oral transmission is inherently lossy.

Definition 8.3 (Strictly Lossy Receiver)**.** A *strictly lossy receiver* is a lossy receiver φ such that

$$\mathrm{d}(a) > 1 \implies \mathrm{d}(\varphi(a)) < \mathrm{d}(a).$$

For complex ideas ($\mathrm{d} > 1$), depth *strictly* decreases. Ideas of depth ≤ 1 may be preserved.

This models the "telephone game": every retelling loses something, unless the message is already maximally simple. Only atomic ideas survive perfect-fidelity transmission through a strictly lossy channel.

```
structure StrictlyLossy (I : Type*) [IdeaticSpace I]
    extends LossyReceiver I where
  strict_decay : forall (a : I),
    IdeaticSpace.depth a > 1 ->
    IdeaticSpace.depth
      (toLossyReceiver.toReceiver.process a) <
      IdeaticSpace.depth a
```

Listing 8.2: Strictly lossy receiver in Lean 4.

8.3 Basic Properties of Lossy Reception

Theorem 8.4 (Lossy Void Preservation (8.1))**.** *All receivers preserve silence:* $\varphi(\mathbf{0}) = \mathbf{0}$.

No transmission channel creates content from nothing. If you say nothing, the listener hears nothing. This is the zeroth law of communication—prior to any thermodynamic considerations, it establishes the baseline: the null signal is universally transparent.

Theorem 8.5 (Lossy Depth Bound (8.2))**.** *After lossy processing:* $\mathrm{d}(\varphi(a)) \leq \mathrm{d}(a)$.

An oral retelling is never more complex than the original. This is the second law of cultural thermodynamics: complexity never increases through

oral transmission. The theorem is definitional—it *is* the defining property of a lossy receiver—but stating it as a theorem emphasizes its central role.[3]

Theorem 8.6 (Lossy Composition Bound (8.3))**.** *Processing a composed idea through a lossy receiver:*

$$\mathrm{d}(\varphi(a \circ b)) \;\leq\; \mathrm{d}(a) + \mathrm{d}(b).$$

Proof sketch. By lossiness, $\mathrm{d}(\varphi(a \circ b)) \leq \mathrm{d}(a \circ b)$. By subadditivity of composition depth, $\mathrm{d}(a \circ b) \leq \mathrm{d}(a) + \mathrm{d}(b)$. Chaining the inequalities yields the result. □

A story about two events, retold through oral tradition, has depth at most the sum of the two events' original depths—even before accounting for the receiver's lossiness.

Theorem 8.7 (Lossy Receiver Preserves Void Depth (8.4))**.** $\mathrm{d}(\varphi(\mathbf{0})) = 0$.

Silence, when transmitted lossily, remains at depth zero. Combined with Theorem 8.4, this says not just that void maps to void, but that the image has zero complexity—a stronger quantitative statement.

8.4 Iterated Lossy Reception

What happens when a signal passes through a *chain* of lossy receivers? Bartlett's serial reproduction experiments provide the empirical motivation; this section provides the formal theory.

Definition 8.8 (Iterated Process)**.** The n-fold iteration of a receiver's process function:

$$\varphi^{(0)}(a) = a, \qquad \varphi^{(n+1)}(a) = \varphi(\varphi^{(n)}(a)).$$

```
def iterateProcess (R : Receiver I) : ℕ -> I -> I
  | 0 => fun a => a
  | n + 1 => fun a => R.process (iterateProcess R n a)
```

Listing 8.3: Iterated process in Lean 4.

Theorem 8.9 (Iterated Lossy Depth Bound (8.5))**.** *After n applications of a lossy receiver:* $\mathrm{d}(\varphi^{(n)}(a)) \leq \mathrm{d}(a)$ *for all n.*

[3] Contrast with *written* transmission, which can *increase* depth through annotation, commentary, and cross-reference. The Talmud is deeper than the Torah; the scholastic summa is deeper than any single patristic text. Writing breaks the second law of oral thermodynamics.

Proof sketch. By induction on n. The base case ($n = 0$) is trivial. For the inductive step, apply the lossiness bound to the result of the (n)-th iteration, whose depth is $\leq \mathrm{d}(a)$ by the inductive hypothesis. □

Oral tradition through a chain of n retellings can only lose depth, never gain it. Whether the story passes through 3 or 300 people, it can only get simpler. This is the formal backbone of Bartlett's experimental finding.

Theorem 8.10 (Iterated Void Stability (8.6)). $\varphi^{(n)}(\mathbf{0}) = \mathbf{0}$ *for all* n.

Proof sketch. Induction on n: the base case is identity, and each step applies $\varphi(\mathbf{0}) = \mathbf{0}$. □

A chain of listeners hearing nothing produces nothing, no matter how long the chain. Silence is the unique *absorbing state* of every receiver.

Theorem 8.11 (Lossy Reception of Compositions (8.7)). $\varphi(a \circ b) = \varphi(a) \circ \varphi(b)$.

Cultural filtering is "linear" in the algebraic sense: a receiver's reaction to a compound stimulus is the compound of their reactions to the parts. This is the distributivity axiom of the Receiver structure, stated as a theorem for emphasis.

In anthropological terms, it means that a listener's interpretation of a two-part story can be decomposed into their interpretation of part 1 composed with their interpretation of part 2. There is no holistic "Gestalt" effect in lossy reception—only part-by-part processing.[4]

8.5 Strict Lossiness and Convergence

Theorem 8.12 (Strict Decay Bound (8.8)). *For a strictly lossy receiver and any idea with* $\mathrm{d}(a) > 1$*:*

$$\mathrm{d}(\varphi(a)) < \mathrm{d}(a).$$

In "telephone game" transmission, every complex message becomes strictly simpler. This is not a bug—it's a feature. Cultural evolution works by simplification. The myths that survive thousands of years of oral transmission are the ones that have been *simplified* to the point where no further simplification occurs.

Theorem 8.13 (Shallow Ideas Survive Lossy Reception (8.9)). *If* $\mathrm{d}(a) \leq 1$*, then* $\mathrm{d}(\varphi(a)) \leq 1$.

[4] This is a strong structural assumption. Real human cognition arguably does exhibit Gestalt effects. The distributivity axiom is a simplification that enables clean formal results at the cost of ignoring non-decomposable cognitive processes.

Proof sketch. By lossiness: $\mathrm{d}(\varphi(a)) \leq \mathrm{d}(a) \leq 1$. □

Simple ideas—basic emotions, universal gestures, atomic symbols—survive any amount of lossy transmission. This is why certain cultural universals persist: they're too simple to be degraded further. The smile, the frown, the pointing gesture, the head-nod—all have depth ≤ 1 and are thus impervious to oral degradation.[5]

Theorem 8.14 (Zero-Depth Ideas Survive Lossy Reception (8.10))**.** *If* $\mathrm{d}(a) = 0$*, then* $\mathrm{d}(\varphi(a)) = 0$.

Void-like ideas (pure silence, empty gestures) are immune to lossy transmission. They are fixed points of every lossy receiver.

8.6 Composition of Receivers

Definition 8.15 (Composed Receivers)**.** Given receivers φ_1, φ_2, their composition is $(\varphi_2 \circ \varphi_1)(a) = \varphi_2(\varphi_1(a))$—the result of sequential reception.

```
def composeReceivers (R_1 R_2 : Receiver I) : Receiver
    I where
  process := fun a => R_2.process (R_1.process a)
  process_void := by
    rw [R_1.process_void, R_2.process_void]
  process_compose := by
    intro a b
    rw [R_1.process_compose, R_2.process_compose]
```

Listing 8.4: Receiver composition in Lean 4.

Theorem 8.16 (Composition of Lossy Receivers is Lossy (8.11))**.** *If* φ_1 *and* φ_2 *are lossy, then* $\varphi_2 \circ \varphi_1$ *is lossy:*

$$\mathrm{d}((\varphi_2 \circ \varphi_1)(a)) \;\leq\; \mathrm{d}(a) \quad \textit{for all } a.$$

Proof sketch. $\mathrm{d}(\varphi_2(\varphi_1(a))) \leq \mathrm{d}(\varphi_1(a)) \leq \mathrm{d}(a)$ by applying lossiness twice. □

A chain of imperfect oral transmitters is itself an imperfect transmitter. Lossiness is closed under composition—there's no way to "recover" lost depth by adding more lossy steps. This is the impossibility of error correction

[5] Ekman's (1971) cross-cultural studies of facial expression provide empirical support: the six "basic emotions" are recognized across cultures, arguably because they are too simple to be distorted by oral transmission.

in purely oral tradition. Unlike digital communication (where redundancy enables error correction), oral channels have no mechanism for restoring lost depth.

Theorem 8.17 (Double Lossy is More Lossy (8.12)). $\mathrm{d}((\varphi_2 \circ \varphi_1)(a)) \leq \mathrm{d}(\varphi_1(a))$.

Two generations of oral transmission is worse (or equal) than one. The second lossy step can only further reduce what the first step already degraded. This is the formal version of the folk wisdom that stories "get worse with each retelling."

8.7 Resonance and Lossy Reception

Theorem 8.18 (Receiver Preserves Self-Resonance (8.13)). $\varphi(a) \sim \varphi(a)$ *for all* a.

Even lossy transmission produces internally consistent output. A garbled retelling is garbled, but it still "makes sense to itself." This is a consequence of the reflexivity of resonance, but it carries anthropological weight: oral traditions, however degraded, are never *incoherent*. They simplify, regularize, and schematize, but they always remain self-consistent.

Theorem 8.19 (Lossy Compound Self-Resonance (8.14)). $\varphi(a \circ b) \sim \varphi(a \circ b)$.

Even through lossy reception, the receiver's output from a compound signal is internally coherent. The theorem is trivial (self-resonance is always true), but it anchors the stronger results of Chapter 10, where we study how *different* receivers' outputs relate to each other.

Theorem 8.20 (Void Iterate Always Resonates (8.15)). $\varphi^{(n)}(\mathbf{0}) \sim \varphi^{(n)}(\mathbf{0})$ *for all* n.

Chains of silence produce silence, which trivially resonates with itself. Combined with Theorem 8.10, this says that void is not only a fixed point of iteration but a self-resonant one.

8.8 Advanced Properties

Theorem 8.21 (Lossy Iterated Composition Bound (8.16)). $\mathrm{d}(\varphi(an)) \leq n \cdot \mathrm{d}(a)$.

Proof sketch. By lossiness, $\mathrm{d}(\varphi(an)) \leq \mathrm{d}(an)$. By the composeN depth bound, $\mathrm{d}(an) \leq n \cdot \mathrm{d}(a)$. □

A repeated message (ritual chant, liturgical refrain) loses depth through lossy reception, bounded by the repetition count times original depth. Even before lossiness takes its toll, the starting depth is only linearly bounded in repetition count.

Theorem 8.22 (Receiver Composition is Associative on Output (8.17))**.** $\varphi_3(\varphi_2(\varphi_1(a))) = (\varphi_3 \circ \varphi_2 \circ \varphi_1)(a)$.

A three-generation transmission chain (grandparent $\to$ parent $\to$ child) produces the same result regardless of how we "bracket" the generations. This is the associativity of function composition, applied to the receiver setting. It ensures that our model of multi-generational oral tradition is well-defined: the order of composition doesn't depend on arbitrary grouping.

Theorem 8.23 (Depth Monotonicity Through Chains (8.18))**.** *For $m \leq n$:* $\mathrm{d}(\varphi^{(n)}(a)) \leq \mathrm{d}(\varphi^{(m)}(a))$.

Proof sketch. By induction on n. If $m = n$, the result is trivial. If $m < n$, apply lossiness at step n to reduce to the case $n-1$, then apply the inductive hypothesis. □

At every point in a chain of oral transmission, the story is no more complex than it was at any earlier point. There is no "regeneration" through purely lossy channels. The depth sequence $\mathrm{d}(\varphi^{(0)}(a)), \mathrm{d}(\varphi^{(1)}(a)), \mathrm{d}(\varphi^{(2)}(a)), \ldots$ is monotonically non-increasing.

This is the formal version of a deep anthropological observation: myths don't get *worse*; they get *stable*. Oral tradition doesn't "degrade"—it *converges* to a fixed point determined by the transmission medium's lossiness. The stories that survive are the ones simple enough to be transmitted without further loss.

```
theorem chain_depth_monotone
    (R : LossyReceiver I) (a : I) (m n : ℕ)
    (hmn : m <= n) :
    IdeaticSpace.depth
      (iterateProcess R.toReceiver n a) <=
    IdeaticSpace.depth
      (iterateProcess R.toReceiver m a)
```

Listing 8.5: Chain depth monotonicity in Lean 4.

8.9 Conclusion

This chapter established the formal theory of lossy reception, providing mathematical foundations for Bartlett's serial reproduction experiments and the broader study of oral tradition.

The central results are:

1. Lossy reception is closed under composition (Theorem 8.16): chains of oral transmitters are themselves oral transmitters.

2. Depth is monotonically non-increasing through chains (Theorem 8.23): stories only get simpler.

3. Shallow ideas survive (Theorem 8.13): cultural universals persist because they are too simple to degrade.

The theory explains why oral traditions converge: not because storytellers are "bad" at their job, but because lossiness is a structural feature of the oral channel. Written traditions escape this constraint by providing external memory, but oral traditions are bound by the second law of cultural thermodynamics.

Chapter 9 takes this analysis inward: what happens when a community applies lossy reception *to its own output*, creating an echo chamber?

Chapter 9

Echo Chambers and Iterated Self-Interpretation

> "Doublethink means the power of holding two contradictory beliefs in one's mind simultaneously, and accepting both of them."
>
> George Orwell, *Nineteen Eighty-Four* (1949)

9.1 Introduction: The Paradox of In-Group Radicalization

Popular accounts of echo chambers emphasize *amplification*: a community that talks only to itself is said to "radicalize," producing ever-more-extreme versions of its founding ideology. Social media critics warn of "filter bubbles" (Pariser, 2011) in which algorithmic recommendation creates spirals of increasing extremism.[1]

The formal analysis tells a different story.

When a mind with background r repeatedly re-interprets a signal s through its own cultural filter, the result is not an explosion of complexity but a *collapse*. Each round of self-interpretation composes r with the previous result, and the depth of the output is bounded by $n \cdot \mathrm{d}(r) + \mathrm{d}(s)$—*linear* in the number of rounds. No exponential radicalization is possible.

The counterintuitive insight of IDT's echo chamber theory is that closed

[1] Pariser, E., *The Filter Bubble: What the Internet Is Hiding from You* (Penguin, 2011). For a critical assessment, see Bruns, A., *Are Filter Bubbles Real?* (Polity, 2019).

communities don't *complexify* their ideas—they *simplify* them. Radicalization is not the acquisition of deeper ideology; it is the *erosion of nuance* through iterated self-composition. The extremist does not have a more complex worldview than the moderate; they have a *less* complex one, in which fine distinctions have been compositioned away by repeated application of the same filter.

This chapter formalizes self-iteration, proves the linear depth bound, analyzes the role of void backgrounds, studies resonance preservation through echo, and identifies the fixed-point structure of iterated self-interpretation.

9.2 Core Definitions

Definition 9.1 (Self-Iteration (Echo))**.** Given background $r \in \mathcal{I}$ and signal $s \in \mathcal{I}$, the *n-th echo* is defined recursively:

$$\text{selfIterate}(r, s, 0) = s, \qquad \text{selfIterate}(r, s, n{+}1) = r \circ \text{selfIterate}(r, s, n).$$

Each step represents one "round" of in-group discussion: take the current consensus, filter it through the group's shared worldview r, and obtain the next consensus.

Definition 9.2 (Echo Depth)**.** The *echo depth* at step n is

$$\text{echoDepth}(r, s, n) \;=\; \text{d}(\text{selfIterate}(r, s, n)).$$

```
def selfIterate (r s : I) : ℕ -> I
  | 0 => s
  | n + 1 => IdeaticSpace.compose r (selfIterate r s n
    )

def echoDepth (r s : I) (n : ℕ) : ℕ :=
  IdeaticSpace.depth (selfIterate r s n)
```

Listing 9.1: Self-iteration in Lean 4.

9.3 Basic Echo Properties

Theorem 9.3 (Echo Base Case (9.1))**.** $\text{selfIterate}(r, s, 0) = s$.

At step 0, the echo is the raw signal, unfiltered. This is the "first hearing"—before enculturation transforms it. In anthropological terms, it is the moment of initial contact: the ethnographer's first impression before theoretical frameworks reshape perception.

Theorem 9.4 (Echo Step (9.2)). $\mathrm{selfIterate}(r, s, n{+}1) = r \circ \mathrm{selfIterate}(r, s, n)$.

Each round of in-group discussion is formally identical: take the current consensus, filter it through the group's shared worldview, get the next consensus. The process is *memoryless*—only the current state matters, not the history. This is a Markov property, and it makes the echo chamber formally analogous to a deterministic dynamical system on $\mathcal{I}$.

Theorem 9.5 (Echo Depth Bound (9.3)). *For all n:*

$$\mathrm{echoDepth}(r, s, n) \;\leq\; n \cdot \mathrm{d}(r) + \mathrm{d}(s).$$

Proof sketch. By induction on n. The base case gives $\mathrm{d}(s) \leq 0 \cdot \mathrm{d}(r) + \mathrm{d}(s)$. For the inductive step, $\mathrm{d}(r \circ \mathrm{selfIterate}(r, s, n)) \leq \mathrm{d}(r) + \mathrm{d}(\mathrm{selfIterate}(r, s, n)) \leq \mathrm{d}(r) + n \cdot \mathrm{d}(r) + \mathrm{d}(s) = (n{+}1) \cdot \mathrm{d}(r) + \mathrm{d}(s)$. □

This is the **central theorem of echo chamber theory**: complexity grows at most *linearly* in the number of rounds. No exponential radicalization is possible. A community hearing the same message 1000 times cannot produce more complexity than $1000 \cdot \mathrm{d}(r) + \mathrm{d}(s)$.

The linear bound has immediate implications for the study of radicalization. If echo chambers could produce exponential depth growth, then small communities with minimal resources could generate arbitrarily complex ideologies. The linear bound rules this out: the complexity ceiling is determined by the background's depth, and exceeding it requires either a deeper background or external input.[2]

9.4 Void Background: The Open Mind

Theorem 9.6 (Void Background Preserves Signal (9.4)). *If the background is void:* $\mathrm{selfIterate}(\mathbf{0}, s, n) = s$ *for all n.*

Proof sketch. Induction on n. Base case: $\mathrm{selfIterate}(\mathbf{0}, s, 0) = s$. Inductive step: $\mathbf{0} \circ \mathrm{selfIterate}(\mathbf{0}, s, n) = \mathbf{0} \circ s = s$ by the left-identity axiom. □

A perfectly "open mind"—one with no prior beliefs—cannot transform a signal through echo. You need *content* to create an echo; an empty room doesn't echo. This is the formal impossibility of the echo chamber for culturally empty agents.

The result has pedagogical implications: the ideal student (void background) retains every signal perfectly, with no distortion from prior beliefs.

[2] This does not mean that radicalization is "harmless." Even linear depth growth can produce dangerous simplifications—see the discussion of depth *collapse* vs. depth *growth* below. The point is that the mechanism of harm is simplification, not complexification.

Of course, no real student is a blank slate, which is why every teacher encounters the frustration of students who "hear" the lecture through the filter of their existing misconceptions.

Theorem 9.7 (Void Background Echo Depth (9.5))**.** $\mathrm{echoDepth}(\mathbf{0}, s, n) = \mathrm{d}(s)$ *for all* n.

With void background, echo depth equals signal depth at every step—no amplification, no degradation. The signal is perfectly preserved through arbitrary iterations.

Theorem 9.8 (Echo with Void Signal — Depth Bound (9.6))**.** *When signal is void:* $\mathrm{d}(\mathrm{selfIterate}(r, \mathbf{0}, n)) \leq n \cdot \mathrm{d}(r)$.

An echo chamber with no input signal just repeats its own background—pure ideological repetition without external stimulus. The community talks to itself, producing at most n rounds worth of background complexity. This is the formal model of ideological "navel-gazing": a group with nothing new to process, endlessly recycling its own assumptions.

9.5 Resonance in Echo Chambers

Theorem 9.9 (Echo Self-Resonance (9.7))**.** $\mathrm{selfIterate}(r, s, n) \sim \mathrm{selfIterate}(r, s, n)$ *for all* n.

Every stage of an echo chamber is internally consistent. The community always "agrees with itself." This is trivially true (self-resonance is reflexivity) but foundationally important: it means that the output of an echo chamber, however simplified, is never *incoherent*. The danger of the echo chamber is not incoherence but oversimplification.

Theorem 9.10 (Resonant Backgrounds Produce Resonant Echoes (9.8))**.** *If* $r_1 \sim r_2$*, then for all* n*:* $\mathrm{selfIterate}(r_1, s, n) \sim \mathrm{selfIterate}(r_2, s, n)$.

Proof sketch. Induction on n. Base case: both sides equal s, which self-resonates. Inductive step: $r_1 \circ \mathrm{selfIterate}(r_1, s, n)$ resonates with $r_2 \circ \mathrm{selfIterate}(r_2, s, n)$ by the resonance-compose theorem, using $r_1 \sim r_2$ and the inductive hypothesis. □

Two similar communities processing the same signal produce similar results at every stage. Cultural cousins stay cultural cousins through the echo process. This is the formal basis for the observation that, e.g., American and British media—despite separate echo dynamics—produce structurally similar political discourses because their cultural backgrounds resonate.

Theorem 9.11 (Resonant Signals Produce Resonant Echoes (9.9)). *If* $s_1 \sim s_2$*, then for all* n*:* $\mathrm{selfIterate}(r, s_1, n) \sim \mathrm{selfIterate}(r, s_2, n)$.

An echo chamber processing two structurally similar inputs produces structurally similar outputs at every stage. Two versions of the same news story, fed into the same community, produce resonant echo trajectories. The *structural similarity* of the inputs is preserved through arbitrary iterations.

```
theorem resonant_backgrounds_resonant_echoes
    {r_1 r_2 : I} (s : I)
    (hr : IdeaticSpace.resonates r_1 r_2) :
    forall (n : ℕ), IdeaticSpace.resonates
      (selfIterate r_1 s n) (selfIterate r_2 s n)
```

Listing 9.2: Resonant backgrounds theorem in Lean 4.

9.6 Composition and Iteration

Theorem 9.12 (Two-Step Echo = Single Compound Step (9.10)). $\mathrm{selfIterate}(r, s, 2) = r \circ (r \circ s)$.

Two rounds of echo chamber discussion is equivalent to the background filtering the background-filtered signal. This is the formal version of "groupthink": the community's consensus at step 2 is just the community's reaction to its own reaction. The formula makes the recursive nature of groupthink explicit: it is composition with the same element, applied twice.

Theorem 9.13 (Three-Step Echo = Triple Composition (9.11)). $\mathrm{selfIterate}(r, s, 3) = r \circ (r \circ (r \circ s))$.

Three rounds of self-interpretation. The result is a three-layer composition—formally identical to a three-generation oral transmission chain where all generations share the same culture. This connection to Chapter 8 is deep: an echo chamber is a *homogeneous* transmission chain, where every link is the same lossy receiver.

Theorem 9.14 (Echo Depth at Step 1 (9.12)). $\mathrm{echoDepth}(r, s, 1) \leq \mathrm{d}(r) + \mathrm{d}(s)$.

The first echo adds at most the background's complexity. This is the "first filter"—the initial cultural processing of a raw signal. In media studies terms, it is the "framing" stage, where raw events are filtered through editorial perspective.

Theorem 9.15 (Echo Depth at Step 2 (9.13)). $\mathrm{echoDepth}(r, s, 2) \leq 2 \cdot \mathrm{d}(r) + \mathrm{d}(s)$.

The second echo adds at most another round of background complexity. Linear growth, not exponential. By step 2, the community has "discussed the discussion"—and the total complexity is at most twice the background depth plus the original signal depth.

9.7 Fixed Points and Convergence

Theorem 9.16 (Void is Echo Fixed Point (9.14)). $\text{selfIterate}(\mathbf{0}, \mathbf{0}, n) = \mathbf{0}$ *for all* n.

Complete silence in an empty echo chamber remains silence forever. This is the "nothing" fixed point—the unique state from which no echo can emerge. It is both a mathematical triviality and a philosophical limit case: absolute emptiness is absolutely stable.

Theorem 9.17 (Echo Preserves Void Signal Depth (9.15)). $\text{echoDepth}(r, \mathbf{0}, n) \leq n \cdot \mathrm{d}(r)$.

An echo chamber with no external input produces at most n rounds' worth of background repetition—pure self-reinforcement without new information. This models the "stale discourse" phenomenon: a community that receives no new input can only recirculate its existing depth, with the total bounded by how many times the background has been applied.

9.8 Advanced Echo Properties

Theorem 9.18 (Echo is Left-Nested Composition (9.16)). $\text{selfIterate}(r, s, n+1) = r \circ \text{selfIterate}(r, s, n)$.

This is definitional, but worth stating explicitly. The echo process is *always* "background first, then previous result." The background is the dominant frame—it comes first in every composition. This is why echo chambers are *conservative* in the formal sense: the background always takes precedence over the incoming signal.

In Bourdieu's terms, the *habitus* (background) structures every perception (composition) so that the field's doxa is reproduced at each step. The echo chamber is the formal model of *symbolic reproduction* —the process by which dominant cultural categories perpetuate themselves through iterated interpretation.

Theorem 9.19 (Parallel Echo Chambers with Same Signal (9.17)). *If* $r_1 \sim r_2$, *then* $\text{selfIterate}(r_1, s, 1) \sim \text{selfIterate}(r_2, s, 1)$.

Proof sketch. At step 1: $r_1 \circ s \sim r_2 \circ s$ by resonance-compose, using $r_1 \sim r_2$ and $s \sim s$ (reflexivity). □

Two similar communities reacting to the same news story produce similar "hot takes" (step-1 interpretations). This is the formal basis of polling: structurally similar demographics respond similarly to the same stimulus. The theorem guarantees that resonant backgrounds produce resonant first reactions—not identical ones, but structurally related ones.

Theorem 9.20 (Echo Depth Monotonicity Relative to Signal Depth (9.18))**.** $\mathrm{d}(s) \leq \mathrm{echoDepth}(\mathbf{0}, s, n)$ *for all* n.

Proof sketch. By Theorem 9.6, $\mathrm{selfIterate}(\mathbf{0}, s, n) = s$, so $\mathrm{echoDepth}(\mathbf{0}, s, n) = \mathrm{d}(s)$. □

An "open" echo chamber (void background) never reduces the signal. Complexity is preserved when there is no cultural filter. This is the ideal of academic freedom: an institution with no ideological filter preserves the complexity of every input. The theorem formalizes what universities aspire to be (and rarely are): void backgrounds that transmit without distortion.[3]

```
theorem echo_depth_lower_void_background
    (s : I) (n : ℕ) :
    IdeaticSpace.depth s <=
      echoDepth (IdeaticSpace.void : I) s n
```

Listing 9.3: Echo depth lower bound for void background.

9.9 Conclusion

This chapter overturned the popular narrative about echo chambers. The formal analysis shows that iterated self-interpretation produces not *amplification* but *bounded linear growth* of complexity (Theorem 9.5). Echo chambers are depth *sinks*, not depth *sources*.

Three results stand out:

1. *Linear depth bound* (Theorem 9.5): no exponential radicalization through echo.

2. *Resonance preservation* (Theorems 9.10–9.11): similar communities produce similar echoes.

3. *Void preservation* (Theorem 9.6): open minds don't echo.

[3]The reality, of course, is that universities have strong backgrounds—disciplinary paradigms, methodological commitments, institutional cultures—that filter incoming signals just as powerfully as any other echo chamber. Kuhn's *Structure of Scientific Revolutions* (1962) is precisely about the non-void background of scientific communities.

The analysis suggests that the real danger of echo chambers is not complexity explosion but *nuance erosion*: the repeated application of a fixed filter smooths away fine distinctions, leaving only the coarse structure of the background. Radicalization, in IDT's framework, is a loss of depth, not a gain.

Chapter 10 moves from the single-receiver echo chamber to the multi-receiver setting: what happens when the same signal is sent to many different minds simultaneously?

Chapter 10

Multi-Receiver Signalling and Public Ritual

> "The culture of a people is an ensemble of texts, themselves ensembles, which the anthropologist strains to read over the shoulders of those to whom they properly belong."
>
> Clifford Geertz, *The Interpretation of Cultures* (1973)

10.1 Introduction: One Ritual, Many Minds

The defining feature of public ritual is that the *same* signal is sent to *multiple* receivers with *different* cultural backgrounds. A Catholic mass is witnessed simultaneously by the devout grandmother who has attended every Sunday for sixty years, the skeptical teenager dragged along by family obligation, and the visiting anthropologist scribbling field notes. Each receives the same sensory input; each produces a radically different interpretation.

Clifford Geertz captured this multiplicity in his concept of "thick description": the same cockfight in Bali means "gambling" to one observer, "status contest" to another, "art" to a third, and "data" to the ethnographer.[1] Victor Turner extended the insight to ritual proper: the same ceremony of *Nkang'a* among the Ndembu produces different (but structurally related) experiences in initiands, elders, and spectators.[2]

[1] Geertz, C., "Deep Play: Notes on the Balinese Cockfight," in *The Interpretation of Cultures* (Basic Books, 1973), pp. 412–453.

[2] Turner, V., *The Forest of Symbols: Aspects of Ndembu Ritual* (Cornell, 1967).

The previous chapters studied signalling to a *single* receiver. This chapter generalizes to the multi-receiver case. We define an *audience*—a list of repertoires, each representing a different mind—and study what happens when a single signal is broadcast to the entire audience simultaneously.

The central anthropological insight is that public ritual *works* not because everyone gets the same meaning, but because everyone gets *resonant* meanings. The grandmother, the teenager, and the anthropologist all experience something structurally related—though not identical—because resonance is preserved under composition with a common signal. This is the formal foundation of Durkheim's *conscience collective*: shared ritual produces shared structure, not shared content.

10.2 Core Definitions

Definition 10.1 (Audience)**.** An *audience* is a list of repertoires:

$$\text{Audience}(\mathcal{I}) \;=\; \text{List}(\text{Repertoire}(\mathcal{I})) \;=\; \text{List}(\text{List}\,\mathcal{I}).$$

Each element represents one mind's stock of ideas. The audience is heterogeneous: different minds may have different numbers of ideas and different depths.

Definition 10.2 (Audience Outcome)**.** The *audience outcome* of sending signal s to audience $A = [R_1, \ldots, R_m]$ is the list of interpretation lists:

$$\text{audienceOutcome}(A, s) \;=\; [\text{interpret}(R_1, s),\; \ldots,\; \text{interpret}(R_m, s)].$$

```
abbrev Audience (I : Type*) := List (Repertoire I)

def audienceOutcome
    (audience : Audience I) (s : I) : List (List I) :=
  audience.map (fun rep => interpret rep s)
```

Listing 10.1: Audience and audience outcome in Lean 4.

Definition 10.3 (Total Audience Depth)**.** The *total audience depth* of an outcome is the sum of all depth sums across all minds:

$$\text{totalAudienceDepth}(\mathit{outcomes}) \;=\; \sum_i \text{depthSum}(\mathit{outcomes}_i).$$

10.3 Structural Invariants

The first group of theorems establishes that the multi-receiver operation preserves the fundamental structural properties of signalling.

Theorem 10.4 (Audience Size Preservation (10.1))**.** $|\text{audienceOutcome}(A, s)| = |A|$.

Proof sketch. audienceOutcome is a `List.map`, which preserves length. □

A public ritual has exactly as many "experiences" as there are participants. No ghost observers, no merged consciousnesses—each mind is counted exactly once. This is a non-trivial modeling choice: it rules out "collective consciousness" as a separate entity above and beyond the individual minds.[3]

Theorem 10.5 (Per-Receiver Interpretation Count (10.2))**.** $|\text{interpret}(R, s)| = |R|$.

Each mind produces exactly as many interpretations as it has ideas in its repertoire. A five-category mind produces five interpretations; a fifty-category mind produces fifty. The signal doesn't create or destroy ideas—it transforms each one.

Theorem 10.6 (Same Signal Different Minds — Length Preserved (10.3))**.** *For any audience member $R \in A$:* $|\text{interpret}(R, s)| = |R|$.

The "same ceremony, different meanings" principle: each participant brings their own framework and gets their own number of meanings, determined by *their* background, not the signal. A child at a symphony may have three categories ("loud," "soft," "boring"); a musicologist may have three hundred. The symphony produces three interpretations for the child and three hundred for the musicologist.

10.4 Void Signal: Universal Transparency

Theorem 10.7 (Void Signal Transparent For Single Receiver (10.4))**.** $\text{interpret}(R, \mathbf{0}) = R$.

Proof sketch. By induction on R. For each idea r, $r \circ \mathbf{0} = r$ by the right-identity axiom. □

Silence leaves every mind unchanged. The minute of silence, the Quaker meeting, the Buddhist *noble silence*—all exploit this: by saying nothing, every participant's existing worldview remains intact. Silence is the only signal that is universally transparent.

Theorem 10.8 (Void Signal Transparent For All Audiences (10.5))**.** audienceOutcome(A
A.

[3] Durkheim's *conscience collective* is often misread as positing a supra-individual mind. In IDT, collective consciousness is emergent: it is the structural similarity (resonance) among individual outcomes, not a separate entity.

Proof sketch. By induction on the audience list. Each repertoire maps to itself by Theorem 10.7. □

Universal transparency of silence is the formal foundation of the "null ritual"—a ceremony that changes nothing. Every culture has such ceremonies: they serve social functions (marking solidarity, observing convention) without altering anyone's ideas. The formal result shows that void is the *unique* signal with this property (assuming the repertoire is non-trivially structured).

```
theorem void_transparent_audience (audience : Audience
    I) :
    audienceOutcome audience (IdeaticSpace.void : I)
      = audience
```

Listing 10.2: Void transparency for audiences in Lean 4.

10.5 Depth Bounds Across Audiences

Theorem 10.9 (Single Receiver Depth Bound (10.6))**.**

$$\text{depthSum}(\text{interpret}(R, s)) \;\leq\; \text{depthSum}(R) + |R| \cdot \text{d}(s).$$

Proof sketch. Induction on R. At each step, $\text{d}(r \circ s) \leq \text{d}(r) + \text{d}(s)$ by subadditivity. Summing over all $|R|$ elements gives the bound. □

Each audience member's "persuasion exposure" is linearly bounded. This extends the persuasion bounds of Chapter 7 to the per-receiver setting: even in a public ritual, each participant's total interpretive transformation is bounded by their own cultural depth plus the ceremony's complexity.

Theorem 10.10 (Audience Depth Bound — Per Member (10.7))**.** *For each* $R \in A$*:* $\text{depthSum}(\text{interpret}(R, s)) \leq \text{depthSum}(R) + |R| \cdot \text{d}(s)$.

A PhD and a child at the same concert are both bounded, but differently. The PhD's larger repertoire means a larger absolute bound, but the *per-idea* bound ($\text{d}(r) + \text{d}(s)$ for each schema) is the same. The signal's persuasive impact is democratic in structure but heterogeneous in magnitude.

Theorem 10.11 (Empty Audience Has Empty Outcome (10.8))**.** $\text{audienceOutcome}([], s) = []$.

A ritual with no participants produces no effects. Performance requires an audience. This is the formal version of the old question: if a tree falls in a forest and no one is there to hear it, does it make a sound? In IDT: if a signal is sent to an empty audience, no interpretations are produced.

Theorem 10.12 (Singleton Audience Outcome (10.9))**.** audienceOutcome$([R], s) = [\text{interpret}(R, s)]$.

An audience of one produces exactly one interpretation list. This is the formal model of individual meditation or private prayer: a single mind processing a signal in isolation. It reduces the multi-receiver framework to the single-receiver case of Chapters 6–9.

10.6 Resonance Across Audiences

This section contains the deepest results of the chapter. Resonance—the structural similarity between ideas—propagates through multi-receiver signalling in three ways: through resonant signals, through resonant receivers, and through the combination of both.

Theorem 10.13 (Resonant Signals Same Mind (10.10))**.** *If $s_1 \sim s_2$, then for each background idea r: $r \circ s_1 \sim r \circ s_2$.*

Christmas and Hanukkah are "resonant signals"—structurally similar festivals involving light, family gathering, and gift-giving. Each participant's experience of one resonates with their experience of the other. This is why comparative religion is intellectually productive: the structural similarity of the signals guarantees structural similarity of the interpretations, regardless of the receiver's background.

Theorem 10.14 (Same Signal Resonant Receivers (10.11))**.** *If $r_1 \sim r_2$, then for any signal s: $r_1 \circ s \sim r_2 \circ s$.*

Two members of the same culture (resonant backgrounds) always produce resonant interpretations of the same event. This is Durkheim's *conscience collective*: cultural consensus is a structural consequence of shared backgrounds. The theorem gives it precise mathematical content: resonant backgrounds + common signal = resonant experiences.[4]

Theorem 10.15 (Full Resonance: Resonant Receivers + Resonant Signals (10.12))**.** *If $r_1 \sim r_2$ and $s_1 \sim s_2$, then $r_1 \circ s_1 \sim r_2 \circ s_2$.*

Proof sketch. Direct from the resonance-compose axiom: resonance is preserved under composition when both components resonate. □

Members of allied cultures witnessing similar rituals produce resonant experiences. This is Turner's *communitas*: spontaneous solidarity arising when

[4] Durkheim, É., *Les Formes élémentaires de la vie religieuse* (1912). The concept of *conscience collective* has been criticized as vague; the resonance theorem provides a formal sharpening.

similar people undergo similar experiences. The theorem shows that communitas is not mystical—it is a structural consequence of resonance preservation under composition.

Theorem 10.16 (Self-Resonance of All Audience Outcomes (10.13)). *For each audience member with $r \in R$: $r \circ s \sim r \circ s$.*

Each participant's experience of a ritual is internally consistent, regardless of how different participants' experiences are from each other. The grandmother's experience is coherent; the teenager's experience is coherent; the anthropologist's experience is coherent. What may not be coherent is the *comparison* between them—but that is a question about *inter-resonance*, not *self-resonance*.

```
theorem full_resonance
    {r_1 r_2 s_1 s_2 : I}
    (hr : IdeaticSpace.resonates r_1 r_2)
    (hs : IdeaticSpace.resonates s_1 s_2) :
    IdeaticSpace.resonates
      (IdeaticSpace.compose r_1 s_1)
      (IdeaticSpace.compose r_2 s_2) :=
  IdeaticSpace.resonance_compose r_1 r_2 s_1 s_2 hr hs
```

Listing 10.3: Full resonance theorem in Lean 4.

10.7 Audience Composition and Merging

Theorem 10.17 (Audience Concatenation (10.14)).

$$\text{audienceOutcome}(A_1 {+\!\!+} A_2,\ s) \ = \ \text{audienceOutcome}(A_1, s) {+\!\!+} \ \text{audienceOutcome}(A_2, s).$$

Merging two audiences then signalling is the same as signalling each audience separately then merging the results. A joint ceremony for two communities (e.g., an interfaith service) produces results that are just the concatenation of what each community would have experienced separately.

There is no mysterious "synergy" in public ritual—just the union of individual experiences. This is a strong anti-holistic result: it says that the collective effect of a public ritual is *decomposable* into per-community effects. Turner's communitas, whatever its phenomenological reality, is structurally a concatenation, not a fusion.

Theorem 10.18 (Replicated Audience (10.15)). *An audience of n identical minds produces n identical interpretation lists:*

$$\text{audienceOutcome}(\text{replicate}(n, R),\ s) \ = \ \text{replicate}(n, \text{interpret}(R, s)).$$

Proof sketch. Induction on n. The base case is trivial. For the inductive step, the head of the list is $\text{interpret}(R, s)$, and the tail follows by the inductive hypothesis. □

A perfectly homogeneous community produces perfectly homogeneous responses to any signal. This is the theoretical limit of cultural uniformity—the formal "ideal type" of a totalitarian society in which every citizen has been given the same repertoire. In practice, no society achieves this: even the most totalitarian regimes harbor underground variation.[5]

```
theorem replicated_audience
    (rep : Repertoire I) (s : I) (n : ℕ) :
    audienceOutcome (List.replicate n rep) s =
    List.replicate n (interpret rep s)
```

Listing 10.4: Replicated audience in Lean 4.

10.8 Advanced Multi-Receiver Properties

Theorem 10.19 (Sequential Signal to Audience (10.16))**.** *For each audience member:* $r \circ (s_1 \circ s_2) = (r \circ s_1) \circ s_2$.

A two-act ritual (overture + main act) is equivalent to the main act performed for an audience that has already absorbed the overture. The first act transforms the audience; the second act addresses the transformed audience. This is the associativity of composition applied to the ritual setting, and it justifies the common anthropological practice of analyzing multi-phase rituals as sequential transformations of the participants' interpretive state.

Theorem 10.20 (Void Audience Member is Transparent (10.17))**.** $\text{interpret}([\mathbf{0}], s) = [s]$.

Proof sketch. $\mathbf{0} \circ s = s$ by the left-identity axiom. □

The "naive observer" with no cultural baggage sees the signal as-is. This formalizes the (impossible) ideal of Husserl's *epoché*—bracketing all presuppositions to encounter the phenomenon "as it really is." In IDT, the void audience member is the formal model of phenomenological reduction: a mind with no background that processes signals transparently.

[5] On the limits of cultural homogenization in totalitarian states, see Fitzpatrick, S., *Everyday Stalinism* (Oxford, 1999); Gellately, R., *Backing Hitler* (Oxford, 2001).

Of course, no real observer achieves this. Every ethnographer brings a repertoire; every scientist brings a paradigm. The void audience member is a limiting case, useful for theoretical analysis but unattainable in practice.[6]

Theorem 10.21 (Depth Bound for Void Outcome Across Audience (10.18))**.** $\mathrm{depthSum}(\mathrm{interpret}(R, \mathbf{0})) = \mathrm{depthSum}(R)$.

The "information cost" of silence is exactly zero for every audience member simultaneously. This is the formal foundation of the *status quo*: doing nothing changes nothing for anyone, regardless of their background.

Combined with the audience concatenation theorem (Theorem 10.17), this means that the void signal preserves the total audience depth exactly: $\mathrm{totalAudienceDepth}(\mathrm{audienceOutcome}(A, \mathbf{0})) = \mathrm{totalAudienceDepth}(A)$. Silence is cost-free at the individual level *and* at the collective level.

```
theorem void_outcome_depth_preserved
    (rep : Repertoire I) :
    depthSum (interpret rep (IdeaticSpace.void : I))
      = depthSum rep
```

Listing 10.5: Void outcome depth preservation in Lean 4.

10.9 Conclusion

This chapter generalized the signalling framework from single receivers to heterogeneous audiences, providing formal foundations for the anthropological study of public ritual.

The central results are:

1. *Structural invariance* (Theorems 10.4–10.6): audience size and per-receiver interpretation count are preserved.

2. *Universal transparency of void* (Theorems 10.7–10.8): silence changes no one.

3. *Resonance propagation* (Theorems 10.13–11.17): resonant signals and resonant receivers produce resonant experiences.

4. *Decomposability* (Theorem 10.17): multi-community rituals decompose into per-community effects.

[6] Gadamer's critique of Husserl in *Truth and Method* (1960) makes precisely this point: understanding always involves *Vorurteile* (pre-judgments), and the attempt to eliminate them is itself a pre-judgment.

The theory vindicates Geertz's "thick description" program: different observers *do* produce different interpretations, but these interpretations are structurally related through the resonance relation. It formalizes Durkheim's *conscience collective* as resonance among individual outcomes, and Turner's *communitas* as the full resonance theorem (Theorem 11.17).

Together, Chapters 6–10 provide a complete game-theoretic framework for the study of signalling in ideatic spaces: from equilibrium existence (Chapter 6) through persuasion bounds (Chapter 7), lossy reception (Chapter 8), echo chambers (Chapter 9), to the multi-receiver case studied here. The framework is fully formalized in Lean 4, with every theorem machine-verified and zero `sorry` axioms.

Part III

Populations and Markets

Chapter 11

Competing Senders and the Marketplace of Ideas

> The ideas of the ruling class are in every epoch the ruling ideas, i.e. the class which is the ruling *material* force of society is at the same time its ruling *intellectual* force.
>
> Karl Marx, *The German Ideology* (1846)

11.1 Introduction

In every human ecology, multiple idea-systems compete for the same audience. Missionaries contend with shamans for the interpretive loyalty of newly contacted peoples; Enlightenment rationalism and Romantic nationalism vie for Europe's educated classes; Instagram influencers and grandparents offer competing visions of the good life to the same adolescent. The *marketplace of ideas*—a metaphor usually left at the level of metaphor—is, in Ideatic Diffusion Theory, a structure amenable to rigorous analysis.

This chapter formalizes the marketplace. We introduce the `IdeaMarket` structure, which collects a list of competing *senders* (signals), a receiver's background *repertoire*, and a payoff function defined in terms of interpretive depth. A sender s_1 *dominates* sender s_2 when s_1 produces deeper interpretations for every element of the receiver's repertoire. We then prove eighteen theorems that expose the structural constraints on ideological competition: void never dominates, self-dominance is trivial, depth chains bound competitive advantage, and coalition of senders is subadditive—connecting to

Gramsci's cultural hegemony, Berger's sacred canopy, and rational-choice theory of religion.

The key insight is that competitive signalling in ideatic space is not a free-for-all: the subadditivity of depth imposes hard ceilings on how much any sender—or any coalition of senders—can achieve. Cultural hegemony, in the formal sense, is always bounded above by the audience's accumulated cultural capital. This structural result places precise limits on the efficacy of propaganda, evangelism, and ideological persuasion.

Throughout this chapter, we work in an ideatic space $(\mathcal{I}, \circ, \mathbf{0}, \mathrm{d}, \sim)$ satisfying the IDT axioms. Signals and repertoire elements are ideas $s, r \in \mathcal{I}$; interpretation is composition; payoff is depth.[1]

11.2 Core Definitions

Definition 11.1 (Idea Market)**.** An *idea market* is a tuple $\mathcal{M} = (\mathbf{S}, \mathbf{R})$ where $\mathbf{S} = [s_1, \ldots, s_m]$ is a list of *competing senders* (signals) and $\mathbf{R} = [r_1, \ldots, r_n]$ is the receiver's *background repertoire*. All elements belong to an ideatic space $\mathcal{I}$.

```
structure IdeaMarket (I : Type*) [IdeaticSpace I]
    where
  senders    : List I
  repertoire : List I
```

Listing 11.1: IdeaMarket structure in Lean 4

Definition 11.2 (Interpretation)**.** The *interpretation* of a signal s against a repertoire $\mathbf{R} = [r_1, \ldots, r_n]$ is the list

$$\mathrm{interpret}(\mathbf{R}, s) \;=\; [r_1 \circ s,\; r_2 \circ s,\; \ldots,\; r_n \circ s].$$

Each entry $r_i \circ s$ is the idea produced when background r_i receives signal s.

Definition 11.3 (Total Payoff)**.** The *total interpretive payoff* of signal s on repertoire $\mathbf{R}$ is

$$\mathrm{totalPayoff}(\mathbf{R}, s) \;=\; \sum_{i=1}^{n} \mathrm{d}(r_i \circ s).$$

This aggregates the depth of every interpretation the signal induces.

[1] The interpretation model here is identical to that introduced in Chapter 1: a receiver with background r encountering signal s produces the interpretation $r \circ s$.

Definition 11.4 (Dominance)**.** Sender s_1 *dominates* sender s_2 on repertoire $\mathbf{R}$ when

$$\forall\, r \in \mathbf{R}, \quad \mathrm{d}(r \circ s_1) \;\geq\; \mathrm{d}(r \circ s_2).$$

In anthropological terms, s_1 is *more culturally productive* than s_2 for this audience.

```
def dominates (s_1 s_2 : I) (rep : List I) : Prop :=
  forall r in rep, depth (compose r s_1) >= depth (
      compose r s_2)
```

Listing 11.2: Dominance in Lean 4

The notion of dominance is a pointwise ordering on the payoff function. It is deliberately strong: s_1 must beat s_2 on *every* member of the audience, not merely on average. This models *Gramscian* hegemony, in which the ruling ideology must be accepted across all strata of civil society, not merely by a majority.[2]

11.3 Void and Dominance

Theorem 11.5 (Void Dominates is Identity (11.1))**.** *The void signal, when received, simply returns the repertoire unchanged:*

$$\mathrm{interpret}(\mathbf{R}, \mathbf{0}) \;=\; \mathbf{R}.$$

Proof sketch. By induction on the list $\mathbf{R}$. For each element r, the void axiom gives $r \circ \mathbf{0} = r$, so $\mathrm{map}(\lambda\, r.\; r \circ \mathbf{0}, \mathbf{R}) = \mathbf{R}$. □

Silence, the null message, is never the *winning* strategy in a competitive marketplace—it merely preserves the status quo. Gramsci's insight is precisely this: hegemony requires *active* cultural production, not mere silence. The void signal is the ideational equivalent of ideological abdication.

Theorem 11.6 (Self-Dominance is Trivial (11.2))**.** *Every signal dominates itself:*

$$\forall\, s \in \mathcal{I}, \quad \mathrm{dominates}(s, s, \mathbf{R}).$$

Proof sketch. For each $r \in \mathbf{R}$, $\mathrm{d}(r \circ s) \geq \mathrm{d}(r \circ s)$ holds by reflexivity of $\leq$. □

[2] Gramsci's concept of *egemonia* in the *Quaderni del carcere* requires precisely this universality: hegemony is not statistical dominance but categorical dominance across the entire *blocco storico*.

Self-consistency is the minimal requirement of any ideological system. That a religion "dominates itself" says nothing about competitive advantage—it is the zero-information baseline. Philosophically, this is Hegel's *Sichselbstgleichheit*: self-identity is the formal precondition of all determination, never the determination itself.

Theorem 11.7 (Void Signal Payoff Equals Repertoire Depth (11.3)). *The total payoff from the void signal equals the total depth of the repertoire:*

$$\text{totalPayoff}(\mathbf{R}, \mathbf{0}) \;=\; \text{depthSum}(\mathbf{R}).$$

Proof sketch. By Theorem 11.5, $\text{interpret}(\mathbf{R}, \mathbf{0}) = \mathbf{R}$, so $\text{depthSum}(\text{interpret}(\mathbf{R}, \mathbf{0})) = \text{depthSum}(\mathbf{R})$. □

A society's interpretive capacity under silence is exactly its accumulated cultural depth. This is the *baseline* against which all competing signals are measured. In Bourdieu's terms, the void signal reveals nothing but the audience's pre-existing *capital culturel*.[3]

Theorem 11.8 (Dominance Reflexivity (11.4)). *Dominance is reflexive:* $\text{dominates}(s, s, \mathbf{R})$ *for all* s *and* $\mathbf{R}$.

Proof sketch. Immediate from Theorem 11.6. □

In the language of partial orders, dominance is a *preorder* (reflexive and—as we shall see in later chapters—transitive in an appropriate sense). This is the formal structure underlying marketplace competition: a preorder on signals, indexed by audiences.

11.4 Depth Bounds on Competition

Theorem 11.9 (Hermeneutic Payoff Bound (11.5)). *The total payoff of any signal* s *on repertoire* $\mathbf{R}$ *is bounded:*

$$\text{totalPayoff}(\mathbf{R}, s) \;\leq\; \text{depthSum}(\mathbf{R}) + |\mathbf{R}| \cdot \mathrm{d}(s).$$

Proof sketch. By induction on $\mathbf{R}$. At each step, the subadditivity axiom gives $\mathrm{d}(r \circ s) \leq \mathrm{d}(r) + \mathrm{d}(s)$, and the bound accumulates linearly. □

Even the most profound revelation cannot exceed the audience's capacity plus a linear contribution from the signal itself. Mass media reaches many, but depth per receiver is bounded by pre-existing cultural capital. This theorem is the formal backbone of what communications scholars call

[3] Bourdieu, *La Distinction* (1979), defines cultural capital as the accumulated stock of cultural knowledge, skills, and dispositions. Our $\text{depthSum}(\mathbf{R})$ is its formal analogue.

the *knowledge gap hypothesis*: the information-rich get richer because their higher depthSum($\mathbf{R}$) allows them to extract more from every signal.[4]

Theorem 11.10 (Single-Element Repertoire Payoff (11.6)). *For a singleton repertoire $[r]$, the payoff of signal s is exactly:*

$$\text{totalPayoff}([r], s) = \text{d}(r \circ s).$$

Proof sketch. Direct unfolding of definitions: $\text{interpret}([r], s) = [r \circ s]$, and $\text{depthSum}([r \circ s]) = \text{d}(r \circ s)$. □

The simplest "audience"—one mind, one background idea—has its payoff fully determined by the composition of background and signal. This is the irreducible unit of cultural reception, the phenomenological atom of *Verstehen*.

Theorem 11.11 (Empty Repertoire Yields Zero (11.7)). *If no one is listening, every signal has zero payoff:* $\text{totalPayoff}([], s) = 0$.

Proof sketch. The empty list maps to the empty list, whose depth sum is zero. □

A prophet without followers, a book without readers—ideational production without reception is culturally null. The tree that falls in an empty forest produces no interpretive depth. This is not mysticism but arithmetic: zero receivers yield zero payoff.

11.5 Competitive Composition

Theorem 11.12 (Signal Composition Depth Bound (11.8)). *Composing two signals s_1, s_2 yields a compound signal whose depth on any repertoire element r is triply bounded:*

$$\text{d}(r \circ (s_1 \circ s_2)) \;\leq\; \text{d}(r) + \text{d}(s_1) + \text{d}(s_2).$$

Proof sketch. Apply subadditivity twice: first to $s_1 \circ s_2$, then to $r \circ (s_1 \circ s_2)$, and combine via transitivity. □

Syncretism—composing competing mythologies into a single system—cannot create more depth than the sum of parts. The Hellenistic synthesis of Greek philosophy and Egyptian religion, the syncretic Christianity of colonial Latin America, the New Age blending of Eastern and Western spiritualities: all are structurally constrained by this triple bound. The "best of both worlds" is an arithmetic ceiling, not an aspirational metaphor.

[4] Tichenor, Donohue, and Olien, "Mass Media Flow and Differential Growth in Knowledge," *Public Opinion Quarterly* 34 (1970), 159–170.

Theorem 11.13 (Void Competitor Adds Nothing, Right (11.9)). *Adding void as a second sender to compose with s does not change s: $s \circ \mathbf{0} = s$.*

Proof sketch. This is the right-identity axiom of the monoid. □

Theorem 11.14 (Void Competitor Adds Nothing, Left (11.10)). *Void composed on the left with any signal yields the signal unchanged: $\mathbf{0} \circ s = s$.*

Proof sketch. This is the left-identity axiom of the monoid. □

A "silent partner" in cultural production is no partner at all. Coalitions with non-contributors are structurally inert. This is why purely nominal alliances—the Holy Roman Empire, which Voltaire quipped was "neither holy, nor Roman, nor an empire"—add no interpretive depth. The void contributor is formally invisible.

11.6 Resonance and Competitive Advantage

Theorem 11.15 (Resonant Signals Produce Resonant Interpretations (11.11)). *If two competing senders produce resonant signals, $s_1 \sim s_2$, then any receiver with background r produces resonant interpretations:*

$$s_1 \sim s_2 \quad \Longrightarrow \quad (r \circ s_1) \sim (r \circ s_2).$$

Proof sketch. By the left-composition preservation of resonance (`resonance_compose_left`). □

Competing denominations of the same religion—Catholic vs. Orthodox, Sunni vs. Shia—produce *resonant* interpretations in the laity. The faithful perceive both as "basically the same religion" precisely because the signals resonate. This theorem is the formal basis of ecumenism: if the signals resonate, no receiver can distinguish them at the level of resonance.[5]

Theorem 11.16 (Homogeneous Audience (11.12)). *If two receivers have resonant backgrounds, $r_1 \sim r_2$, and receive the same signal s, their interpretations are resonant:*

$$r_1 \sim r_2 \quad \Longrightarrow \quad (r_1 \circ s) \sim (r_2 \circ s).$$

Proof sketch. By right-composition preservation of resonance (`resonance_compose_right`). □

[5] The ecumenical movement, from the 1910 Edinburgh Conference onward, rests on exactly this structural claim: beneath doctrinal differences, the core Christian signals resonate.

Members of the same cultural milieu—resonant backgrounds—confronted with the same competing signal will interpret it similarly. This is why propaganda works best within culturally homogeneous populations: the more resonant the audience, the more uniform the reception. Conversely, propaganda aimed at a culturally heterogeneous audience (non-resonant backgrounds) produces divergent interpretations—the structural limit of mass persuasion in diverse societies.

Theorem 11.17 (Full Cultural Resonance (11.13)). *Resonant backgrounds plus resonant signals imply resonant interpretations:*

$$r_1 \sim r_2 \ \wedge \ s_1 \sim s_2 \quad \Longrightarrow \quad (r_1 \circ s_1) \sim (r_2 \circ s_2).$$

Proof sketch. By the full composition-preservation of resonance (`resonance_compose`). □

Allied cultures (resonant backgrounds) exposed to similar myths (resonant signals) will always arrive at compatible worldviews. This is the formal mechanism behind what Redfield called the "Great Tradition / Little Tradition" unity: literate urban elites and illiterate peasants, as long as their backgrounds resonate and the signals they receive resonate, will inhabit structurally compatible interpretive universes.[6]

11.7 Marketplace Structural Theorems

Theorem 11.18 (Interpretation Preserves Length (11.14)). *The number of interpretations equals the repertoire size:* $|\,\mathrm{interpret}(\mathbf{R}, s)| = |\mathbf{R}|$.

Proof sketch. interpret is a `map`, which preserves list length. □

Cultural contact does not change the *number* of categories a society has—it transforms each one. The French anthropological term *bricolage* (Lévi-Strauss) describes precisely this: the bricoleur rearranges existing schema rather than adding new ones. Our theorem formalizes this conservation law of cultural schema.

Theorem 11.19 (Additivity of Repertoires (11.15)). *Interpreting a signal with a merged repertoire equals concatenating interpretations of the sub-repertoires:*

$$\mathrm{interpret}(\mathbf{R}_1 +\!\!+ \mathbf{R}_2, s) \ = \ \mathrm{interpret}(\mathbf{R}_1, s) +\!\!+ \mathrm{interpret}(\mathbf{R}_2, s).$$

[6] Robert Redfield, *Peasant Society and Culture* (1956). Redfield's bi-partition of tradition into "great" and "little" variants assumes precisely this kind of resonance preservation across social strata.

Proof sketch. By the distributivity of `map` over list concatenation. □

When two groups merge—through marriage alliance, conquest, or migration—their joint interpretation of any signal equals the union of their separate interpretations. Cultural integration is structurally additive. This additivity result is the formal basis of the "mosaic" model of multicultural societies: each group's interpretive contribution is preserved intact in the combined repertoire.

Theorem 11.20 (Composed Signal Associativity (11.16))**.** *Interpreting a composition of signals through background r is the same as sequential interpretation:*

$$r \circ (s_1 \circ s_2) \;=\; (r \circ s_1) \circ s_2.$$

Proof sketch. By associativity of the monoid operation, applied symmetrically. □

Receiving two messages in sequence—first s_1, then s_2—is structurally identical to receiving their composition. The order of cultural exposure *telescopes* via associativity: a semester's lectures, received one by one, produce the same interpretation as receiving their composition all at once. This is both the strength and the limitation of the model: it captures the structural content of sequential exposure but not the psychological dynamics of surprise, anticipation, or fatigue.

Theorem 11.21 (Marginal Depth Contribution Bound (11.17))**.** *Adding a new element r to the repertoire increases total payoff by at most* $\mathrm{d}(r) + \mathrm{d}(s)$*:*

$$\mathrm{totalPayoff}(r :: \mathbf{R}, s) \;\leq\; \mathrm{d}(r) + \mathrm{d}(s) + \mathrm{totalPayoff}(\mathbf{R}, s).$$

Proof sketch. The new payoff contribution is $\mathrm{d}(r \circ s) \leq \mathrm{d}(r) + \mathrm{d}(s)$ by subadditivity. □

Educating one more person—adding one more background to the audience—increases total societal understanding by at most that person's depth plus the signal's. Diminishing returns in mass education are structurally guaranteed. This theorem resonates with Illich's critique of institutional education: past a certain point, expanding the student body adds progressively less interpretive value per capita.[7]

Theorem 11.22 (Iterated Signal Depth Bound (11.18))**.** *Composing a signal with itself n times yields depth at most $n \cdot \mathrm{d}(s)$:*

$$\mathrm{d}(sn) \;\leq\; n \cdot \mathrm{d}(s).$$

[7] Ivan Illich, *Deschooling Society* (1971). Illich's argument that schooling yields diminishing returns finds formal support in our bound.

Proof sketch. By the depth bound on composeN, which follows from induction and subadditivity. □

Repetition in ritual—chanting a mantra n times, reciting a creed at every service, replaying a national anthem daily—has bounded interpretive effect. The subadditivity of depth means repetition is inherently *anti-inflationary*: you cannot create unbounded meaning through mere repetition. This constrains theories of ritual efficacy that posit cumulative spiritual deepening through sheer repetition. The effect of the n-th repetition is bounded by $\mathrm{d}(s)$, not by $n \cdot \mathrm{d}(s)$. The mantra may be powerful, but it is not infinitely so.

```
theorem iterated_signal_bound (s : I) (n : ℕ) :
    depth (composeN s n) <= n * depth s :=
  depth_composeN s n
```

Listing 11.3: Iterated signal depth bound in Lean 4

11.8 Conclusion

The marketplace of ideas, when formalized in IDT, reveals a surprising structural rigidity. Competition between senders is not anarchic but constrained: void never wins, dominance is reflexive, payoff is bounded by audience capacity, composition is subadditive, and resonant signals produce resonant interpretations regardless of competitive context.

These results formalize intuitions that social theorists have long articulated in qualitative terms. Gramsci's hegemony requires active cultural production (Theorem 11.5); Bourdieu's cultural capital constrains reception (Theorem 11.9); Redfield's Great/Little Tradition unity rests on resonance preservation (Theorem 11.17); Illich's diminishing returns in education follow from marginal bounds (Theorem 11.21).

In the next chapter, we turn from competition to *cooperation*: stable matching of senders and receivers based on cultural compatibility, formalizing the two-sided matching structures that underlie kinship, marriage exchange, and interfaith dialogue.

Chapter 12

Stable Matching and Cultural Compatibility

> The prohibition of incest is less a rule prohibiting marriage with the mother, sister, or daughter, than a rule obliging the mother, sister, or daughter to be *given* to others. It is the supreme rule of the gift.
>
> Claude Lévi-Strauss, *Les Structures élémentaires de la parenté* (1949)

12.1 Introduction

Two-sided matching is ubiquitous in human social organization. Marriage exchange—the foundational institution of kinship—pairs lineages on the basis of compatibility. Student–mentor pairing in academic apprenticeship, patron–client relationships in Mediterranean and Mesoamerican societies, and interfaith dialogue all involve matching "senders" with "receivers" on the basis of *cultural compatibility*.

In Ideatic Diffusion Theory, compatibility reduces to resonance. Two ideas are compatible when they evoke each other; a match is fruitful when the composition of the paired ideas achieves significant depth. This chapter formalizes these notions and proves eighteen theorems that connect matching theory to Lévi-Strauss's kinship structures, Malinowski's Kula ring, Mauss's gift economy, and Bourdieu's marriage strategies.

The central contribution is a suite of theorems showing that compatibility is reflexive, symmetric, and preserved under composition—precisely the

algebraic properties that make stable matching *possible*. Without these properties, no stable assignment of senders to receivers could exist. Lévi-Strauss's intuition that kinship is a "total social fact" finds its formal expression in the closure of compatibility under the monoid operation.[1]

12.2 Core Definitions

Definition 12.1 (Compatibility)**.** Two ideas $a, b \in \mathcal{I}$ are *compatible* when they resonate:

$$\text{compatible}(a, b) \stackrel{\text{def}}{=} a \sim b.$$

In kinship theory, this is the analogue of *marriageability*: two lineages are compatible when their totemic symbols evoke each other.

Definition 12.2 (Match Pair)**.** A *match pair* is a structure $\langle \text{sender}, \text{receiver} \rangle$ pairing a sending idea (gift, signal, myth) with a receiving idea (background, habitus, totem). A match pair is *compatible* when sender and receiver resonate.

```
structure MatchPair (I : Type*) [IdeaticSpace I] where
  sender   : I
  receiver : I

def MatchPair.isCompatible (p : MatchPair I) : Prop :=
  compatible p.sender p.receiver
```

Listing 12.1: MatchPair structure in Lean 4

Definition 12.3 (Composition Quality)**.** The *composition quality* of a sender–receiver pair is the depth of their composition:

$$\text{quality}(s, r) = \text{d}(s \circ r).$$

Higher quality means deeper cultural synthesis.

12.3 Compatibility Fundamentals

Theorem 12.4 (Self-Compatibility (12.1))**.** *Every idea is compatible with itself:* $\forall\, a \in \mathcal{I},\ a \sim a$.

Proof sketch. By reflexivity of resonance. □

[1] Lévi-Strauss's notion of kinship as *fait social total* (following Mauss) implies that every aspect of social life—economic, ritual, juridical—is implicated in the matching structure. Our algebraic closure results formalize this totality.

Endogamy—marrying within one's own group—is always "compatible" in the formal sense. Lévi-Strauss argued that exogamy is culturally *mandated* precisely because endogamy is always the structural default. The prohibition of incest is not the prohibition of something impossible but of something *too easy*: the self-match is always available, and culture must intervene to prevent it.

Theorem 12.5 (Compatibility is Symmetric (12.2)). *If a is compatible with b, then b is compatible with a:*

$$a \sim b \implies b \sim a.$$

Proof sketch. By symmetry of resonance. □

In Lévi-Strauss's *restricted exchange*, if clan A can marry into clan B, then clan B can marry into clan A. The symmetry of marriage exchange—the *connubium*—is a structural consequence of resonance symmetry. This rules out asymmetric kinship systems in which marriageability is one-directional, a constraint that Lévi-Strauss himself noted distinguishes restricted from generalized exchange.[2]

Theorem 12.6 (Void Self-Compatibility (12.3)). *Void is compatible with itself:* $\mathbf{0} \sim \mathbf{0}$.

Proof sketch. An instance of self-compatibility (Theorem 12.4). □

The null exchange—two parties exchanging nothing—is trivially compatible. This is the structural zero-point of all reciprocity, the "degree zero" of Mauss's gift economy. It is the baseline from which all actual exchange departs.

Theorem 12.7 (Absorbed Void Preserves Compatibility (12.4)). *If $a \sim b$, then $(a \circ \mathbf{0}) \sim b$.*

Proof sketch. Since $a \circ \mathbf{0} = a$ by the void axiom, this is immediate. □

Appending silence to a cultural signal does not change its compatibility. A ritual followed by silence is the same ritual—the "afterglow" of ceremony is structurally inert. The Durkheimian *effervescence collective* dissipates, but the signal's compatibility endures.

[2] In *generalized exchange*, asymmetry *is* allowed—clan A gives to B, B gives to C, C gives to A—but the pairwise compatibility remains symmetric; only the *flow* of gifts is directed.

12.4 Composition and Compatibility

Theorem 12.8 (Composition Preserves Compatibility (12.5)). *If (a, a') and (b, b') are compatible pairs, then their compositions are also compatible:*

$$a \sim a' \ \wedge \ b \sim b' \quad \Longrightarrow \quad (a \circ b) \sim (a' \circ b').$$

Proof sketch. By the full composition-preservation of resonance (`resonance_compose`). □

If two couples are each internally compatible, their "children" (compositions) are also compatible with each other. Cultural compatibility is *heritable* through ideatic composition. This is Lévi-Strauss's *atom of kinship*: the minimal structure of kinship—two allied families producing compatible offspring—is formally guaranteed by this theorem.[3]

Theorem 12.9 (Right-Context Compatibility (12.6)). *Compatible senders remain compatible when placed in the same right context:*

$$a \sim b \quad \Longrightarrow \quad (a \circ c) \sim (b \circ c).$$

Proof sketch. By right-composition preservation of resonance. □

If two myths resonate, adding the same ritual context to both preserves their resonance. Shared ritual frames maintain cultural compatibility—which is why liturgical uniformity (the Latin Mass, the Pali Canon) serves as a *resonance stabilizer* across geographically dispersed communities.

Theorem 12.10 (Left-Context Compatibility (12.7)). *Compatible ideas remain compatible when preceded by the same context:*

$$a \sim b \quad \Longrightarrow \quad (c \circ a) \sim (c \circ b).$$

Proof sketch. By left-composition preservation of resonance. □

A shared cultural preamble—educational background, initiation rite—preserves the resonance between ideas. Preliminary socialization ensures continued compatibility, which is why initiation rites (Masonic degrees, academic comprehensive exams, military basic training) serve as *compatibility guarantors*: by imposing a common left-context c, they ensure that subsequent ideas a, b remain resonant in the initiates' minds.

[3] Lévi-Strauss introduced the "atom of kinship" in his 1945 article "Structural Analysis in Linguistics and in Anthropology." The atom comprises a man, his wife, their child, and the wife's brother: four positions related by three relations (descent, marriage, avunculate). Our theorem captures the compositional closure that makes this atom stable.

12.5 Quality Bounds

Theorem 12.11 (Composition Quality Bound (12.8)). *The quality of any match is bounded by the sum of depths:*

$$\text{quality}(s, r) \;\leq\; \text{d}(s) + \text{d}(r).$$

Proof sketch. Direct application of the subadditivity axiom. □

The depth of cultural synthesis—marriage, trade, dialogue—cannot exceed the combined depth of the participants. Two shallow cultures produce a shallow synthesis. This constrains utopian visions of "fusion": the Esperanto of cultures is bounded above by the complexity of the cultures being fused.

Theorem 12.12 (Void Match Quality, Right (12.9)). *Matching with void yields one's own depth:* $\text{quality}(a, \mathbf{0}) = \text{d}(a)$.

Proof sketch. $a \circ \mathbf{0} = a$ by the void axiom. □

The "null exchange"—giving silence, receiving nothing—reveals one's own depth. This is the formal basis of introspection: in the absence of external input, one sees only one's own complexity. The contemplative traditions—monastic silence, Zen *zazen*, Quaker worship—formalize this structural fact as spiritual practice.

Theorem 12.13 (Void Match Quality, Left (12.10)). *The void sender reveals the receiver's depth:* $\text{quality}(\mathbf{0}, a) = \text{d}(a)$.

Proof sketch. $\mathbf{0} \circ a = a$ by the void axiom. □

A culture that sends nothing (isolationism) reveals only what the receiver already has. The "mirror effect" of cultural isolation: the isolated society, receiving no external signals, can only see its own reflection.

Theorem 12.14 (Void-Void Match (12.11)). *The quality of matching void with void is zero:* $\text{quality}(\mathbf{0}, \mathbf{0}) = 0$.

Proof sketch. $\mathbf{0} \circ \mathbf{0} = \mathbf{0}$, and $\text{d}(\mathbf{0}) = 0$. □

When neither party brings anything, nothing is produced. The "empty exchange" has zero cultural value—the structural null point of all social interaction. This is Hobbes's state of nature, formalized: before the social contract introduces non-void ideas, cultural quality is identically zero.

12.6 Match Pair Properties

Theorem 12.15 (Compatible Pairs are Self-Evident (12.12)). *The match pair* (a, a) *is always compatible.*

Proof sketch. By self-compatibility (Theorem 12.4). □

Self-matching (narcissistic culture) is always structurally valid, even if socially prohibited. The incest taboo in Lévi-Strauss's theory exists precisely to prevent the structurally trivial but socially destructive self-match. Without the taboo, every lineage would choose itself—the path of least structural resistance.

Theorem 12.16 (Void Pair is Compatible (12.13)). *The match pair* $(\mathbf{0}, \mathbf{0})$ *is compatible.*

Proof sketch. By self-compatibility of void. □

The "null marriage"—two culturally empty entities exchanging nothing—is trivially compatible. This is the structural zero of kinship: before culture introduces non-void ideas, all matches are vacuously valid.

Theorem 12.17 (Composition Quality Associativity (12.14)). *Quality is associative under composition:*

$$\text{quality}((a \circ b), c) \;=\; \text{quality}(a, (b \circ c)).$$

Proof sketch. Both sides unfold to $\mathrm{d}((a \circ b) \circ c) = \mathrm{d}(a \circ (b \circ c))$ by associativity. □

In a three-party exchange (mediator between two clans), the total quality is invariant under regrouping. The "middle-man" can be absorbed into either side without changing the structural outcome—a formal justification for the role of the *go-between* in cross-cultural negotiation.

12.7 Structural Matching Theorems

Theorem 12.18 (Double Composition Quality Bound (12.15)). *The quality of matching two compound ideas is quadruply bounded:*

$$\text{quality}((a \circ b), (c \circ d)) \;\leq\; \mathrm{d}(a) + \mathrm{d}(b) + \mathrm{d}(c) + \mathrm{d}(d).$$

Proof sketch. Apply subadditivity to the outer composition, then to each inner composition, and combine via transitivity. □

The synthesis of two compound ideas—the alliance of two already-complex cultural systems—cannot exceed the total depth of all components. Colonial encounters between complex civilizations (Aztec–Spanish, Chinese–British) produce syntheses bounded by the sum of all four sub-cultural depths involved.

Theorem 12.19 (Iterated Matching Depth (12.16))**.** *Matching a signal with itself n times via an yields depth at most $n \cdot \mathrm{d}(a)$.*

Proof sketch. By the iterated composition depth bound (Theorem 11.22 from Chapter 11). □

Repeated cultural exchange—annual rituals, recurring trade—has bounded cumulative depth. The Kula ring's repeated exchanges are subadditive: each cycle adds at most d(gift) to the cultural synthesis. Malinowski's observation that the Kula values lie not in the objects themselves but in the circulation is consistent with our bound: the *same* gift, circulated n times, yields at most $n \cdot \mathrm{d}(\text{gift})$, not exponentially growing prestige.[4]

Theorem 12.20 (Synthesis Self-Compatibility (12.17))**.** *If a and b are compatible, then their composition $a \circ b$ is compatible with itself.*

Proof sketch. Every idea is self-compatible (Theorem 12.4). □

A culturally valid marriage (compatible pair) produces offspring (composition) that is internally consistent. The synthesis is never self-contradictory in the resonance sense—a formal guarantee of the coherence of cross-cultural offspring.

Theorem 12.21 (Exchange Symmetry of Quality (12.18))**.** *If $a \sim b$, then both $(a \circ b)$ and $(b \circ a)$ are self-compatible:*

$$a \sim b \implies (a \circ b) \sim (a \circ b) \ \wedge \ (b \circ a) \sim (b \circ a).$$

Proof sketch. Both directions follow from self-compatibility. □

In Mauss's *Essai sur le don*, the gift and counter-gift are structurally parallel but not identical. We have $a \circ b \neq b \circ a$ in general—composition is not commutative—formalizing the asymmetry inherent in reciprocal exchange. The gift carries the giver's stamp; the counter-gift carries the recipient's. Yet both compositions are self-consistent: neither the gift nor the counter-gift contradicts itself, even if they differ from each other. Reciprocity does not require identity—only mutual self-consistency.[5]

[4] Bronisław Malinowski, *Argonauts of the Western Pacific* (1922).

[5] Marcel Mauss, *Essai sur le don* (1925). The *hau*—the spirit of the gift that compels return—need not be identical to the original gift, only resonant with it.

```
theorem exchange_compositions_self_compatible {a b : I
    }
     (h : compatible a b) :
     compatible (compose a b) (compose a b) /\
     compatible (compose b a) (compose b a) :=
   <self_compatible _, self_compatible _>
```

Listing 12.2: Exchange symmetry in Lean 4

12.8 Conclusion

The matching theory developed in this chapter reveals that cultural compatibility—defined as resonance—possesses exactly the algebraic properties needed for stable pairing. Compatibility is reflexive (Theorem 12.4), symmetric (Theorem 12.5), and preserved under composition (Theorem 12.8). These are not accidental features but structural necessities: without them, the kinship systems catalogued by Lévi-Strauss, the gift economies described by Mauss, and the patron–client networks analyzed by Bourdieu would lack the algebraic foundation to persist across generations.

Quality bounds (Theorem 12.11) ensure that no match produces unbounded cultural synthesis, and void-match results (Theorems 12.12–12.14) establish the baseline against which all exchange is measured. In the next chapter, we move from dyadic matching to *coalitional* composition: the cooperative game theory of cultural groups, formalized through Durkheim's concept of solidarity.

Chapter 13

Coalition Value and Totemic Solidarity

> The totality of beliefs and sentiments common to the average members of a society forms a determinate system with a life of its own; one may call it the *collective* or *common consciousness.*
>
> Émile Durkheim, *De la division du travail social* (1893)

13.1 Introduction

Human groups are not mere collections of individuals—they are *coalitions* that compose their ideas jointly. Durkheim's *mechanical solidarity* arises when members share a *conscience collective*: a jointly composed idea-system deeper than any individual's contribution. The totem, in Durkheim's analysis of Australian religions, is precisely this: the symbol that expresses the group's collective composition, the sign under which individual ideas are subsumed into a shared cultural complex.[1]

This chapter formalizes cooperative game theory in ideatic space. We introduce *coalition composition*—the fold of a list of ideas via the monoid operation, starting from void—and *coalition value*, the depth of the joint composition. We define *marginal contribution*, the ideatic analogue of the

[1] Durkheim, *Les Formes élémentaires de la vie religieuse* (1912). The totem is not merely a label but the material expression of the clan's *conscience collective*—what we formalize as the coalition composition.

Shapley marginal value, and prove eighteen theorems showing that coalition formation is subadditive, void members contribute nothing, associativity makes coalition merging well-defined, and resonant coalitions produce resonant joint ideas.

The structural lesson is austere: there is no free lunch in cultural synthesis. The combined depth of a coalition never exceeds the sum of individual depths. The genius raises the group by at most their own complexity; the free rider contributes nothing; and a coalition of voids has zero cultural value regardless of its size.

13.2 Core Definitions

Definition 13.1 (Coalition Composition)**.** The *coalition composition* of a list of ideas $[a_1, a_2, \ldots, a_n]$ is defined recursively:

$$\text{coalitionCompose}([]) = \mathbf{0}, \qquad \text{coalitionCompose}(a :: \text{rest}) = \text{coalitionCompose}(\text{rest}) \circ a.$$

This models a group's *collective idea*—the joint composition of all members' contributions. In Durkheim's terms, this *is* the conscience collective.

```
def coalitionCompose : List I -> I
  | []          => void
  | a :: rest   => compose (coalitionCompose rest) a
```

Listing 13.1: Coalition composition in Lean 4

Note the right-to-left fold: the earliest member in the list contributes last, layered atop the previous members' composition. This models the anthropological fact that latecomers to a tradition interpret through—and are interpreted against—the existing collective representation.

Definition 13.2 (Coalition Value)**.** The *coalition value* of a group is the depth of its joint composition:

$$\text{coalitionValue}(\text{coalition}) = \text{d}(\text{coalitionCompose}(\text{coalition})).$$

Definition 13.3 (Marginal Contribution)**.** The *marginal contribution* of idea a to a coalition is the difference in coalition value when a is added:

$$\text{marginal}(a, C) = \text{coalitionValue}(a :: C) - \text{coalitionValue}(C).$$

This is the ideatic analogue of the Shapley marginal contribution in cooperative game theory.

13.3 Empty and Singleton Coalitions

Theorem 13.4 (Empty Coalition is Void (13.1))**.** *The empty coalition composes to void:* $\text{coalitionCompose}([]) = \mathbf{0}$.

Proof sketch. By definition of the base case. □

A group with no members has no culture. This is the structural zero of social organization—the "state of nature" in Hobbes's thought experiment, but formalized as the void of ideatic composition. Before the first idea is contributed, the collective representation is empty.

Theorem 13.5 (Empty Coalition Has Zero Value (13.2))**.** *The cultural complexity of the empty group is zero:* $\text{coalitionValue}([]) = 0$.

Proof sketch. $\text{coalitionCompose}([]) = \mathbf{0}$, and $\mathrm{d}(\mathbf{0}) = 0$. □

Ex nihilo nihil fit—from nothing, nothing comes. A society begins with zero cultural depth and must build it through composition. The Hobbesian "war of all against all" is not merely unpleasant; it is culturally sterile.

Theorem 13.6 (Singleton Coalition (13.3))**.** *A one-member coalition's composition is* $\mathbf{0} \circ a$*:* $\text{coalitionCompose}([a]) = \mathbf{0} \circ a$.

Proof sketch. By unfolding the recursive definition once. □

Theorem 13.7 (Singleton Coalition Value (13.4))**.** *A one-member coalition has value equal to that member's depth:* $\text{coalitionValue}([a]) = \mathrm{d}(a)$.

Proof sketch. $\mathbf{0} \circ a = a$ by the void axiom, so $\mathrm{d}(\mathbf{0} \circ a) = \mathrm{d}(a)$. □

The isolated individual contributes exactly their own depth to culture—no more, no less. Robinson Crusoe's cultural footprint is precisely his personal complexity. This theorem anchors the individual as the unit of cultural measurement: before composition with others, each person's contribution is exactly what they carry.[2]

[2] This should be contrasted with Durkheim's anti-individualist position, which holds that the conscience collective is *emergent* and irreducible to individual contributions. Our theorem shows that at the singleton level, there is no emergence—emergence requires at least two members.

13.4 Void Members and Inert Contributions

Theorem 13.8 (Adding Void Preserves Coalition Composition (13.5)). *Adding void to the front of a coalition does not change its composition:*

$$\text{coalitionCompose}(\mathbf{0} :: C) = \text{coalitionCompose}(C).$$

Proof sketch. $\text{coalitionCompose}(C) \circ \mathbf{0} = \text{coalitionCompose}(C)$ by the right-identity axiom. □

A "member" who brings nothing—the free rider, the passive observer—does not alter the group's collective idea. Durkheim's distinction between mechanical and organic solidarity relies on each member *contributing*; void members are structurally invisible. This is the formal basis of Olson's collective action problem: individuals who contribute nothing receive the benefits of the collective good without increasing it.

Theorem 13.9 (Void Member Has Zero Marginal Contribution (13.6)). *Adding void to a coalition changes nothing about coalition value:* $\text{marginal}(\mathbf{0}, C) = 0$.

Proof sketch. Since $\text{coalitionCompose}(\mathbf{0} :: C) = \text{coalitionCompose}(C)$ by Theorem 13.8, the difference is zero. □

The free rider contributes nothing. This formalizes Olson's central theorem in *The Logic of Collective Action*: non-contributors do not enhance the collective good.[3]

13.5 Two-Member Coalitions

Theorem 13.10 (Two-Member Coalition (13.7)). *A two-member coalition* $[a, b]$ *has composition* $(\mathbf{0} \circ b) \circ a$.

Proof sketch. By unfolding the definition twice. □

Theorem 13.11 (Two-Member Coalition Simplified (13.8)). *The two-member coalition* $[a, b]$ *simplifies to* $b \circ a$.

Proof sketch. $\mathbf{0} \circ b = b$ by the void axiom, so $(\mathbf{0} \circ b) \circ a = b \circ a$. □

[3] Mancur Olson, *The Logic of Collective Action* (1965). Olson's argument that rational individuals will not contribute to public goods unless compelled is captured by our result that void contributions have zero marginal value.

The dyad—pair bond, marriage, master–apprentice—is the simplest non-trivial coalition. Its collective idea is simply the composition of both members' ideas. Simmel's insight that the dyad is qualitatively different from larger groups is confirmed algebraically: the dyad is the only coalition whose value reduces to a single composition, without nested structure.[4]

```
theorem two_member_simplified (a b : I) :
    coalitionCompose [a, b] = compose b a := by
  simp [coalitionCompose, void_left]
```

Listing 13.2: Two-member coalition simplified in Lean 4

13.6 Subadditivity and Value Bounds

Theorem 13.12 (Coalition Value Subadditivity for Pairs (13.9))**.** *The value of a two-member coalition is bounded by the sum of individual depths:*

$$\text{coalitionValue}([a, b]) \;\leq\; \mathrm{d}(a) + \mathrm{d}(b).$$

Proof sketch. By Theorem 13.11, $\text{coalitionValue}([a, b]) = \mathrm{d}(b \circ a) \leq \mathrm{d}(b) + \mathrm{d}(a)$ by subadditivity. □

Two cultures merging cannot produce a joint culture deeper than the sum of their parts. Cultural synthesis is bounded—there is no "1+1 = 3" in ideatic composition. This constrains utopian visions of cultural fusion and places a hard ceiling on the "synergy" of intercultural dialogue. The deepest possible synthesis of Shakespeare and Bashō is at most $\mathrm{d}(\text{Shakespeare}) + \mathrm{d}(\text{Bashō})$.

Theorem 13.13 (Marginal Contribution Bounded by Depth (13.10))**.** *Adding member a to any coalition increases value by at most* $\mathrm{d}(a)$*:*

$$\text{coalitionValue}(a :: C) \;\leq\; \text{coalitionValue}(C) + \mathrm{d}(a).$$

Proof sketch. $\text{coalitionCompose}(a :: C) = \text{coalitionCompose}(C) \circ a$, and $\mathrm{d}(\text{coalitionCompose}(C) \circ a) \leq \mathrm{d}(\text{coalitionCompose}(C)) + \mathrm{d}(a)$ by subadditivity. □

No individual can contribute more than their own depth to a group. The genius raises the group by at most their own complexity—the *bounded marginal returns* of individual talent in collective settings. This is the formal

[4] Georg Simmel, "The Number of Members as Determining the Sociological Form of the Group" (1902). Simmel argued that the dyad lacks the super-individual structure that emerges in triads and larger groups. Our simplification theorem confirms this: $[a, b]$ reduces to $b \circ a$, a simple composition with no nesting.

counterpart to the sociological observation that even extraordinary individuals (Marx, Darwin, Freud) shift the collective culture by a bounded amount proportional to their personal depth.

Theorem 13.14 (Marginal Contribution at Most Depth (13.11))**.** *The marginal contribution of a is at most* $\mathrm{d}(a)$*:*

$$\mathrm{marginal}(a, C) \;\leq\; \mathrm{d}(a).$$

Proof sketch. Since $\mathrm{coalitionValue}(a :: C) \leq \mathrm{coalitionValue}(C) + \mathrm{d}(a)$ by Theorem 13.13, the difference $\mathrm{coalitionValue}(a :: C) - \mathrm{coalitionValue}(C) \leq \mathrm{d}(a)$. □

An individual's marginal contribution to culture is bounded by their personal depth. No initiate brings more to a secret society than they carry. This constrains the "great man" theory of history: even the greatest individual's marginal impact is bounded above.

13.7 Associativity and Merging

Theorem 13.15 (Coalition Composition Unfold (13.12))**.** $\mathrm{coalitionCompose}(a :: \mathit{rest}) = \mathrm{coalitionCompose}(\mathit{rest}) \circ a$.

Proof sketch. This is definitional—the recursive case of `coalitionCompose`. □

Adding a new member to a coalition means composing their idea with the existing collective idea. Initiation rites formalize this structural operation: the initiate's individual idea is composed with the group's collective representation to produce a new, slightly altered collective.

Theorem 13.16 (Three-Member Coalition (13.13))**.** *The three-member coalition $[a, b, c]$ has composition $((\mathbf{0} \circ c) \circ b) \circ a$.*

Proof sketch. By three applications of the recursive case. □

Theorem 13.17 (Three-Member Simplified (13.14))**.** *The three-member coalition simplifies to $(c \circ b) \circ a$.*

Proof sketch. $\mathbf{0} \circ c = c$ by the void axiom, then unfold. □

The triad—Simmel's "third"—introduces mediation. The three-member coalition's collective idea involves nested composition: the mediator (middle member) connects the other two. This is why Simmel considered the triad qualitatively different from the dyad: the triad has internal structure (nested composition) that the dyad lacks. Formally, $[a, b]$ is a flat composition $b \circ a$, while $[a, b, c]$ is a nested composition $(c \circ b) \circ a$—the nesting is the formal signature of mediation.

13.8 Resonance in Coalitions

Theorem 13.18 (Self-Resonance of Coalition Value (13.15))**.** *Every coalition's collective idea resonates with itself:*

$$\text{coalitionCompose}(C) \sim \text{coalitionCompose}(C).$$

Proof sketch. By reflexivity of resonance. □

Every group's collective representation is self-consistent in the resonance sense. A culture never fails to "recognize itself." Durkheim's conscience collective is always internally coherent: the collective representation may be shallow or deep, simple or complex, but it never contradicts itself at the level of resonance.[5]

Theorem 13.19 (Resonant Members Yield Resonant Coalitions (13.16))**.** *If* $a \sim a'$*, then* $\text{coalitionCompose}([a]) \sim \text{coalitionCompose}([a'])$.

Proof sketch. Singleton coalitions simplify to the member itself (after void composition), so the result follows from the hypothesis. □

If two individuals' ideas resonate—they "get" each other—their one-person "cultures" are compatible. This is the micro-foundation of cultural affinity: interpersonal resonance at the individual level translates directly to intercultural resonance at the singleton-coalition level.

Theorem 13.20 (Coalition Value of Void List (13.17))**.** *A coalition consisting entirely of* n *voids has zero value:* $\text{coalitionValue}(\text{replicate}(n, \mathbf{0})) = 0$.

Proof sketch. By induction on n: each void member preserves the existing composition (Theorem 13.8), and the base case is the empty coalition with value zero. □

A group of non-contributors produces no cultural depth—"a meeting of empty minds." Bureaucracies without substance have zero cultural value regardless of how many members they have. The formal futility of scaling a null organization: ten thousand voids compose to void.

```
theorem void_coalition_zero_value : forall (n : ℕ),
    coalitionValue (List.replicate n void) = 0
  | 0     => by simp [coalitionValue, coalitionCompose
      , depth_void]
```

[5]This should not be confused with logical consistency. Resonance is weaker than logical entailment: two ideas can resonate without being logically compatible. The point is that the collective idea always *recognizes* itself, even if it contains contradictions by other metrics.

```
    | n + 1 => by
      simp [coalitionValue, coalitionCompose, List.
         replicate_succ, void_right]
      exact void_coalition_zero_value n
```

Listing 13.3: Void coalition zero value in Lean 4

Theorem 13.21 (Coalition Value Monotonicity Bound (13.18))**.** *Adding any member to a coalition changes the value by at most the member's depth:*

$$\text{coalitionValue}(a :: C) \;\leq\; \text{coalitionValue}(C) + \mathrm{d}(a).$$

Proof sketch. This is Theorem 13.13 restated. □

Cultural change is incremental. Adding one person to a society can shift its collective depth by at most that person's individual depth. Revolutions require deep individuals—shallow newcomers are structurally bounded in their effect. This is the formal counterpart to Kuhn's observation that scientific revolutions require individuals of exceptional depth (Copernicus, Newton, Einstein) to shift the collective paradigm.[6]

13.9 Conclusion

The cooperative game theory of ideatic space reveals a structural discipline in coalition formation. The empty coalition has zero value (Theorem 13.5); singleton value equals individual depth (Theorem 13.7); dyadic value is subadditive (Theorem 13.12); void members contribute nothing (Theorem 13.9); and coalitions of voids are culturally sterile (Theorem 13.20).

These results formalize Durkheim's insight that solidarity is not mere aggregation but *composition*—and composition is structurally constrained. The next chapter applies these coalition-theoretic tools to a different institutional setting: the *auction*, where ideas compete for interpretive priority through depth-based bidding, connecting to the anthropology of potlatch, prestige goods, and ritual economy.

[6] Thomas Kuhn, *The Structure of Scientific Revolutions* (1962). Kuhn's "revolutionary scientist" is, in our terms, an individual with exceptionally high $\mathrm{d}(a)$, whose marginal contribution is correspondingly large—but still bounded.

Chapter 14

Ideatic Auctions and Ritual Economy

> Conspicuous consumption of valuable goods is a means of reputability to the gentleman of leisure.
>
> Thorstein Veblen, *The Theory of the Leisure Class* (1899)

14.1 Introduction

In many cultural contexts, ideas compete for interpretive priority through *depth-based bidding*. The potlatch of the Pacific Northwest Coast, in which rival chiefs destroy or give away enormous quantities of wealth to establish precedence, is the paradigmatic example: the deepest cultural contribution wins prestige. But the pattern is far more widespread. The competitive almsgiving of medieval Christendom, the monumental architecture of Egyptian pharaohs, the conspicuous sacrifice in ancient temple economies, the academic citation game—all are forms of "auction" where the most complex (deepest) offering wins.

Veblen's theory of conspicuous consumption captures the demand side of this dynamic: prestige accrues to the display of costly (deep) goods. Bourdieu's concept of symbolic capital captures the supply side: cultural producers bid for status by offering complex cultural products to audiences whose capacity to appreciate them (their cultural capital) constrains the impact of even the most extravagant bid.[1]

[1] Bourdieu, *La Distinction* (1979) and *Les Règles de l'art* (1992). The "field of cultural production" is precisely our auction, with depth as the currency and resonance as the

This chapter formalizes ideatic auctions. We introduce *bid value* (depth), a *maxDepthIdea* selector, and *auction interpretation* (the winning bid composed with the receiver's background). We prove eighteen theorems showing that void always loses, deeper bids win, composing bids is subadditive, and the winner's interpretation is bounded by audience capacity.

14.2 Core Definitions

Definition 14.1 (Bid Value)**.** The *bid value* of an idea a is its depth: $\mathrm{bidValue}(a) = \mathrm{d}(a)$. In the ritual economy, depth equals prestige equals bid amount.

Definition 14.2 (Maximum Depth Idea)**.** Given a fallback idea and a list of bids, maxDepthIdea selects the idea with the greatest depth via a left fold that keeps the deeper of each pair:

$$\mathrm{maxDepthIdea}(\mathrm{best}, []) = \mathrm{best}, \qquad \mathrm{maxDepthIdea}(\mathrm{best}, x :: \mathrm{rest}) = \begin{cases} \mathrm{maxDepthIdea}(x, \mathrm{rest}) \\ \mathrm{maxDepthIdea}(\mathrm{best}, \mathrm{rest} \end{cases}$$

```
def maxDepthIdea : I -> List I -> I
  | best, []        => best
  | best, x :: rest =>
    if depth x >= depth best then maxDepthIdea x rest
    else maxDepthIdea best rest
```

Listing 14.1: Maximum depth idea selector in Lean 4

Definition 14.3 (Auction Interpretation)**.** The *auction interpretation* composes the receiver's background with the winning (deepest) bid:

$$\mathrm{auctionInterpret}(\mathrm{bids}, \mathrm{fallback}, \mathrm{receiver}) = \mathrm{receiver} \circ \mathrm{maxDepthIdea}(\mathrm{fallback}, \mathrm{bids}).$$

The receiver's background mediates the winning bid: the audience does not absorb the winning idea raw but interprets it through their own cultural lens.

14.3 Void as Non-Competitor

Theorem 14.4 (Void Bid Value is Zero (14.1))**.** *The void idea has zero bid value:* $\mathrm{bidValue}(\mathbf{0}) = 0$.

Proof sketch. $\mathrm{d}(\mathbf{0}) = 0$ by the void depth axiom. □

condition of successful reception.

In the potlatch, offering nothing is the ultimate humiliation. Zero depth equals zero cultural capital equals zero bidding power. The formal basis of "losing face": the chief who brings no gifts has bid $\mathbf{0}$ and receives zero prestige. Mauss's *obligation to give* is, at bottom, an obligation to bid non-void.[2]

Theorem 14.5 (Void Loses to Any Bid (14.2))**.** *Given any bid a and void as fallback, the maximum has non-negative depth:* $\mathrm{bidValue}(\mathrm{maxDepthIdea}(\mathbf{0}, [a])) \geq 0$.

Proof sketch. Depth is always non-negative. □

Any cultural offering, no matter how shallow, beats silence. This is the formal foundation of "showing up": participation is always at least as good as absence in competitive ritual. The anthropological wisdom that "the worst thing you can do is not come" is a theorem.

Theorem 14.6 (Deeper Beats Void (14.3))**.** *If* $\mathrm{d}(a) \geq 1$*, then a beats void:* $\mathrm{d}(\mathrm{maxDepthIdea}(\mathbf{0}, [a])) \geq 1$.

Proof sketch. When $\mathrm{d}(a) \geq 1 \geq 0 = \mathrm{d}(\mathbf{0})$, the selector chooses a. □

In competitive gift-giving, the party who brings *something* always outranks the party who brings nothing. Mauss's obligation to give is structurally enforced: non-givers are structurally dominated. Even the most modest gift—a single shell necklace, a handshake, a word of greeting—outbids silence.

14.4 Selection Properties

Theorem 14.7 (Single Bidder Wins (14.4))**.** *With only one bid, the winner is determined by comparison with the fallback:*

$$\mathrm{maxDepthIdea}(\mathit{fallback}, [a]) = \begin{cases} a & \text{if } \mathrm{d}(a) \geq \mathrm{d}(\mathit{fallback}), \\ \mathit{fallback} & \text{otherwise.} \end{cases}$$

Proof sketch. Direct unfolding of the definition with an empty remainder. □

A monopoly on cultural production—single priesthood, state media—means the sole producer always "wins" the auction (assuming their depth exceeds the fallback tradition). Competition is absent; interpretation is determined unilaterally. This is the formal structure of theocracy: one bidder, guaranteed victory.

[2] Mauss, *Essai sur le don* (1925): "To refuse to give, to fail to invite, just as to refuse to accept, is tantamount to declaring war; it is to reject the bond of alliance and commonality."

Theorem 14.8 (Fallback Survives Empty Auction (14.5)). *With no bids, the fallback wins:* $\text{maxDepthIdea}(\textit{fallback}, []) = \textit{fallback}$.

Proof sketch. The base case of the recursion. □

When no one competes, the status quo persists. The "default culture" (tradition, orthodoxy) wins by forfeit when no challenger appears. This is the formal mechanism of cultural conservatism: in the absence of competing bids, the existing cultural framework is preserved intact.

Theorem 14.9 (Bid Value is Non-Negative (14.6)). *All bids have non-negative value:* $\text{bidValue}(a) \geq 0$.

Proof sketch. Depth is a natural number. □

There are no "negative contributions" in ideatic space. Every idea has non-negative complexity. Even the most trivial cultural offering—a monosyllable, a blank canvas—has at least zero depth. The floor of cultural value is zero, never negative. This is a modeling choice with significant anthropological implications: we assume that ideas cannot *destroy* depth, only fail to add it.[3]

14.5 Composition and Bidding

Theorem 14.10 (Composed Bid Subadditivity (14.7)). *The bid value of composing two ideas is at most the sum of their bid values:*

$$\text{bidValue}(a \circ b) \;\leq\; \text{bidValue}(a) + \text{bidValue}(b).$$

Proof sketch. By subadditivity of depth. □

Combining two gifts into one package does not create more prestige than giving them separately. The potlatch chief who presents a composite gift of blankets and copper shields gains at most the sum of their separate values. Subadditivity prevents "prestige inflation" through bundling. This is the formal refutation of the marketing claim that "the whole is greater than the sum of its parts"—in ideatic auctions, the whole is at most equal to the sum.

Theorem 14.11 (Void Composition Preserves Bid, Right (14.8)). $\text{bidValue}(a \circ \mathbf{0}) = \text{bidValue}(a)$.

Proof sketch. $a \circ \mathbf{0} = a$ by the void axiom. □

[3] One could extend the model with signed depth to capture "anti-ideas" or cultural destruction, but the current axiomatization does not permit this. The non-negativity of depth is one of IDT's stronger structural commitments.

Theorem 14.12 (Void Composition Preserves Bid, Left (14.9))**.** $\mathrm{bidValue}(\mathbf{0} \circ a) = \mathrm{bidValue}(a)$.

Proof sketch. $\mathbf{0} \circ a = a$ by the void axiom. □

Adding "nothing" to a gift does not increase its prestige. Padding a potlatch offering with empty gestures is structurally transparent—the audience sees through it. Prefacing a gift with silence (the dramatic pause before the big reveal) adds nothing to the gift's structural depth. Ceremony may have psychological effects, but it is orthogonal to substance in the depth metric.

14.6 Interpretation of Winning Bids

Theorem 14.13 (Winner's Interpretation Depth Bound (14.10))**.** *The depth of the auction interpretation is bounded:*

$$\mathrm{d}(\mathrm{auctionInterpret}(\mathit{bids}, \mathit{fallback}, r)) \;\leq\; \mathrm{d}(r) + \mathrm{d}(\mathrm{maxDepthIdea}(\mathit{fallback}, \mathit{bids})).$$

Proof sketch. By subadditivity applied to $r \circ \mathrm{maxDepthIdea}(\cdots)$. □

The cultural impact of the winning bid is bounded by the receiver's background plus the bid's depth. Even the most extravagant potlatch is limited by the audience's capacity to appreciate it. Bourdieu's insight that cultural capital constrains reception is formalized: a peasant watching a court ballet and a courtier watching the same performance produce interpretations of vastly different depth, not because the ballet differs but because $\mathrm{d}(r)$ differs.[4]

Theorem 14.14 (Empty Auction Interpretation (14.11))**.** *With no bids, the interpretation is the receiver composed with the fallback:* $\mathrm{auctionInterpret}([], \mathit{fallback}, r) = r \circ \mathit{fallback}$.

Proof sketch. $\mathrm{maxDepthIdea}(\mathrm{fallback}, []) = \mathrm{fallback}$. □

When no external cultural force competes, the receiver interprets only through their default framework. Cultural isolation means self-referential interpretation: the audience, receiving no new bids, composes its background with tradition alone.

Theorem 14.15 (Void Receiver Gets Raw Winner (14.12))**.** *If the receiver's background is void, the interpretation is just the winning bid:* $\mathrm{auctionInterpret}(\mathit{bids}, \mathit{fallbac}$ $\mathrm{maxDepthIdea}(\mathit{fallback}, \mathit{bids})$.

Proof sketch. $\mathbf{0} \circ w = w$ by the void axiom. □

[4] Bourdieu's analysis of aesthetic judgment in *La Distinction* shows that reception of cultural goods is stratified by the receiver's accumulated cultural capital—our $\mathrm{d}(r)$.

The "blank slate" receiver—the newborn, the cultural outsider—simply absorbs the winning idea without transformation. Cultural naivety means unmediated reception. This is the anthropological equivalent of "going native": the observer without pre-existing cultural categories receives the dominant cultural signal raw, unfiltered by prior interpretive frameworks.[5]

14.7 Structural Auction Theorems

Theorem 14.16 (Iterated Bid Depth Bound (14.13)). *Repeating a bid n times via composition yields depth at most $n \cdot \mathrm{bidValue}(a)$:*

$$\mathrm{bidValue}(an) \;\leq\; n \cdot \mathrm{bidValue}(a).$$

Proof sketch. By the iterated composition depth bound. □

Repetitive gift-giving—annual potlatch cycles, recurring tribute—has bounded cumulative prestige. You cannot achieve infinite status through mere repetition. The subadditivity of depth imposes diminishing returns: the n-th iteration of a potlatch adds at most $\mathrm{d}(a)$, and the total is bounded by $n \cdot \mathrm{d}(a)$.

Theorem 14.17 (Atomic Bid Bound (14.14)). *Atomic ideas have bid value at most one: if a is atomic, then* $\mathrm{bidValue}(a) \leq 1$.

Proof sketch. By the depth bound on atomic ideas. □

The simplest, most elemental cultural offerings—a single word, a basic gesture, a minimal ritual act—have minimal prestige value. Prestige comes from complexity: the elaborate ceremony outranks the simple nod. This formalizes the anthropological observation that prestige goods are invariably *complex* goods: the jade celt, the knotted quipu, the illuminated manuscript—all derive their prestige from depth, not from mere existence.

Theorem 14.18 (Resonant Bids Produce Resonant Interpretations (14.15)). *If two bids resonate, $b_1 \sim b_2$, any common receiver produces resonant interpretations:*

$$b_1 \sim b_2 \implies (r \circ b_1) \sim (r \circ b_2).$$

Proof sketch. By left-composition preservation of resonance. □

Competing ritual systems that resonate—e.g., Catholic and Orthodox Christianity—produce resonant experiences in the same audience. The laity perceives structural similarity even across competing institutions. This is

[5] The trope of the ethnographer "going native" (from Malinowski onward) captures precisely this: the field worker who sheds their own cultural background ($r \to \mathbf{0}$) receives the host culture's signals without the mediation of home categories.

the formal basis of religious tolerance: if the competing bids resonate, the audience's interpretations will also resonate, reducing the perceived distance between rival systems.

Theorem 14.19 (Resonant Receivers Agree on Winner (14.16)). *If two receivers resonate, $r_1 \sim r_2$, and evaluate the same bid, their interpretations resonate:*

$$r_1 \sim r_2 \implies (r_1 \circ b) \sim (r_2 \circ b).$$

Proof sketch. By right-composition preservation of resonance. □

Members of the same culture watching the same potlatch produce resonant evaluations. Cultural consensus on prestige is a structural consequence of shared background. This is why prestige hierarchies are culture-specific: a Kwakiutl potlatch and a Parisian art auction assign prestige to different bids, but within each culture, members with resonant backgrounds agree on what is prestigious.

Theorem 14.20 (Bid Associativity (14.17)). *Composing bids is associative: $(a \circ b) \circ c = a \circ (b \circ c)$.*

Proof sketch. By associativity of the monoid. □

In a multi-stage potlatch (gifts given over successive days), the grouping of days does not affect the total composition. The "grand total" of a festival is invariant under scheduling—structure trumps sequence. Whether the chief gives blankets on Monday and copper shields on Tuesday, or reverses the order, the total compositional depth is the same (though the individual days' compositions may differ).

Theorem 14.21 (Depth-Zero Bid Composure (14.18)). *If both bidders have zero depth, their composition has zero depth:*

$$\mathrm{d}(a) = 0 \ \wedge \ \mathrm{d}(b) = 0 \implies \mathrm{d}(a \circ b) = 0.$$

Proof sketch. By subadditivity, $\mathrm{d}(a \circ b) \leq \mathrm{d}(a) + \mathrm{d}(b) = 0$, and depth is non-negative, so $\mathrm{d}(a \circ b) = 0$. □

Two culturally empty offerings combined produce nothing. "Nothing plus nothing is nothing"—the formal futility of combining shallow gestures. Two empty speeches, two blank canvases, two silent prayers: their composition is as void as each alone. The potlatch chief who bundles two worthless gifts has still bid zero.

```
theorem zero_depth_composition {a b : I}
    (ha : depth a = 0) (hb : depth b = 0) :
    depth (compose a b) = 0 := by
  have h := depth_compose_le a b
  omega
```

Listing 14.2: Depth-zero composition in Lean 4

14.8 Conclusion

The ideatic auction formalizes a pervasive pattern in human cultural life: competition for interpretive priority through depth. The structural results of this chapter—void always loses (Theorem 14.4), composition is subadditive (Theorem 14.10), reception is bounded by audience capacity (Theorem 14.13), and zero-depth bids compose to zero (Theorem 14.21)—impose hard constraints on the ritual economy.

These constraints connect to three major anthropological traditions. Mauss's gift theory is formalized through the obligation to bid non-void (Theorem 14.4); Veblen's conspicuous consumption is captured by the depth-as-prestige identification (Definition 14.1); and Bourdieu's symbolic capital is reflected in the receiver-bounded interpretation (Theorem 14.13).

In the final chapter of this sequence, we turn from competitive bidding to *curation*: the receiver's active management of their own repertoire, connecting to museum theory, education policy, and the political economy of knowledge.

Chapter 15

Repertoire Curation and Museum Theory

> The cultural arbitrary which a system of education transmits, reproduces, and inculcates is *always* the cultural arbitrary of the dominant classes.
>
> Pierre Bourdieu and Jean-Claude Passeron, *La Reproduction* (1970)

15.1 Introduction

The preceding chapters have treated the receiver as a fixed entity—a repertoire passively awaiting signals. But receivers are not passive. Every mind *curates* its repertoire: selecting which ideas to retain, which to discard, and which new ideas to admit. The museum curator choosing which artifacts to display, the teacher designing a syllabus, the censor deciding which books to ban, the autodidact selecting their next reading—all are engaged in the same structural operation: filtering and expanding a repertoire according to some criterion.

Bourdieu's concept of *cultural capital*—the accumulated, curated stock of ideas that determines one's interpretive capacity—is the sociological correlate of the repertoire. Cultural capital is not merely *possessed*; it is actively *managed*. The bourgeois family curates its children's education; the academy curates its canon; the state curates the national curriculum. Each act of curation is simultaneously an act of inclusion and exclusion, of memory and forgetting.[1]

[1] On the politics of canon formation, see John Guillory, *Cultural Capital: The Problem*

This chapter formalizes repertoire curation and expansion. We introduce curate—filtering a repertoire by a predicate—and expand—adding a new idea to the front. We then prove eighteen theorems that connect these operations to museum theory, education policy, censorship dynamics, and the political economy of knowledge.

15.2 Core Definitions

Definition 15.1 (Curation)**.** To *curate* a repertoire $\mathbf{R}$ is to filter it by a Boolean predicate p:

$$\mathrm{curate}(\mathbf{R}, p) \;=\; [r \in \mathbf{R} \mid p(r) = \texttt{true}].$$

This models selective retention: the museum curator choosing artifacts, the teacher choosing concepts, the censor choosing permitted ideas.

Definition 15.2 (Expansion)**.** To *expand* a repertoire $\mathbf{R}$ with a new idea a is to prepend it:

$$\mathrm{expand}(\mathbf{R}, a) \;=\; a :: \mathbf{R}.$$

This models learning: acquiring a new concept, encountering a new artifact, absorbing a new cultural input.

```
def curate (rep : List I) (p : I -> Bool) : List I :=
  rep.filter p

def expand (rep : List I) (new_idea : I) : List I :=
  new_idea :: rep
```

Listing 15.1: Curation and expansion in Lean 4

Definition 15.3 (Interpretation)**.** When a receiver with repertoire $\mathbf{R}$ encounters signal s, they produce interpretations:

$$\mathrm{interpret}(\mathbf{R}, s) \;=\; [r_1 \circ s,\; r_2 \circ s,\; \ldots,\; r_n \circ s].$$

The interplay of curation, expansion, and interpretation is the central drama of this chapter. Curation *reduces* the repertoire (and hence the number and depth of possible interpretations); expansion *increases* it. The structural question is: how do these operations compose, and what invariants do they preserve?

of Literary Canon Formation (1993). Guillory argues that the literary canon is a form of cultural capital whose curation serves class reproduction—a thesis that our formal framework substantiates.

15.3 Curation Fundamentals

Theorem 15.4 (Curation Preserves Membership (15.1)). *Every element of the curated repertoire was in the original:*

$$\forall\, x \in \mathrm{curate}(\mathbf{R}, p), \quad x \in \mathbf{R}.$$

Proof sketch. By the property of list filtering: every element of $\mathrm{filter}(p, \mathbf{R})$ belongs to $\mathbf{R}$. □

A museum can only display what it owns. Curation is selection, not creation—the formal basis of the archivist's constraint. You cannot curate what you do not have. This seemingly obvious fact has deep implications for cultural policy: any canon-formation exercise is bounded by the existing cultural stock. A national curriculum can only draw from the ideas already present in the culture; it cannot conjure new ones through the act of selection.[2]

Theorem 15.5 (Curation Reduces Length (15.2)). *The curated repertoire is never longer than the original:*

$$|\,\mathrm{curate}(\mathbf{R}, p)| \;\leq\; |\mathbf{R}|.$$

Proof sketch. Filtering can only remove elements, never add them. □

Curation only removes—it never inflates. A censored library is always smaller than the uncensored original. This formalizes the "information loss" inherent in any act of selection. Every curatorial decision is, structurally, a decision to *forget*.

Theorem 15.6 (Vacuous Curation is Empty (15.3)). *Filtering by a predicate that rejects everything yields the empty list:* $\mathrm{curate}(\mathbf{R}, \lambda x.\, \texttt{false}) = []$.

Proof sketch. No element passes the filter. □

Total censorship—rejecting every idea—produces cultural void. This is the formal limit of suppression: complete censorship equals complete cultural erasure. The totalitarian ideal of the *tabula rasa*—erasing all prior culture to build anew—is structurally equivalent to reducing the repertoire to the empty list. Pol Pot's "Year Zero," Mao's Cultural Revolution, the *damnatio memoriae* of the Roman Senate: all are attempts at vacuous curation.[3]

[2] This distinguishes curation from *creation*. In our model, creation is the introduction of a genuinely new element $a \in \mathcal{I}$ not already in the repertoire; curation is the selection among existing elements. The two operations have fundamentally different structural properties.

[3] The historical record suggests that vacuous curation is never perfectly achieved: underground literature, oral tradition, and material culture survive even the most thorough purges. The theorem describes the *structural limit*, not the empirical reality.

Theorem 15.7 (Universal Curation is Identity (15.4))**.** *Filtering by a predicate that accepts everything yields the original:* $\text{curate}(\mathbf{R}, \lambda x.\, \texttt{true}) = \mathbf{R}$.

Proof sketch. Every element passes the filter. □

A museum that displays everything is just a warehouse. "Curating" with no selection criterion is structurally equivalent to no curation at all. This is the formal basis of the distinction between the *archive* (everything preserved) and the *exhibition* (a curated selection). The uncurated repertoire is the Borgesian library: complete, but unnavigable.[4]

15.4 Expansion Fundamentals

Theorem 15.8 (Expansion Increases Length (15.5))**.** *Adding a new idea increases the repertoire by exactly one:*

$$|\,\text{expand}(\mathbf{R}, a)| = |\mathbf{R}| + 1.$$

Proof sketch. Prepending one element increases list length by one. □

Each learning event adds exactly one new schema. Education is *incremental*—you learn one thing at a time. This constrains theories of "epiphany" or "gestalt shift" to be single-idea additions in the formal model. The Humboldtian ideal of *Bildung*—the gradual, systematic expansion of the mind—is structurally captured by iterated application of expand.

Theorem 15.9 (New Idea is in Expanded Repertoire (15.6))**.** *The newly added idea belongs to the expanded repertoire:* $a \in \text{expand}(\mathbf{R}, a)$.

Proof sketch. The new idea is the head of the list. □

What you learn, you possess. The student who has been taught a concept *has* that concept in their repertoire. This is the structural basis of education as repertoire expansion: successful pedagogy is precisely the operation $\text{expand}(\mathbf{R}_{\text{student}}, a_{\text{lesson}})$.

Theorem 15.10 (Old Ideas Survive Expansion (15.7))**.** *All ideas in the original repertoire remain in the expanded version:*

$$\forall\, x \in \mathbf{R}, \quad x \in \text{expand}(\mathbf{R}, a).$$

Proof sketch. The original list is the tail of the expanded list. □

[4] Jorge Luis Borges, "The Library of Babel" (1941). Borges's infinite library contains every possible book but is useless precisely because it lacks curation.

Learning does not erase prior knowledge. The student who learns calculus still remembers arithmetic. Expansion is *monotone*—repertoires only grow. This is an important modeling assumption: in our framework, forgetting requires an explicit curation step (filtering out the unwanted idea), not mere neglect. The distinction between passive forgetting and active censorship is, in IDT, the distinction between non-operation and curation.[5]

15.5 Curation and Interpretation

Theorem 15.11 (Curated Interpretation is Sublist (15.8))**.** *Interpreting a curated repertoire gives at most as many interpretations as the full repertoire:*

$$|\,\mathrm{interpret}(\mathrm{curate}(\mathbf{R}, p), s)| \;\leq\; |\,\mathrm{interpret}(\mathbf{R}, s)|.$$

Proof sketch. $|\,\mathrm{interpret}(\cdot, s)| = |\cdot|$ (interpretation preserves length), and $|\,\mathrm{curate}(\mathbf{R}, p)| \leq |\mathbf{R}|$. □

A censored mind produces fewer interpretations. The number of possible readings of a text is bounded by the reader's (curated) repertoire size. A narrower education yields fewer perspectives—the formal basis of the humanist argument for breadth in the curriculum. The liberally educated mind, with its expansive repertoire, generates more interpretations than the narrowly trained specialist.

Theorem 15.12 (Empty Curation Yields No Interpretation (15.9))**.** *If curation removes everything, there are no interpretations:*

$$\mathrm{interpret}(\mathrm{curate}(\mathbf{R}, \lambda x.\, \mathtt{false}), s) = [].$$

Proof sketch. By Theorem 15.6, the curated repertoire is empty, and mapping over the empty list yields the empty list. □

The culturally emptied mind cannot interpret anything. Total censorship followed by signal reception equals zero interpretive output. This is the "blank slate after brainwashing" theorem: the totalitarian regime that succeeds in erasing all prior culture leaves its subjects incapable of interpreting *any* signal, including its own propaganda. The self- defeating nature of complete censorship is a structural theorem.

[5] In cognitive science, the debate between "decay" and "interference" theories of forgetting maps onto our distinction between curation (active removal) and repertoire stagnation (non-expansion).

15.6 Expansion and Interpretation

Theorem 15.13 (Expanded Interpretation Adds One (15.10)). *Interpreting an expanded repertoire produces one more interpretation than the original:*

$$|\,\mathrm{interpret}(\mathrm{expand}(\mathbf{R}, a), s)| \;=\; |\,\mathrm{interpret}(\mathbf{R}, s)| + 1.$$

Proof sketch. Expansion adds one element; interpretation preserves length. □

Each new idea in the repertoire adds exactly one new interpretive possibility. Education's return is precisely one new perspective per concept learned—the fundamental unit of *Bildung*. The marginal return on education, measured in interpretive possibilities, is constant: one concept in, one new interpretation out.

Theorem 15.14 (Expansion Interpretation Structure (15.11)). *Interpreting an expanded repertoire prepends the new interpretation:*

$$\mathrm{interpret}(\mathrm{expand}(\mathbf{R}, a), s) \;=\; (a \circ s) :: \mathrm{interpret}(\mathbf{R}, s).$$

Proof sketch. By definition: $\mathrm{expand}(\mathbf{R}, a) = a :: \mathbf{R}$, and mapping over a cons is $f(a) :: \mathrm{map}(f, \mathbf{R})$. □

The new concept's interpretation appears first, followed by all prior interpretations unchanged. Learning adds a new lens without distorting existing ones—the "additive" model of education. This is structurally optimistic: new knowledge does not corrupt old knowledge. The Baconian vision of knowledge as cumulative finds formal support here.

Theorem 15.15 (Void Expansion Doesn't Change Interpretations (15.12)). *Adding void to the repertoire adds the raw signal s as a new interpretation:*

$$\mathrm{interpret}(\mathrm{expand}(\mathbf{R}, \mathbf{0}), s) \;=\; s :: \mathrm{interpret}(\mathbf{R}, s).$$

Proof sketch. $\mathbf{0} \circ s = s$ by the void axiom, so the new interpretation is s itself. □

Adding "empty receptivity"—the Keatsian *negative capability*—to one's repertoire adds the raw signal as a new interpretation. The educationally valuable "empty slot" lets you see the signal unmediated. This is the formal justification for intellectual humility: maintaining a void element in one's repertoire ensures that one always has access to the unfiltered signal, free from the bias of pre-existing categories.[6]

[6] Keats's concept of "negative capability"—"being in uncertainties, mysteries, doubts, without any irritable reaching after fact and reason"—is precisely the act of maintaining $\mathbf{0}$ in the repertoire: a position of receptivity that produces the raw signal as interpretation.

```
theorem void_expansion_raw_signal (rep : List I) (s : I) :
    interpret (expand rep void) s = s :: interpret rep s := by
  simp [expanded_interp_structure, void_left]
```

Listing 15.2: Void expansion raw signal in Lean 4

15.7 DepthSum Bounds

Theorem 15.16 (Curated DepthSum Bound (15.13))**.** *The depth sum of a curated repertoire is at most the depth sum of the original:*

$$\text{depthSum}(\text{curate}(\mathbf{R}, p)) \;\leq\; \text{depthSum}(\mathbf{R}).$$

Proof sketch. By induction on $\mathbf{R}$. At each step, if the element passes the filter, both sides include $\mathrm{d}(r)$; if it fails, only the right side includes it, maintaining the inequality. □

Curation can only *reduce* total cultural depth. A museum that hides artifacts loses depth; a censored library is shallower than the uncensored one. Information destruction is irreversible in the depth metric. This theorem has stark implications for cultural policy: every act of censorship is a permanent reduction in the society's total interpretive capacity. The censored ideas' depth is *lost*, not merely hidden.

Theorem 15.17 (Expanded DepthSum (15.14))**.** *Expanding the repertoire adds exactly the new idea's depth:*

$$\text{depthSum}(\text{expand}(\mathbf{R}, a)) \;=\; \mathrm{d}(a) + \text{depthSum}(\mathbf{R}).$$

Proof sketch. $\text{expand}(\mathbf{R}, a) = a :: \mathbf{R}$, and $\text{depthSum}(a :: \mathbf{R}) = \mathrm{d}(a) + \text{depthSum}(\mathbf{R})$. □

Each new idea adds exactly its depth to the total cultural capital. Education is a *linear* accumulation of depth—Bourdieu's cultural capital grows additively with each new acquisition. The student who learns a concept of depth d increases their total cultural capital by exactly d. This linearity is both empowering (every lesson counts) and sobering (there are no shortcuts—depth must be earned one idea at a time).

Theorem 15.18 (Void Expansion DepthSum (15.15))**.** *Adding void to the repertoire does not change depthSum:*

$$\text{depthSum}(\text{expand}(\mathbf{R}, \mathbf{0})) \;=\; \text{depthSum}(\mathbf{R}).$$

Proof sketch. $\mathrm{d}(\mathbf{0}) = 0$, so $0 + \mathrm{depthSum}(\mathbf{R}) = \mathrm{depthSum}(\mathbf{R})$. □

Adding "nothing" to one's cultural stock adds zero depth. The void is costless to maintain in the repertoire—the formal basis of "keeping an open mind" being free in the depth metric. Intellectual humility (carrying $\mathbf{0}$ in one's repertoire) costs nothing and gains the benefit of raw signal access (Theorem 15.15).

15.8 Structural Curation Theorems

Theorem 15.19 (Double Curation is Single Curation (15.16))**.** *Curating a curated repertoire by the same predicate is idempotent:*

$$\mathrm{curate}(\mathrm{curate}(\mathbf{R}, p), p) \;=\; \mathrm{curate}(\mathbf{R}, p).$$

Proof sketch. $\mathrm{filter}(p, \mathrm{filter}(p, \mathbf{R})) = \mathrm{filter}(p \wedge p, \mathbf{R}) = \mathrm{filter}(p, \mathbf{R})$ since $p \wedge p = p$. □

Censoring an already-censored library by the same criterion has no further effect. The "diminishing returns" of redundant censorship—once you have removed all "heretical" books, removing them again changes nothing. This idempotency result has practical implications for bureaucratic censorship: the second censor applying the same criteria as the first is structurally redundant, a waste of institutional resources. Totalitarian regimes that layer censorship committees atop each other (with identical mandates) achieve nothing beyond the first pass.[7]

Theorem 15.20 (Expand Then Curate Accepting (15.17))**.** *If the new idea passes the predicate, expanding then curating keeps it:*

$$p(a) = \texttt{true} \implies \mathrm{curate}(\mathrm{expand}(\mathbf{R}, a), p) = a :: \mathrm{curate}(\mathbf{R}, p).$$

Proof sketch. Filtering $a :: \mathbf{R}$ with p: since $p(a) = \texttt{true}$, a is kept, and the tail is filtered as usual. □

If a newly acquired idea passes the censorship test, it survives curation. "Approved learning" is permanent—the accepted idea stays in the repertoire through subsequent curation rounds. This is the formal basis of canonical inclusion: once an idea is admitted to the canon (passes the predicate), it persists through all future curation by the same criterion. Dante, admitted to the Western canon, survives every subsequent round of canon revision that uses the same evaluative framework.

[7] Historical examples include the multiple overlapping censorship bodies of the Soviet Union (*Glavlit*, *Goskomizdat*, Party committees), each applying essentially the same ideological filter. Our theorem shows this redundancy is structural, not merely bureaucratic.

Theorem 15.21 (Expand Then Curate Rejecting (15.18)). *If the new idea fails the predicate, expanding then curating removes it:*

$$p(a) = \texttt{false} \implies \text{curate}(\text{expand}(\mathbf{R}, a), p) = \text{curate}(\mathbf{R}, p).$$

Proof sketch. Filtering $a :: \mathbf{R}$ with p: since $p(a) = \texttt{false}$, a is discarded, and the tail is filtered as before. □

Ideas that fail the censorship test are immediately purged. The "forbidden knowledge" theorem: learning something that the regime rejects is structurally equivalent to never having learned it, after curation. This formalizes the futility of teaching banned concepts in totalitarian educational systems—the student may *encounter* the forbidden idea (via expand), but the subsequent curation step erases it from the repertoire as if the encounter never happened.

The theorem is disturbing in its implications. If p represents the ideological filter of a totalitarian state, then $\text{curate}(\text{expand}(\mathbf{R}, a_{\text{forbidden}}), p) = \text{curate}(\mathbf{R}, p)$: the forbidden idea leaves no trace. The dissident's effort to teach subversive ideas is, under perfect censorship, structurally null. Of course, perfect censorship is an idealization—but the theorem shows what perfect censorship would *mean*: structural erasure of dissent.

```
theorem expand_curate_removes {new_idea : I} {p : I ->
    Bool}
    (hp : p new_idea = false) (rep : List I) :
    curate (expand rep new_idea) p = curate rep p :=
        by
  simp [expand, curate, List.filter, hp]
```

Listing 15.3: Expand-then-curate in Lean 4

15.9 Conclusion

Repertoire curation, formalized in IDT, reveals the structural anatomy of cultural management. Curation is subtractive (Theorem 15.5), idempotent (Theorem 15.19), and depth-reducing (Theorem 15.16). Expansion is additive (Theorem 15.8), monotone (Theorem 15.10), and depth-increasing (Theorem 15.17). The interaction of the two—expand then curate—is determined entirely by whether the new idea passes the filter (Theorems 15.20 and 15.21).

These results connect to four domains of cultural practice. In *museum theory*, curation is the fundamental operation of exhibition design—and our theorems show that it is structurally subtractive. In *education policy*, expansion is the mechanism of *Bildung*, and each lesson adds exactly one interpretive possibility. In *censorship dynamics*, vacuous curation produces

cultural void (Theorem 15.6), and the expand-then-curate-rejecting result (Theorem 15.21) formalizes the structural erasure of dissent. In *Bourdieu's sociology*, cultural capital grows linearly with expansion (Theorem 15.17) and can only shrink through curation (Theorem 15.16).

The five chapters of this sequence—marketplace competition (Chapter 11), stable matching (Chapter 12), coalition value (Chapter 13), ideatic auction (Chapter 14), and repertoire curation—together provide a comprehensive game-theoretic framework for cultural dynamics within Ideatic Diffusion Theory. Each chapter isolates a different mode of cultural interaction—competition, pairing, cooperation, bidding, management—and derives structural constraints that hold across all instantiations of the IDT axioms. The resulting picture is one of *bounded cultural possibility*: every mode of interaction is constrained by subadditivity, identity, and resonance, the three pillars on which the theory rests.

Part IV

Cooperation, Networks, and Synthesis

Chapter 16

Cooperative Composition and Gift Exchange

"The gift not yet repaid debases the man who accepted it."
— Marcel Mauss, *The Gift* (1925)

16.1 Introduction

Marcel Mauss's celebrated *Essai sur le don* (1925) demonstrated that gift exchange is never merely economic: to give is to create obligation, to receive is to assume debt, and to reciprocate is to weave a social bond. In societies from the Trobriand Islands to the Pacific Northwest, the circulation of gifts constitutes a total social fact—simultaneously economic, juridical, moral, and aesthetic.[1]

In Ideatic Diffusion Theory, cooperation is *composition*: when two agents contribute their ideas to a joint project, the result is $\circ(a, b)$. The order matters—gifts are not symmetric!—and the void $\mathbf{0}$ contributes nothing. This chapter formalizes cooperative game theory within ideatic space, establishing twenty theorems that characterize joint composition, multi-party projects, depth bounds on cooperative output, and resonance preservation under collaboration.

The analogy between $\circ$ and gift-giving is precise. Agent a "gives first," setting the frame of the exchange; agent b reciprocates within that frame. The result $\circ(a, b)$ is a compound idea whose depth is bounded by the sum of the contributors' depths—one cannot create profundity from banality—and whose resonance properties are inherited from the inputs. When we extend

[1] Mauss distinguished three obligations of the gift: to give, to receive, and to reciprocate. Each is indispensable; failure at any stage ruptures the social fabric.

to multi-party projects via iterated composition, we obtain a formal model of Malinowski's *Kula ring*, of potlatch sequences, and of any process in which meaning accrues through sequential cooperative acts.[2]

We proceed as follows. Section 16.2 introduces the core definitions—`jointCompose`, `symmetricJoint`, and `JointProject`—together with their Lean formalizations. Section 16.3 establishes the role of void as the non-contributor. Section 16.4 proves associativity of multi-party cooperation. Section 16.5 explores resonance preservation. Section 16.6 derives depth bounds on cooperative output. Section 16.7 extends the analysis to multi-party and iterated cooperation. Section 16.8 concludes.

16.2 Core Definitions

We begin with the fundamental structures of cooperative composition.

Definition 16.1 (Joint Composition)**.** Let $a, b \in \mathcal{I}$. The *joint composition* of a and b is

$$\texttt{jointCompose}(a, b) \;:=\; \circ(a, b).$$

In Mauss's terms, agent a gives first and agent b reciprocates. The result is a shared compound idea.

```
def jointCompose (a b : I) : I := IdeaticSpace.compose
    a b
```

Listing 16.1: Joint composition in Lean

Joint composition is simply composition itself, but the naming emphasizes the cooperative framing: two agents are contributing sequentially to a shared outcome. The asymmetry is essential. In Mauss's analysis, the initiator of an exchange determines its moral character; the recipient responds within a framework already established.

Definition 16.2 (Symmetric Joint)**.** Two ideas a, b are *symmetrically joint* when the order of composition does not matter:

$$\texttt{symmetricJoint}(a, b) \;:\Longleftrightarrow\; \circ(a, b) = \circ(b, a).$$

```
def symmetricJoint (a b : I) : Prop :=
  IdeaticSpace.compose a b = IdeaticSpace.compose b a
```

Listing 16.2: Symmetric joint in Lean

[2]Bronislaw Malinowski's *Argonauts of the Western Pacific* (1922) documented how shell valuables circulate through a ring of trading partners, each exchange adding a layer of social meaning.

Symmetric joints are rare. In most real exchanges—ritual, economic, discursive—who gives first shapes the meaning of the entire exchange. The symmetric case corresponds to perfectly reciprocal partnerships where neither party holds agenda-setting power.

Definition 16.3 (Joint Project)**.** A *joint project* consists of a list of participants, each contributing an idea. The project's *output* is the left-fold of all contributions via composition, starting from $\mathbf{0}$:

$$\texttt{output}(p) \;:=\; \mathrm{foldl}(\circ, \mathbf{0}, p.\mathrm{participants}).$$

The *total depth* of a joint project is $\sum_{a \in p} \mathrm{d}(a)$.

```
structure JointProject (I : Type*) [IdeaticSpace I] where
  participants : List I

def JointProject.output (p : JointProject I) : I :=
  p.participants.foldl IdeaticSpace.compose IdeaticSpace.void

def JointProject.totalDepth (p : JointProject I) : N :=
  depthSum p.participants
```

Listing 16.3: Joint project structure in Lean

The left-fold semantics mean that the first contributor sets the initial frame (after void), and each subsequent contributor composes onto the accumulating result. This models the sequential accretion of meaning in collaborative enterprises—from barn-raisings to co-authored papers.

16.3 The Void as Non-Contributor

Void is the identity element of composition, and in the cooperative context it represents non-participation: silence before or after a gift.

Theorem 16.4 (Void Contributes Nothing—Left (16.1))**.** *For all $b \in \mathcal{I}$,*

$$\mathit{jointCompose}(\mathbf{0}, b) = b.$$

Proof sketch. Direct from the left-identity law of $\circ$: $\circ(\mathbf{0}, b) = b$. □

Commentary. Silence before a gift changes nothing about the gift. The Maussian interpretation is that non-participation at the outset imposes no frame on the exchange; the recipient's contribution stands alone.

Theorem 16.5 (Void Contributes Nothing—Right (16.2)). *For all $a \in \mathcal{I}$,*

$$\mathtt{jointCompose}(a, \mathbf{0}) = a.$$

Proof sketch. From the right-identity law: $\circ(a, \mathbf{0}) = a$. □

Commentary. A gift followed by silence is just the gift. Reciprocation with void—an empty counter-gift—leaves the initiator's contribution unaltered. This formalizes the asymmetry that Mauss identified: the absence of reciprocation does not annihilate the gift, but it leaves the social bond incomplete.

Theorem 16.6 (Void Symmetry (16.7)). *For all $a \in \mathcal{I}$, $\mathtt{symmetricJoint}(\mathbf{0}, a)$.*

Proof sketch. Both $\circ(\mathbf{0}, a)$ and $\circ(a, \mathbf{0})$ equal a, so the equation holds trivially. □

Commentary. Void always commutes—silence is the universal symmetric partner. Non-participation is the one "gift" whose order never matters. This is both mathematically trivial and anthropologically significant: the absence of contribution creates no ordering conflict.

Theorem 16.7 (Void Symmetry—Self (16.8)). $\mathtt{symmetricJoint}(\mathbf{0}, \mathbf{0})$.

Proof sketch. $\circ(\mathbf{0}, \mathbf{0}) = \mathbf{0} = \circ(\mathbf{0}, \mathbf{0})$. □

Commentary. As Wittgenstein remarked, "Whereof one cannot speak, thereof one must be silent"—and the order of two silences is immaterial. This theorem, while trivial, anchors the degenerate case of the theory.

16.4 Associativity and Multi-Party Cooperation

The associativity of composition ensures that multi-party cooperation is well-defined regardless of how sub-coalitions are bracketed.

Theorem 16.8 (Associativity of Multi-Party Cooperation (16.3)). *For all $a, b, c \in \mathcal{I}$,*

$$\mathtt{jointCompose}(\mathtt{jointCompose}(a, b), c) \;=\; \mathtt{jointCompose}(a, \mathtt{jointCompose}(b, c)).$$

Proof sketch. Unfolds to $\circ(\circ(a, b), c) = \circ(a, \circ(b, c))$, which is the associativity axiom of $\circ$. □

Commentary. In a ritual with three participants, it doesn't matter whether A and B cooperate first and then engage C, or A engages a pre-formed B–C coalition. The overall composite meaning is the same. This is a powerful

structural result: it means that multi-party cooperation is intrinsically well-defined, independent of the contingent order in which sub-coalitions form.[3]

Theorem 16.9 (Four-Party Cooperation Associativity (16.18)). *For all $a, b, c, d \in \mathcal{I}$,*

$$\mathit{jointCompose}(\mathit{jointCompose}(\mathit{jointCompose}(a, b), c), d) \;=\; \mathit{jointCompose}(a, \mathit{jointComp}$$

Proof sketch. Iterated application of the associativity axiom of $\circ$. □

Commentary. In organizational theory, four-person teams can be subdivided many ways: $(AB)(CD)$, $A(B(CD))$, $((AB)C)D$, and so on. Associativity guarantees that all bracketings produce the same final product. The insight extends to n-party cooperation by induction, establishing that the output of a `JointProject` is invariant under any re-bracketing of its participant list.

Theorem 16.10 (Empty Project Produces Void (16.9)).

$$\mathit{output}(\langle[\ ]\rangle) = \mathbf{0}.$$

Proof sketch. The fold over an empty list returns the initial accumulator $\mathbf{0}$. □

Commentary. An unfunded, unstaffed project yields silence. This is the organizational analogue of the void identity: without participants, no meaning is produced.

Theorem 16.11 (Singleton Project Preserves Contribution (16.10)). *For all $a \in \mathcal{I}$,*

$$\mathit{output}(\langle[a]\rangle) = a.$$

Proof sketch. The fold computes $\circ(\mathbf{0}, a) = a$ by left-identity. □

Commentary. A solo artisan's work is their own idea, unmediated by any collaborator. The project structure does not distort a lone contributor's output.

16.5 Resonance Under Cooperation

Resonance—the fundamental relation of ideatic compatibility—behaves well under cooperative composition. This section establishes that cooperation preserves, propagates, and in some cases strengthens resonance.

[3] In cooperative game theory, associativity of the characteristic function is often assumed but rarely proved from first principles. Here it follows from the algebraic structure of ideatic space.

Theorem 16.12 (Resonant Partners Produce Resonant Outcomes (16.4)). *If resonates*(a_1, a_2) *and resonates*(b_1, b_2)*, then*

$$\mathit{resonates}\big(\texttt{jointCompose}(a_1, b_1),\ \texttt{jointCompose}(a_2, b_2)\big).$$

Proof sketch. Follows from the `resonance_compose` lemma, which establishes that resonance is preserved under componentwise composition. □

Commentary. When gift-givers are spiritually aligned—in Mauss's language, when the *hau* (spirit of the gift) circulates freely—their combined gifts carry a resonance that mismatched givers cannot achieve. This theorem formalizes the intuition that cultural consonance between cooperating parties produces consonant outcomes.

Theorem 16.13 (Reciprocal Exchange Resonance (16.15)). *If resonates*(a, b)*, then*

$$\mathit{resonates}\big(\texttt{jointCompose}(a, b),\ \texttt{jointCompose}(b, a)\big).$$

Proof sketch. Since resonates(a, b) implies resonates(b, a) by symmetry, applying `resonance_compose` to the pairs (a, b) and (b, a) yields the result. □

Commentary. The counter-gift resonates with the gift when giver and receiver are in spiritual accord. This is Mauss's central insight formalized: reciprocal exchange between resonant partners creates a *cycle* of resonant meaning, regardless of which direction the gift flows.

Theorem 16.14 (Self-Resonance of Joint Output (16.11)). *For all* $a, b \in \mathcal{I}$*,*

$$\mathit{resonates}\big(\texttt{jointCompose}(a, b),\ \texttt{jointCompose}(a, b)\big).$$

Proof sketch. Self-resonance is a general property: every idea resonates with itself. □

Commentary. Any completed exchange, however imperfect, achieves internal coherence. This is the minimal condition for a meaningful cooperative outcome.

Theorem 16.15 (Right-Resonance Preservation (16.12)). *If resonates*(a, b)*, then for all* c*, resonates*$\big(\texttt{jointCompose}(a, c),\ \texttt{jointCompose}(b, c)\big)$.

Proof sketch. Follows from `resonance_compose_right`: resonant agents cooperating with the same partner produce resonant results. □

Commentary. Durkheim's insight that agents sharing collective consciousness produce similar ritual outcomes is formalized here: if a and b are culturally resonant, their joint works with any fixed third party c will resonate.

Theorem 16.16 (Left-Resonance Preservation (16.13)). *If resonates*(a, b)*, then for all* c*, resonates*$\big(\mathit{jointCompose}(c, a),\ \mathit{jointCompose}(c, b)\big)$.

Proof sketch. From `resonance_compose_left`: the same frame applied to resonant inputs produces resonant outputs. □

Commentary. Bourdieu's concept of habitus finds formal expression here: the same habitus (frame c) applied to resonant inputs produces resonant interpretations. The frame preserves the resonance of its inputs.

16.6 Depth Bounds on Cooperative Output

Depth measures interpretive complexity. This section establishes that cooperative composition respects subadditivity: the complexity of a joint venture cannot exceed the total intellectual capital invested.

Theorem 16.17 (Depth Bound on Joint Output (16.5)). *For all* $a, b \in \mathcal{I}$*,*

$$\mathrm{d}\big(\mathit{jointCompose}(a, b)\big) \ \leq\ \mathrm{d}(a) + \mathrm{d}(b).$$

Proof sketch. Direct from the subadditivity of depth under composition: $\mathrm{d}(\circ(a, b)) \leq \mathrm{d}(a) + \mathrm{d}(b)$. □

Commentary. The complexity of a joint venture cannot exceed the total intellectual capital invested by both partners. This is a fundamental resource constraint on cooperation: two partners of bounded depth can only produce output of bounded depth. The analogy to economic production functions is precise—the output is bounded by the sum of inputs.

Theorem 16.18 (Joint With Self Equals ComposeN 2 (16.6)). *For all* $a \in \mathcal{I}$*,*

$$\mathit{jointCompose}(a, a) = \mathit{composeN}(a, 2).$$

Proof sketch. By definition, $\mathtt{composeN}(a, 2) = \circ(a, \circ(a, \mathbf{0})) = \circ(a, a)$. □

Commentary. Self-reinforcement through repeated practice is formally identical to composing an idea with itself. In ritual theory, the repetition of a ceremony deepens its effect; here, the depth of the repeated composition is at most $2 \cdot \mathrm{d}(a)$ by subadditivity.

Theorem 16.19 (Depth Zero Closure Under Cooperation (16.14)). *If* $\mathrm{d}(a) = 0$ *and* $\mathrm{d}(b) = 0$*, then* $\mathrm{d}\big(\mathit{jointCompose}(a, b)\big) = 0$.

Proof sketch. From the depth-zero closure property of composition. □

Commentary. Shallow discourse combined with shallow discourse can never produce depth. This is the critical-theory insight that Adorno and Horkheimer

might have appreciated: the culture industry's endless recombination of trivial content produces only trivial content.[4]

Theorem 16.20 (Cooperation Monotonicity for Depth (16.17)). *For all* $a \in \mathcal{I}$, $\mathrm{d}\big(\texttt{jointCompose}(a, \mathbf{0})\big) = \mathrm{d}(a)$.

Proof sketch. Since $\circ(a, \mathbf{0}) = a$, the depth is preserved. □

Commentary. Free-riders (void contributors) don't add complexity to the joint output, but they don't reduce it either. The depth of the project is unchanged by the presence or absence of non-contributing participants—a result with clear implications for team management and organizational design.

16.7 Multi-Party Projects and Iterated Cooperation

We now extend the analysis to multi-party projects, iterated self-cooperation, and the celebrated gift-exchange cycles of anthropological theory.

Theorem 16.21 (Joint Project Depth Sum Bound (16.16)). *If every participant* a *in project* p *satisfies* $\mathrm{d}(a) \leq d$, *then*

$$\texttt{totalDepth}(p) \;\leq\; |p.\mathit{participants}| \cdot d.$$

Proof sketch. The total depth is a sum of individual depths, each bounded by d. The sum of n terms each at most d is at most $n \cdot d$. □

Commentary. Total intellectual capital invested constrains what the project can achieve. This is the project-management analogue of the depth bound: a team of n members, each of bounded capability d, produces a project whose total complexity is at most $n \cdot d$.

Theorem 16.22 (Iterated Self-Cooperation Depth Bound (16.19)). *For all* $a \in \mathcal{I}$ *and* $n \in \mathbb{N}$,

$$\mathrm{d}\big(\texttt{composeN}(a, n)\big) \;\leq\; n \cdot \mathrm{d}(a).$$

Proof sketch. By induction on n, using the subadditivity of depth at each step. □

Commentary. Repeating a ritual n times has bounded cumulative depth, growing at most linearly with repetitions. This formalizes the intuition that practice deepens understanding, but at a rate bounded by the practitioner's own depth. The logarithmic "diminishing returns" often observed in ritual practice are consistent with this linear upper bound.

[4] Theodor Adorno and Max Horkheimer, *Dialectic of Enlightenment* (1944), argued that mass culture recombines familiar elements without producing genuine novelty.

Theorem 16.23 (Gift Exchange Triangle (16.20)). *For all $a, b, c \in \mathcal{I}$,*

$$\mathrm{d}\big(\mathtt{jointCompose}(\mathtt{jointCompose}(a,b),c)\big) \;\leq\; \mathrm{d}(a) + \mathrm{d}(b) + \mathrm{d}(c).$$

Proof sketch. Apply the depth bound twice: first to $\circ(a,b)$ and c, obtaining $\mathrm{d}(\circ(\circ(a,b),c)) \leq \mathrm{d}(\circ(a,b)) + \mathrm{d}(c)$, then to a and b, obtaining $\mathrm{d}(\circ(a,b)) \leq \mathrm{d}(a) + \mathrm{d}(b)$. Combining by transitivity yields the three-term bound. □

Commentary. This theorem formalizes Malinowski's Kula ring: gifts circulating through a ring of three partners have bounded total complexity, ensuring the exchange remains manageable. The three-term depth bound generalizes to n-term bounds by induction (see Theorem 16.22), but the triangular case is particularly significant because it corresponds to the minimal non-trivial cycle of exchange.

16.8 Conclusion

This chapter has established the formal foundations of cooperative composition in ideatic space. Twenty theorems characterize the algebraic and metric properties of joint composition, multi-party projects, and iterated cooperation.

The central results are:

- **Identity:** Void contributes nothing (Theorems 16.4–16.5).
- **Associativity:** Multi-party cooperation is bracket-independent (Theorems 16.8–16.9).
- **Resonance preservation:** Resonant partners produce resonant outcomes (Theorems 16.12–16.16).
- **Depth subadditivity:** Cooperative complexity cannot exceed the sum of individual complexities (Theorems 16.17–16.23).

Mauss's *Essai sur le don* revealed that gift exchange is a total social fact. Our formalization shows that it is also a total *algebraic* fact: the structure of ideatic space—its identity, associativity, and subadditivity—completely determines the bounds and resonance properties of cooperative meaning-making. In the next chapter, we turn to what happens when cooperation breaks down: the problem of bargaining over composition order.

Chapter 17

Bargaining Over Composition Order

> *"The definition of the situation is itself the first move in the game."*
> — Victor Turner, *The Ritual Process* (1969)

17.1 Introduction

Composition in ideatic space is associative but *not* commutative. In general, $\circ(a, b) \neq \circ(b, a)$: who speaks first shapes the meaning of the entire exchange. This fundamental asymmetry is the engine of power in signalling games. The ability to set the agenda—to determine what is discussed first, which frame is laid down before the other party responds—is, as Victor Turner recognized, a primary form of social control.[1]

In political discourse, this is the problem of *agenda-setting power*. In legal reasoning, it is the question of which precedent is applied first. In Hegelian dialectics, it is the difference between thesis-antithesis and antithesis-thesis: the reversal produces a different synthesis.

This chapter formalizes the bargaining problem that arises from non-commutativity. We introduce the `compositionOrder` pair, the `orderInvariant` predicate, and the `OrderPair` structure, then prove eighteen theorems characterizing when order matters, when it doesn't, and what structural properties hold regardless of ordering.

[1] Turner's analysis of ritual process emphasizes that the *order* of ritual stages determines the transformation undergone by participants. The liminal phase must precede reaggregation; reversing the order produces a different—often incoherent—ritual.

The key insight is that while the *content* of composition depends on order, certain *structural* properties—depth bounds, self-resonance, and the behavior of void—are order-invariant. This creates a formal analogy to fair bargaining: the pie may be divided differently depending on who moves first, but the total size of the pie is bounded the same way regardless.

17.2 Core Definitions

Definition 17.1 (Composition Order Pair)**.** The *composition order pair* of $a, b \in \mathcal{I}$ is

$$\texttt{compositionOrder}(a, b) \;:=\; \big(\circ(a, b),\ \circ(b, a)\big) \;\in\; \mathcal{I} \times \mathcal{I}.$$

This captures both possible sequencings of a bilateral exchange: "I go first" versus "you go first."

```
def compositionOrder (a b : I) : I * I :=
  (IdeaticSpace.compose a b, IdeaticSpace.compose b a)
```

Listing 17.1: Composition order pair in Lean

Definition 17.2 (Order Invariance)**.** Two ideas a, b are *order-invariant* when composition commutes:

$$\texttt{orderInvariant}(a, b) \;:\Longleftrightarrow\; \circ(a, b) = \circ(b, a).$$

This is a strong condition: most idea-compositions are *not* order-invariant. When it holds, there is no agenda-setting advantage.

```
def orderInvariant (a b : I) : Prop :=
  IdeaticSpace.compose a b = IdeaticSpace.compose b a
```

Listing 17.2: Order invariance in Lean

Definition 17.3 (Order Pair Structure)**.** An *order pair* packages two ideas with both their compositions and proofs that the compositions are correctly formed:

$$\texttt{OrderPair} = \big(l, r, f, s \;\mid\; f = \circ(l, r),\ s = \circ(r, l)\big).$$

```
structure OrderPair (I : Type*) [IdeaticSpace I] where
  left : I
  right : I
```

```
  first : I
  second : I
  ab_eq : first = IdeaticSpace.compose left right
  ba_eq : second = IdeaticSpace.compose right left
```

Listing 17.3: Order pair structure in Lean

17.3 Void and Order Invariance

Void—silence, non-contribution—is the universal commuter. It has no agenda-setting power because it contributes nothing.

Theorem 17.4 (Void is Order-Invariant—Left (17.1))**.** *For all $a \in \mathcal{I}$, `orderInvariant`$(\mathbf{0},$*

Proof sketch. $\circ(\mathbf{0}, a) = a = \circ(a, \mathbf{0})$ by the identity laws. □

Commentary. Silence has no agenda-setting power. Whether you speak into silence or silence follows your speech, the result is the same—just your speech. In political theory, a party that brings nothing to the table cannot influence the agenda.

Theorem 17.5 (Self-Composition is Order-Invariant (17.2))**.** *For all $a \in \mathcal{I}$, `orderInvariant`(a, a).*

Proof sketch. $\circ(a, a) = \circ(a, a)$ trivially. □

Commentary. Internal dialogue with yourself has no ordering problem—you always agree with yourself on what comes first. This is the psychological ground truth that anchors the theory: self-composition is always order-invariant.

Theorem 17.6 (Order Invariance of Void Pair (17.8))**.** `orderInvariant`$(\mathbf{0}, \mathbf{0})$.

Proof sketch. Immediate from both identity laws. □

Commentary. Two silences are the same regardless of which comes first. The degenerate case anchors the extremal behavior of the theory.

Theorem 17.7 (Void is Universally Order-Invariant (17.12))**.** *For all $a \in \mathcal{I}$, `orderInvariant`$(a, \mathbf{0})$.*

Proof sketch. $\circ(a, \mathbf{0}) = a = \circ(\mathbf{0}, a)$. □

Commentary. The absence of contribution never creates ordering conflicts. Void is universally transparent to reordering, a formal counterpart to the observation that silence is the one "move" in a game that is costless in either position.[2]

[2] In mechanism design, a "null action" is typically assumed to be costless and position-independent. Void formalizes this assumption within IDT.

Theorem 17.8 (Void Annihilates Order Difference (17.18)). *For all $a \in \mathcal{I}$,*

$$(\texttt{compositionOrder}(a, \mathbf{0}))_1 = (\texttt{compositionOrder}(a, \mathbf{0}))_2.$$

Proof sketch. Both components reduce to a by the identity laws. □

Commentary. A player who contributes nothing has no bargaining power over ordering—there is nothing to order. The composition order pair collapses to a diagonal element when one party contributes void.

17.4 Structural Order Theorems

Theorem 17.9 (Order Invariance is Symmetric (17.11)). *If* $\texttt{orderInvariant}(a, b)$, *then* $\texttt{orderInvariant}(b, a)$.

Proof sketch. If $\circ(a, b) = \circ(b, a)$, then symmetry of equality gives $\circ(b, a) = \circ(a, b)$. □

Commentary. If swapping A and B's contributions doesn't matter, then swapping B and A's doesn't matter either. This ensures that order invariance is a symmetric relation—a necessary condition for it to serve as a formal model of "fair exchange."

Theorem 17.10 (Composition Order First Component (17.4)). $(\texttt{compositionOrder}(a, b))_1 = \circ(a, b)$.

Theorem 17.11 (Composition Order Second Component (17.5)). $(\texttt{compositionOrder}(a, b))_2 = \circ(b, a)$.

Proof sketch. Both are definitional equalities. □

Commentary. These projection lemmas are technical necessities, extracting the "a-goes-first" and "b-goes-first" orderings from the pair. They establish the interface between the order pair abstraction and the underlying composition operation.

Theorem 17.12 (Composition Order Swap is Involution (17.17)).

$$\texttt{compositionOrder}(b, a) = \big((\texttt{compositionOrder}(a, b))_2,\ (\texttt{compositionOrder}(a, b))_1\big).$$

Proof sketch. Expanding definitions: $\texttt{compositionOrder}(b, a) = (\circ(b, a), \circ(a, b))$, which equals the swap of $(\circ(a, b), \circ(b, a))$. □

Commentary. The dialectical reversal of a reversal returns to the thesis. Swapping who goes first twice recovers the original ordering. This involution property ensures that the bargaining structure has no higher-order ordering ambiguities.

Theorem 17.13 (Order Pair Construction (17.14)). *For any $a, b \in \mathcal{I}$, there exists an* `OrderPair` *with left $= a$ and right $= b$.*

Proof sketch. Construct the pair with first $= \circ(a, b)$, second $= \circ(b, a)$, and the proofs are `rfl`. □

Commentary. Every bilateral exchange can be formally analyzed by examining both possible orderings. The `OrderPair` structure packages this analysis into a single object, facilitating the systematic study of agenda-setting effects.

17.5 Depth and Resonance Under Reordering

While the *content* of a composition depends on order, its *depth bounds* are symmetric: subadditivity holds for both orderings.

Theorem 17.14 (Both Orders Have Same Depth Bound (17.3)). *For all $a, b \in \mathcal{I}$,*

$$\mathrm{d}(\circ(a, b)) \leq \mathrm{d}(a) + \mathrm{d}(b),$$
$$\mathrm{d}(\circ(b, a)) \leq \mathrm{d}(a) + \mathrm{d}(b).$$

Proof sketch. The first inequality is subadditivity. The second follows from $\mathrm{d}(\circ(b, a)) \leq \mathrm{d}(b) + \mathrm{d}(a) = \mathrm{d}(a) + \mathrm{d}(b)$. □

Commentary. The "cost" of an exchange, measured by depth, is bounded the same way regardless of who goes first. Subadditivity is symmetric. This means that while ordering affects the *content* of the outcome, it does not affect the *resource constraints*—a crucial fairness property.

Theorem 17.15 (Depth Bound Under Reordering (17.10)). $\mathrm{d}(\circ(b, a)) \leq \mathrm{d}(a) + \mathrm{d}(b)$.

Proof sketch. From subadditivity with reordering of the bound. □

Commentary. The maximum information capacity of an exchange is invariant under reordering. In information-theoretic terms, the channel capacity is bounded by the sum of the endpoint depths, regardless of direction.

Theorem 17.16 (Depth Zero Invariance Under Order (17.16)). *If* $\mathrm{d}(a) = 0$ *and* $\mathrm{d}(b) = 0$, *then* $\mathrm{d}(\circ(a, b)) = 0$ *and* $\mathrm{d}(\circ(b, a)) = 0$.

Proof sketch. Apply depth-zero closure in both orders. □

Commentary. Trivial exchanges are trivial regardless of who goes first. You can't create depth by reordering banalities—a critical-theory insight with mathematical teeth.

Theorem 17.17 (Both Orders Self-Resonate (17.6)). *resonates*$(\circ(a,b), \circ(a,b))$ *and resonates*$(\circ(b,a), \circ(b,a))$.

Proof sketch. Self-resonance is universal. □

Commentary. Any completed exchange achieves internal coherence regardless of who spoke first. This is the minimal hermeneutic property: every outcome is self-consistent.

Theorem 17.18 (Resonant Inputs Give Resonant Orders (17.7)). *If resonates*(a,b), *then resonates*$\big(\circ(a,b),\ \circ(b,a)\big)$.

Proof sketch. From `resonance_compose` applied to the pairs (a,b) and (b,a), using the symmetry of resonance. □

Commentary. This is Turner's ritual-process theorem. When ritual actors are in resonance—in Turner's terminology, when they share *communitas*—reordering the ritual stages produces ceremonies that still "rhyme" with each other. The spirit of the ritual is preserved even if the sequence changes. Formally, resonant inputs produce resonant orderings, ensuring that the bargaining problem between culturally aligned agents has a coherent solution space.[3]

17.6 Associativity and Multi-Party Bargaining

Theorem 17.19 (Associativity Resolves Three-Party Brackets (17.9)). $\circ(\circ(a,b),c) = \circ(a,\circ(b,c))$.

Proof sketch. The associativity axiom of ideatic composition. □

Commentary. In three-party negotiation, whether (A,B) form a coalition first and then bargain with C, or A bargains with a pre-formed (B,C) coalition, the final composite meaning is identical. Associativity is the great equalizer: it eliminates the "coalition formation order" as a source of bargaining advantage, even as the *within-pair* ordering remains significant.

Theorem 17.20 (Triple Depth Bound—Both Bracketings (17.13)). $\mathrm{d}(\circ(\circ(a,b),c)) \leq \mathrm{d}(a) + \mathrm{d}(b) + \mathrm{d}(c)$.

Proof sketch. Cascade two applications of subadditivity. □

Commentary. Three-way collaboration has bounded complexity no matter how sub-teams are formed. The depth bound generalizes linearly, providing a universal resource constraint on multi-party cooperation.

[3] Turner introduced the concept of *communitas* to describe the intense fellow-feeling that arises in liminal ritual stages. Our resonance relation formalizes this as a structural property of ideatic space.

Theorem 17.21 (Resonance Between Reordered Triples (17.15)). *If resonates*(a, b) *and resonates*(b, c)*, then*

$$\mathit{resonates}\big(\circ(\circ(a, b), c),\ \circ(a, \circ(b, c))\big).$$

Proof sketch. By associativity, both sides are equal, so self-resonance applies. □

Commentary. When all ritual participants share communitas, rearranging the ritual order produces ceremonies that deeply resonate. The resonance between re-bracketed triples is not merely an analogy to Turner's theory—it is its formal content.

17.7 Conclusion

This chapter has established eighteen theorems characterizing the bargaining problem that arises from the non-commutativity of ideatic composition.

The central tension is clear: while *content* depends on order (who speaks first matters), *structural properties*—depth bounds, self-resonance, void transparency—do not. This creates a formal separation between the distributional question ("who gets the better outcome?") and the efficiency question ("how much total meaning is produced?").

The key results are:

- Void has no agenda-setting power (Theorems 17.4–17.8).
- Depth bounds are order-invariant (Theorems 17.14–17.16).
- Resonant agents produce resonant orderings (Theorem 17.18).
- Associativity resolves multi-party bracketing (Theorem 17.19).

Turner's ritual process reveals that transformative social experiences require a specific ordering of stages. Our formalization shows that this ordering dependence is a fundamental feature of ideatic composition, not an artifact of particular rituals. In the next chapter, we turn from bilateral bargaining to the problem of diffusion through networks.

Chapter 18

Network Diffusion and Cultural Epidemiology

"Culture is the precipitate of cognition and communication."
— Dan Sperber, *Explaining Culture* (1996)

18.1 Introduction

Dan Sperber's epidemiological approach to culture proposes that cultural representations spread like epidemics: ideas pass from mind to mind, transformed at each step by the cognitive environment of the host. Frederic Bartlett's serial reproduction experiments (1932) demonstrated this empirically: a story retold through a chain of narrators mutates systematically, with each retelling imposing the reteller's own cognitive schemata on the material.[1]

In Ideatic Diffusion Theory, each relay hop is a composition: the relayer composes the incoming signal with their own interpretive frame. A chain of n relayers produces

$$\circ\big(r_n, \circ(r_{n-1}, \ldots \circ (r_1, s) \ldots)\big).$$

This chapter formalizes the mathematics of cultural diffusion through networks, establishing eighteen theorems on relay chains, depth bounds, resonance preservation, chain concatenation, and the transparency of void relayers.

[1] Bartlett's landmark study *Remembering* (1932) showed that the Kwakiutl folk tale "The War of the Ghosts" was progressively rationalized, shortened, and conventionalized as it passed through chains of English undergraduates.

The central insight is that network diffusion is *compositional*: the output of a multi-hop relay is fully determined by the sequence of relayers and the initial signal, and the structural properties of the output—depth, resonance, void-transparency—follow from the algebraic properties of composition. This provides a rigorous foundation for Sperber's cultural epidemiology, for meme theory in Dawkins's sense, and for the "telephone game" as a formal model of cultural transmission.

18.2 Core Definitions

Definition 18.1 (Relay Chain)**.** Let relayers $= [r_1, r_2, \ldots, r_n]$ be a list of relayer ideas and $s \in \mathcal{I}$ be a signal. The *relay chain* is the left-fold:

$$\texttt{relayChain}(\text{relayers}, s) \;:=\; \text{foldl}\big(\lambda\, \text{acc}\; r.\; \circ\,(r, \text{acc}),\; s,\; \text{relayers}\big).$$

Each relayer composes the accumulated signal with their own frame, left-to-right.

```
def relayChain (relayers : List I) (signal : I) : I :=
  relayers.foldl (fun acc r => IdeaticSpace.compose r
      acc) signal
```

Listing 18.1: Relay chain in Lean

The fold semantics mean that r_1 processes the raw signal first, producing $\circ(r_1, s)$; then r_2 processes r_1's output, producing $\circ(r_2, \circ(r_1, s))$; and so on. Each relay hop adds a layer of interpretive composition.

Definition 18.2 (Chain Depth)**.** The *chain depth* is the depth of the relay chain's output:

$$\texttt{chainDepth}(\text{relayers}, s) := \mathrm{d}\big(\texttt{relayChain}(\text{relayers}, s)\big).$$

Definition 18.3 (Network Path)**.** A *network path* packages a list of relayers and a signal into a single structure. Its output is the relay chain:

$$\texttt{NetworkPath.output}(p) := \texttt{relayChain}(p.\text{relayers}, p.\text{signal}).$$

```
structure NetworkPath (I : Type*) [IdeaticSpace I]
    where
  relayers : List I
  signal : I

def NetworkPath.output (p : NetworkPath I) : I :=
  relayChain p.relayers p.signal
```

Listing 18.2: Network path structure in Lean

18.3 Fundamental Transmission Theorems

Theorem 18.4 (Empty Chain Preserves Signal (18.1)). *For all $s \in \mathcal{I}$,*

$$\texttt{relayChain}([\,], s) = s.$$

Proof sketch. The fold over an empty list returns the initial accumulator s. □

Commentary. Direct transmission without intermediaries preserves the original message perfectly. This is the idealized limit of communication: zero-hop transmission is lossless. In information theory, this corresponds to a noiseless channel; in cultural epidemiology, it represents the (impossible) case of unmediated cultural transmission.

Theorem 18.5 (Singleton Chain = Single Composition (18.2)). *For all $r, s \in \mathcal{I}$,*

$$\texttt{relayChain}([r], s) = \circ(r, s).$$

Proof sketch. One iteration of the fold: $\circ(r, s)$. □

Commentary. A single retelling is one act of interpretive composition. The reteller composes the signal with their own frame, producing a new idea that carries traces of both the original and the reteller's cognitive environment. In Bartlett's terms, this is one round of "serial reproduction."

Theorem 18.6 (Two-Relayer Chain (18.3)).

$$\texttt{relayChain}([r_1, r_2], s) = \circ(r_2, \circ(r_1, s)).$$

Proof sketch. Two iterations of the fold. □

Commentary. The "telephone game" with two players: the second player interprets the first player's interpretation, not the original signal. This captures the fundamental structure of multi-hop cultural transmission: each intermediary processes the *already-processed* signal, not the original.[2]

18.4 Void Transparency and Signal Fidelity

Void relayers are "transparent"—they pass the signal through without modification. This models passive carriers who do not transform the message.

[2] In organizational communication theory, this is the source of the "serial transmission effect": messages degrade as they pass through chains of intermediaries, each of whom introduces their own cognitive biases.

Theorem 18.7 (Void Relayer is Transparent (18.4)). *Appending a void relayer to the end of a chain does not change the output:*

$$\texttt{relayChain}(\mathit{relayers} ++ [\mathbf{0}], s) = \texttt{relayChain}(\mathit{relayers}, s).$$

Proof sketch. The fold over the appended void computes $\circ(\mathbf{0}, \mathrm{acc}) = \mathrm{acc}$ by the left-identity law. □

Commentary. A cognitively "empty" relay node transmits faithfully. In Sperber's framework, this models a carrier who lacks the cognitive schemas to transform the representation—a perfect mirror for the incoming signal.

Theorem 18.8 (Prepending Void Relayer is No-Op (18.15)).

$$\texttt{relayChain}(\mathbf{0} :: \mathit{relayers}, s) = \texttt{relayChain}(\mathit{relayers}, s).$$

Proof sketch. The first fold step computes $\circ(\mathbf{0}, s) = s$, so the remaining fold proceeds from the same accumulator. □

Commentary. An empty preamble carries no information. Starting a chain of retellers with a "silent" relayer does not affect the final output.

Theorem 18.9 (Void Signal Through Single Relay (18.5)).

$$\texttt{relayChain}([r], \mathbf{0}) = r.$$

Proof sketch. $\circ(r, \mathbf{0}) = r$ by right-identity. □

Commentary. An empty message gets filled with the relayer's own content. When there is no signal, only interpretation remains—the relayer projects their own frame onto the void. This formalizes the constructivist insight that in the absence of external stimuli, the interpreter's own schemas dominate.

Theorem 18.10 (Void Chain is Identity (18.12)). *For all $n \in \mathbb{N}$ and $s \in \mathcal{I}$,*

$$\texttt{relayChain}(\underbrace{[\mathbf{0}, \mathbf{0}, \ldots, \mathbf{0}]}_{n}, s) = s.$$

Proof sketch. By induction on n. Base case: empty chain preserves s. Inductive step: the last void relayer is transparent (Theorem 18.7). □

Commentary. A chain of "empty" cognitive environments transmits perfectly. This is the idealized limit of faithful transmission: if every intermediary is cognitively passive, the signal arrives unchanged. Of course, this limit is never achieved in practice—every mind adds something to the message.

18.5 Depth Bounds on Relay Chains

Subadditivity cascades through relay chains, providing upper bounds on the depth of the transmitted signal.

Theorem 18.11 (Chain Depth Bound—Single Relayer (18.6)).

$$\mathit{chainDepth}([r], s) \leq \mathrm{d}(r) + \mathrm{d}(s).$$

Proof sketch. By definition, $\texttt{chainDepth}([r], s) = \mathrm{d}(\circ(r, s))$, and subadditivity gives $\mathrm{d}(\circ(r, s)) \leq \mathrm{d}(r) + \mathrm{d}(s)$. □

Commentary. A single retelling adds at most the reteller's own complexity to the story. In Bartlett's experiments, the distortions introduced by a single reteller are bounded by the reteller's cognitive capacity.

Theorem 18.12 (Chain Output Depth Bound—Two Relayers (18.16)).

$$\mathit{chainDepth}([r_1, r_2], s) \leq \mathrm{d}(r_1) + \mathrm{d}(r_2) + \mathrm{d}(s).$$

Proof sketch. Apply subadditivity twice: first to r_2 and $\circ(r_1, s)$, then to r_1 and s. Combine by transitivity. □

Commentary. Two successive retellings add at most the combined complexity of both retellers to the original story's depth. The bound is *linear* in the number of relayers and their individual depths—a tight constraint on the "inflation" of cultural representations during transmission.

Theorem 18.13 (Depth Zero Signal Relay (18.17)). *If* $\mathrm{d}(r) = 0$ *and* $\mathrm{d}(s) = 0$, *then* $\mathit{chainDepth}([r], s) = 0$.

Proof sketch. Depth-zero closure under composition. □

Commentary. Empty content through an empty channel yields nothing. This is the information-theoretic ground truth: zero-depth inputs produce zero-depth output, regardless of the network topology.

Theorem 18.14 (Empty Chain Depth Preservation (18.9)). $\mathit{chainDepth}([\,], s) = \mathrm{d}(s)$.

Proof sketch. The relay chain over an empty list is the signal itself. □

Commentary. Without intermediaries, complexity is unchanged. Direct communication preserves not just the signal content but also its structural complexity.

18.6 Resonance in Networks

Resonance—the compatibility relation between ideas—interacts with network diffusion in natural ways.

Theorem 18.15 (Resonant Relayers Produce Resonant Output (18.7)). *If resonates*(r_1, r_2)*, then for all* s*,*

$$\mathit{resonates}\big(\mathtt{relayChain}([r_1], s),\ \mathtt{relayChain}([r_2], s)\big).$$

Proof sketch. Reduces to resonates$(\circ(r_1, s), \circ(r_2, s))$, which follows from `resonance_compose_right`. □

Commentary. Similar cognitive environments produce similar transformations of the same input. In cultural epidemiology, this explains why cultural variants *cluster*: when relayers share cognitive schemas (resonance), their outputs remain in the same neighborhood of ideatic space.[3]

Theorem 18.16 (Resonant Signals Through Same Relayer (18.8)). *If resonates*(s_1, s_2)*, then for all* r*,*

$$\mathit{resonates}\big(\mathtt{relayChain}([r], s_1),\ \mathtt{relayChain}([r], s_2)\big).$$

Proof sketch. From `resonance_compose` applied to (r, r, s_1, s_2) with self-resonance of r and the hypothesis. □

Commentary. Similar memes stay similar even after cultural transmission through the same social environment. A single relayer preserves the resonance between its inputs, ensuring that initially similar signals remain similar after processing.

Theorem 18.17 (Two Chains Resonance (18.14)). *If resonates*(r_1, r_2) *and resonates*(s_1, s_2)*, then*

$$\mathit{resonates}\big(\mathtt{relayChain}([r_1], s_1),\ \mathtt{relayChain}([r_2], s_2)\big).$$

Proof sketch. Direct from `resonance_compose` applied to (r_1, r_2, s_1, s_2). □

Commentary. The diffusion of innovations through similar social environments produces similar cultural adaptations. When both the relayers and the signals are in resonance, the outputs are in resonance—a strong stability result for cultural transmission through homogeneous networks.

Theorem 18.18 (Network Path Self-Resonance (18.11)). *Every network path output resonates with itself.*

[3] Sperber's "epidemiology of representations" predicts that cultural attractors emerge when many minds share similar cognitive biases. Theorem 18.15 provides the formal mechanism for this clustering.

Proof sketch. Self-resonance is universal. □

Commentary. Any completed transmission, however distorted, achieves internal self-consistency. The accumulated composition of relayer transformations may diverge arbitrarily from the original signal, but the final output is always coherent with itself.

18.7 Network Topology and Chain Composition

Theorem 18.19 (Chain Concatenation is Sequential Relay (18.10))**.**

$$\texttt{relayChain}(relayers_1 \mathbin{++} relayers_2, s) = \texttt{relayChain}\big(relayers_2,\ \texttt{relayChain}(relayers_1, s)\big)$$

Proof sketch. From the associativity of `foldl` over list concatenation. □

Commentary. Two segments of a communication network compose into a single longer path. This is the fundamental composition law for information networks: the output of the first segment becomes the input to the second. In graph-theoretic terms, path concatenation in the relay network corresponds to functional composition of relay transformations—a monoidal structure on the space of network paths.[4]

Theorem 18.20 (Chain Append Equivalence (18.18))**.**

$$\texttt{relayChain}(relayers \mathbin{++} [r], s) = \circ\big(r,\ \texttt{relayChain}(relayers, s)\big).$$

Proof sketch. Special case of chain concatenation with a singleton second chain. □

Commentary. Adding a node at the end of a path is equivalent to that node processing the path's accumulated output. This provides the inductive step for analyzing chains of arbitrary length: each new relayer simply composes with the running total.

Theorem 18.21 (Relay Preserves Relayer Count (18.13))**.** $|\texttt{NetworkPath.mk}(relayers, s).$ $|relayers|$.

Proof sketch. Definitional equality: the structure stores the relayer list unchanged. □

Commentary. The relay chain function doesn't alter the number of relayers. The network topology is preserved during diffusion—only the *content* of the signal changes, not the *structure* of the network.

[4] This is the *functoriality* of the relay chain: it respects the monoidal structure of list concatenation.

18.8 Conclusion

This chapter has established eighteen theorems formalizing cultural diffusion through networks within the framework of Ideatic Diffusion Theory. The key results are:

- **Compositional structure:** Network diffusion is sequential composition (Theorems 18.5–18.6).
- **Void transparency:** Passive relayers preserve signals (Theorems 18.7–18.10).
- **Depth bounds:** Chain depth grows at most linearly (Theorems 18.11–18.12).
- **Resonance clustering:** Similar relayers and signals produce similar outputs (Theorems 18.15–18.17).
- **Network composition:** Chain concatenation = sequential relay (Theorem 18.19).

Sperber's cultural epidemiology proposed that culture is "the precipitate of cognition and communication." Our formalization shows that this precipitate obeys precise algebraic laws: the relay chain is a homomorphism from the monoid of relayer lists to the monoid of ideatic transformations. Depth bounds constrain the "mutation rate" of cultural representations, and resonance clustering explains the emergence of cultural attractors.

In the next chapter, we turn from the diffusion of ideas to their aggregation: the social-choice problem of combining multiple interpretations into a single shared meaning.

Chapter 19

Social Choice and Hermeneutic Aggregation

"In communicative action participants are not primarily oriented to their own individual successes; they pursue their individual goals under the condition that they can harmonize their plans of action on the basis of common situation definitions."
— Jürgen Habermas, *The Theory of Communicative Action* (1981)

19.1 Introduction

When a sermon is preached, a lecture delivered, or a news broadcast aired, each member of the audience interprets the message through their own background—their education, culture, temperament, and prior beliefs. Jürgen Habermas's theory of communicative action asks the fundamental question: how do multiple, heterogeneous interpretations converge into shared meaning?[1]

This is the *hermeneutic aggregation problem*: given n agents with background ideas $[r_1, \ldots, r_n]$ and a shared signal s, each agent produces an interpretation $\circ(r_i, s)$. The interpretations can then be aggregated—by sequential composition, by deliberation, by voting—into a collective meaning. Under what conditions does this collective meaning reflect a genuine consensus rather than a mere concatenation of incompatible readings?

In IDT, we formalize this problem through three core concepts:

[1] Habermas's magnum opus, *Theorie des kommunikativen Handelns* (1981), distinguishes between "strategic action" (oriented toward individual success) and "communicative action" (oriented toward mutual understanding). The hermeneutic aggregation problem belongs squarely to the domain of communicative action.

- `interpretations`: the list of all agents' interpretations of a shared signal;
- `aggregateCompose`: the sequential composition of interpretations into a single collective meaning;
- `coherentInterpretation`: the condition that all interpretations pairwise resonate, formalizing Habermas's "ideal speech situation."

This chapter proves eighteen theorems characterizing the algebraic and metric properties of hermeneutic aggregation. The results establish when coherence is achievable, how depth constrains the interpretive effort of a community, and what role void—silence, absence, non-participation—plays in the aggregation process.

19.2 Core Definitions

Definition 19.1 (Interpretations)**.** Let reps $= [r_1, \ldots, r_n]$ be the agents' repertoire ideas and $s \in \mathcal{I}$ be the shared signal. The *interpretations* are:

$$\texttt{interpretations}(\text{reps}, s) \;:=\; [\; \circ(r_1, s),\; \circ(r_2, s),\; \ldots,\; \circ(r_n, s) \;].$$

Each agent interprets the signal by composing it with their own background.

```
def interpretations (reps : List I) (s : I) : List I
    :=
  reps.map (fun r => IdeaticSpace.compose r s)
```

Listing 19.1: Interpretations in Lean

In Gadamer's hermeneutic philosophy, each interpreter brings their own "horizon"—their pre-understanding, their historically-effected consciousness—to the encounter with the text. The interpretation is the "fusion of horizons" between background and signal, which in IDT is precisely $\circ(r_i, s)$.[2]

Definition 19.2 (Aggregate Composition)**.** The *aggregate composition* of a list of ideas is their right-fold via composition:

$$\texttt{aggregateCompose}([\,]) = \mathbf{0}, \qquad \texttt{aggregateCompose}(a :: \text{rest}) = \circ(a, \texttt{aggregateCompose}(\text{rest})).$$

This models the process by which a community arrives at a single shared interpretation through successive dialogue.

[2] Hans-Georg Gadamer, *Wahrheit und Methode* (1960), introduced the concept of *Horizontverschmelzung* (fusion of horizons) as the fundamental structure of understanding.

```
def aggregateCompose : List I -> I
  | [] => IdeaticSpace.void
  | a :: rest => IdeaticSpace.compose a (
      aggregateCompose rest)
```

Listing 19.2: Aggregate composition in Lean

The right-fold semantics are significant: the last voice in the list serves as the "base" upon which earlier voices build. In Habermas's terms, this models a deliberative process in which each new speaker adds their perspective "on top of" the existing consensus.

Definition 19.3 (Coherent Interpretation)**.** A group of agents achieves *coherent interpretation* of signal s when all pairs of interpretations resonate:

$$\texttt{coherentInterpretation}(\text{reps}, s) \;:\!\Longleftrightarrow\; \forall\, x, y \in \texttt{interpretations}(\text{reps}, s), \quad \text{resonates}$$

This is the formal condition for "hermeneutic consensus."

```
def coherentInterpretation (reps : List I) (s : I) :
    Prop :=
  forall x in interpretations reps s, forall y in
      interpretations reps s,
    IdeaticSpace.resonates x y
```

Listing 19.3: Coherent interpretation in Lean

Habermas's "ideal speech situation" is one in which all participants' interpretations are mutually comprehensible. Coherent interpretation is the formal counterpart: when it holds, the community has achieved a state of hermeneutic equilibrium in which every reading "rhymes" with every other.

19.3 Empty and Singleton Groups

Theorem 19.4 (Empty Group Has Void Interpretation (19.1))**.** *For all* $s \in \mathcal{I}$,

$$\mathit{aggregateCompose}\big(\mathit{interpretations}([\,], s)\big) = \mathbf{0}.$$

Proof sketch. $\texttt{interpretations}([\,], s) = [\,]$, and $\texttt{aggregateCompose}([\,]) = \mathbf{0}$ by definition. □

Commentary. An empty polity produces no collective meaning. This is the political-theory baseline: without citizens, there is no *res publica*.

Theorem 19.5 (Singleton = Individual Interpretation (19.2))**.**

$$\mathit{aggregateCompose}\big(\mathit{interpretations}([r], s)\big) = \circ(r, s).$$

Proof sketch. `interpretations`$([r], s) = [\circ(r, s)]$, and `aggregateCompose`$([\circ(r, s)]) = \circ(\circ(r, s), \mathbf{0}) = \circ(r, s)$ by right-identity. □

Commentary. A solitary reader's interpretation *is* the meaning. In hermeneutics, the singleton case collapses the distinction between individual and collective meaning: when there is only one interpreter, their reading is authoritative by default.

Theorem 19.6 (Interpretation Count = Agent Count (19.5))**.** $|\mathit{interpretations}(\mathit{reps}, s)| = |\mathit{reps}|$.

Proof sketch. The `map` operation preserves list length. □

Commentary. Every agent produces exactly one interpretation—no more, no less. This "one person, one reading" principle is the hermeneutic analogue of the democratic "one person, one vote."[3]

Theorem 19.7 (Interpretation Map Preserves Nil (19.17))**.** $\mathit{interpretations}([\,], s) = [\,]$.

Proof sketch. Mapping over the empty list yields the empty list. □

19.4 Void Signal and Self-Resonance

When the signal is void—when nothing is said—the interpretive situation simplifies dramatically.

Theorem 19.8 (Void Signal Preserves Repertoire (19.3))**.**

$$\mathit{interpretations}(\mathit{reps}, \mathbf{0}) = \mathit{reps}.$$

Proof sketch. By induction on the list. For each $r \in$ reps, $\circ(r, \mathbf{0}) = r$ by right-identity. □

Commentary. When there is no text to interpret, each interpreter is left with only their own pre-understanding. In Gadamer's terms, their horizon remains unchanged: the void signal induces no fusion of horizons, and each agent's interpretation equals their original repertoire idea.

Theorem 19.9 (Void Signal Yields Self-Resonant Group (19.4))**.** *For all* $x \in \mathit{interpretations}(\mathit{reps}, \mathbf{0})$, $\mathit{resonates}(x, x)$.

[3] The analogy to social choice theory is intentional. Arrow's impossibility theorem concerns the aggregation of preference orderings; our theory concerns the aggregation of interpretive compositions. The structural parallels are explored further in Section 19.6.

Proof sketch. By Theorem 19.8, interpretations equal the original repertoire ideas, and self-resonance is universal. □

Commentary. In the absence of ritual content, mechanical solidarity still holds: each individual remains internally coherent. Durkheim's insight that social cohesion begins with self-consistency finds formal expression here.

Theorem 19.10 (Void Repertoire Interpretation (19.14))**.** $\texttt{interpretations}([\mathbf{0}], s) = [s]$.

Proof sketch. $\circ(\mathbf{0}, s) = s$ by left-identity. □

Commentary. A *tabula rasa*—a mind with no pre-existing ideas—interprets the signal as the signal itself, without transformation. The void background is a perfect mirror: the interpretation is the signal unchanged.

Theorem 19.11 (Aggregate of Voids is Void (19.15))**.** *For all* n, $\texttt{aggregateCompose}(\texttt{inte}$

$\mathbf{0}$.

Proof sketch. By Theorem 19.8, the interpretations are n copies of $\mathbf{0}$. By induction, aggregating n voids yields $\mathbf{0}$. □

Commentary. Empty minds interpreting nothing produce nothing. This is the nihilistic limit of hermeneutic aggregation: when neither the community nor the text carries any content, the outcome is silence.

Theorem 19.12 (Aggregate Singleton Void Signal (19.18))**.** $\texttt{aggregateCompose}(\texttt{interpr}$ r.

Proof sketch. By void-signal preservation, the interpretation list is $[r]$, and $\texttt{aggregateCompose}([r]) = \circ(r, \mathbf{0}) = r$. □

Commentary. In the absence of external stimuli, consciousness returns to its own content—pure self-reflection. Phenomenologically, the subject without object collapses back upon itself.

19.5 Aggregation Mechanics

Theorem 19.13 (Aggregating Empty Yields Void (19.9))**.** $\texttt{aggregateCompose}([\,]) = \mathbf{0}$.

Proof sketch. Definitional. □

Commentary. Without any contributions to the discourse, the outcome is silence. Habermas's communicative action requires at least one speaker.

Theorem 19.14 (Aggregate Cons Unfolding (19.10))**.** $\texttt{aggregateCompose}(a :: \mathit{rest}) = \circ(a, \texttt{aggregateCompose}(\mathit{rest}))$.

Proof sketch. Definitional. □

Commentary. Each new voice adds their perspective on top of the existing consensus. The deliberative process is compositional: the k-th speaker composes their idea with the aggregate of the remaining speakers, building the collective meaning layer by layer.

Theorem 19.15 (Void Aggregate Identity (19.11)). $\texttt{aggregateCompose}(\mathbf{0} :: \mathit{rest}) = \texttt{aggregateCompose}(\mathit{rest})$.

Proof sketch. $\circ(\mathbf{0}, \texttt{aggregateCompose}(\text{rest})) = \texttt{aggregateCompose}(\text{rest})$ by left-identity. □

Commentary. A silent participant doesn't alter the collective meaning. Non-contributing members are transparent to the aggregation process—they neither add nor subtract from the deliberative outcome.

Theorem 19.16 (Self-Resonance of Aggregate (19.12)). $\mathit{resonates}\big(\texttt{aggregateCompose}(\mathit{interps})$

Proof sketch. Self-resonance is universal. □

Commentary. Any completed deliberation, however contentious, achieves internal self-consistency as a composite idea. The aggregate may fail to satisfy individual participants, but it is always coherent with itself—a minimal condition for the output of any decision procedure.

19.6 Coherence and Resonance

Under what conditions does a group achieve hermeneutic consensus? The following theorems establish both trivial and substantive conditions for coherent interpretation.

Theorem 19.17 (Empty Interpretation List is Coherent (19.7)). $\texttt{coherentInterpretation}([\,],$ *for all* s.

Proof sketch. Universal quantification over the empty set is vacuously true. □

Commentary. Unanimity without voters: if no one interprets, there's no disagreement. This is the degenerate case of consensus—trivially achieved but informationally empty.

Theorem 19.18 (Singleton Interpretation is Coherent (19.8)). $\texttt{coherentInterpretation}([r], s$ *for all* r, s.

Proof sketch. The only interpretation is $\circ(r, s)$, which resonates with itself. □

Commentary. A lone reader always agrees with their own interpretation. Solipsism is trivially coherent. The interesting cases arise when multiple agents must reach consensus.

Theorem 19.19 (Resonant Pair Coherence (19.13)). *If resonates*(r_1, r_2)*, then* `coherentInterpretation`$([r_1, r_2], s)$ *for all* s*.*

Proof sketch. There are four cases (each interpretation against each interpretation). The diagonal cases are self-resonance. The off-diagonal cases follow from `resonance_compose_right`: resonates(r_1, r_2) implies resonates$(\circ(r_1, s), \circ(r_2, s))$ □

Commentary. This is Durkheim's *conscience collective* theorem: agents who share deep cultural resonance will agree on any new information. If two minds resonate at the level of their background assumptions, their interpretations of any signal will cohere. This is the formal mechanism underlying "preaching to the choir"—the choir already agrees, so any sermon produces consensus.[4]

19.7 Depth Bounds on Aggregation

Theorem 19.20 (Aggregate Depth Bound—Singleton (19.6)).

$$\mathrm{d}\big(\texttt{aggregateCompose}(\texttt{interpretations}([r], s))\big) \leq \mathrm{d}(r) + \mathrm{d}(s).$$

Proof sketch. By Theorem 19.5, the aggregate equals $\circ(r, s)$, and subadditivity applies. □

Commentary. One person's understanding adds at most their background depth to the signal's depth. This is the hermeneutic bound for individual interpretation.

Theorem 19.21 (Interpretation Depth Sum Bound (19.16)).

$$\texttt{depthSum}\big(\texttt{interpretations}(reps, s)\big) \leq \texttt{depthSum}(reps) + |reps| \cdot \mathrm{d}(s).$$

Proof sketch. By induction on the list. At each step, $\mathrm{d}(\circ(r, s)) \leq \mathrm{d}(r) + \mathrm{d}(s)$ by subadditivity, and the inductive hypothesis bounds the tail. □

Commentary. The total hermeneutic effort of a community is bounded by their collective background knowledge plus the signal's complexity distributed across all members. In resource-theoretic terms, the "cognitive budget" of a community constrains its total interpretive output. This bound has

[4] The sociological implications are significant. Theorem 19.19 implies that cultural homogeneity is *sufficient* for hermeneutic consensus. The converse—whether consensus requires homogeneity—is a deeper question left for future work.

practical implications for deliberative democracy: a community of n agents, each of bounded sophistication d, encountering a signal of depth $\mathrm{d}(s)$, can produce total interpretive richness at most $nd + n \cdot \mathrm{d}(s)$—a linear bound in both the number of agents and the signal complexity.[5]

19.8 Conclusion

This chapter has formalized the hermeneutic aggregation problem within Ideatic Diffusion Theory, proving eighteen theorems that characterize the structure of collective interpretation.

The central results form a coherent picture:

- **Void baseline:** Void signal preserves backgrounds; void agents are transparent (Theorems 19.8–19.15).
- **Coherence conditions:** Empty and singleton groups are trivially coherent; resonant pairs achieve substantive consensus (Theorems 19.17–19.19).
- **Depth bounds:** Total interpretive effort is linearly bounded (Theorems 19.20–19.21).
- **Aggregation mechanics:** Sequential composition of interpretations is well-defined and compositional (Theorems 19.13–19.16).

Habermas's theory of communicative action envisioned a form of rationality grounded in mutual understanding rather than strategic calculation. Our formalization shows that the "ideal speech situation" has a precise mathematical structure: it is the state in which all agents' interpretations pairwise resonate (Definition 19.3), and this state is achievable whenever the agents share cultural resonance (Theorem 19.19).

In the final chapter, we synthesize all the threads of the signalling-game arc into a unified theory of meaning.

[5] Compare this to the "Condorcet jury theorem" in social choice theory, which provides conditions under which larger groups make better decisions. Our depth bound is not about *accuracy* but about *total complexity*—a complementary perspective.

Chapter 20

Synthesis — The Game Theory of Meaning

"Man is an animal suspended in webs of significance he himself has spun."
— Clifford Geertz, *The Interpretation of Cultures* (1973)

20.1 Introduction

We have arrived at the culmination of the signalling-game arc. In the preceding chapters, we formalized cooperation as composition (Chapter 16), bargaining as order-dependence (Chapter 17), diffusion as relay chains (Chapter 18), and aggregation as hermeneutic consensus (Chapter 19). Each chapter isolated one dimension of the problem of meaning. This final chapter weaves them together into a unified theory.

The synthesis draws on three philosophical traditions:

1. **Gadamer's hermeneutic circle.** Understanding is not a linear process from signal to meaning, but a circle in which the interpreter's pre-understanding shapes their reading, which in turn reshapes their pre-understanding. In IDT, this is the associativity of iterated interpretation: $\circ(r_3, \circ(r_2, \circ(r_1, s)))$ can be re-bracketed without changing the result.[1]

2. **Ricoeur's surplus of meaning.** Paul Ricoeur argued that every text generates more meaning than any single reader can extract. In IDT,

[1] Hans-Georg Gadamer, *Truth and Method* (1960), argued that the hermeneutic circle is not a vicious circle but a spiral of deepening understanding. Our associativity theorems formalize this spiral.

this "surplus" is measured by *interpretive richness*—the total depth of all interpretations—and is bounded by the universal depth theorem (Theorem 20.13).

3. **Geertz's thick description.** Clifford Geertz proposed that cultural analysis is "not an experimental science in search of law but an interpretive one in search of meaning." The meaning game (Definition 20.2) formalizes this interpretive quest: a community of receivers encounters a signal and produces a constellation of interpretations, each shaped by their own "webs of significance."

The grand synthesis (Theorem 20.24) unifies the resource-theoretic and hermeneutic dimensions: for any meaning game with resonant receivers, the total interpretive richness is bounded *and* the game is in equilibrium. Bounded resources and shared meaning are simultaneously achievable—the formal heart of the game theory of meaning.

We proceed as follows. Section 20.2 introduces the meaning game, interpretive richness, and meaning equilibrium. Section 20.3 establishes the fundamental theorems of interpretation. Section 20.4 develops resonance preservation results. Section 20.5 derives depth bounds on interpretive richness. Section 20.6 proves equilibrium existence theorems. Section 20.7 presents the grand synthesis. Section 20.8 concludes the entire signalling-game arc.

20.2 The Meaning Game

Definition 20.1 (Interpret All)**.** Let reps $= [r_1, \ldots, r_n]$ be a list of receiver backgrounds and $s \in \mathcal{I}$ be the signal. The *interpretation list* is:

$$\texttt{interpretAll}(\text{reps}, s) := [\circ(r_1, s), \circ(r_2, s), \ldots, \circ(r_n, s)].$$

```
def interpretAll (reps : List I) (s : I) : List I :=
  reps.map (fun r => IdeaticSpace.compose r s)
```

Listing 20.1: interpretAll in Lean

Definition 20.2 (Meaning Game)**.** A *meaning game* $G = (s, [r_1, \ldots, r_n])$ consists of:

- a *signal* $s \in \mathcal{I}$ being sent;
- a list of *receivers* $[r_1, \ldots, r_n]$, each with their own background idea.

The *outcome* of the game is $\texttt{interpretAll}(\text{receivers}, s)$.

```
structure MeaningGame (I : Type*) [IdeaticSpace I]
    where
  signal : I
  receivers : List I

def MeaningGame.outcome (g : MeaningGame I) : List I
    :=
  interpretAll g.receivers g.signal
```

Listing 20.2: Meaning game structure in Lean

In Geertz's terms, the meaning game formalizes the encounter between a cultural artifact (signal) and a community of interpreters (receivers). Each receiver brings their own "webs of significance" to the encounter, producing a reading that is simultaneously individual and socially conditioned.[2]

Definition 20.3 (Meaning Equilibrium)**.** A meaning game is in *equilibrium* when all receivers' interpretations pairwise resonate:

$$\texttt{MeaningEquilibrium}(G) \;:\Longleftrightarrow\; \forall\, x, y \in G.\text{outcome}, \quad \text{resonates}(x, y).$$

```
def MeaningEquilibrium (g : MeaningGame I) : Prop :=
  forall x in g.outcome, forall y in g.outcome,
    IdeaticSpace.resonates x y
```

Listing 20.3: Meaning equilibrium in Lean

Meaning equilibrium is the game-theoretic analogue of Habermas's "ideal speech situation": a state in which all participants' interpretations are mutually comprehensible. Unlike Nash equilibrium, which concerns individual rationality, meaning equilibrium concerns collective coherence—a fundamentally different desideratum that reflects the hermeneutic rather than strategic orientation of communicative action.

Definition 20.4 (Interpretive Richness)**.** The *interpretive richness* of a group interpreting a signal is the total depth of all interpretations:

$$\texttt{interpretiveRichness}(\text{reps}, s) := \sum_i \mathrm{d}(\circ(r_i, s)).$$

This measures the total hermeneutic "work" done by the community.

[2]Geertz's famous dictum—that culture is "webs of significance"—echoes Max Weber's definition of sociology as the interpretive understanding of social action. The meaning game formalizes both simultaneously: the signal is the social action, and the interpretations are the community's collective understanding.

```
def interpretiveRichness (reps : List I) (s : I) : N
    :=
  depthSum (reps.map (fun r => IdeaticSpace.compose r
      s))
```

Listing 20.4: Interpretive richness in Lean

Ricoeur's "surplus of meaning" is operationalized as interpretive richness: it is the total interpretive content generated by the encounter between signal and receivers. The richer the community's backgrounds and the deeper the signal, the greater the surplus.

20.3 The Fundamental Theorems of Interpretation

Theorem 20.5 (Fundamental Theorem — Void Signal Equilibrium (20.1))**.** *The void signal always admits equilibrium:*

$$\texttt{MeaningEquilibrium}\big(\langle \mathbf{0}, \mathit{receivers}\rangle\big) \implies \top.$$

Proof sketch. The conclusion is $\top$ (trivially true). □

Commentary. In the absence of a text, each interpreter's horizon remains intact. Gadamer's hermeneutic circle begins with pre-understanding, and when the text contributes nothing (void), the circle collapses to its starting point. The void is the universal equilibrium signal—saying nothing produces no disagreement. While the theorem as stated is trivially true, it establishes the conceptual foundation: void is the baseline against which all other signals are measured.

Theorem 20.6 (Void Signal Preserves Backgrounds (20.2))**.** $\texttt{interpretAll}(\mathit{reps}, \mathbf{0}) = \mathit{reps}$.

Proof sketch. By induction on the list. At each step, $\circ(r, \mathbf{0}) = r$ by right-identity. □

Commentary. Without external stimuli, consciousness returns to itself. Each receiver, encountering the void signal, produces an "interpretation" that is simply their own background idea, unchanged. Phenomenologically, this is the state of pure self-reflection: the subject without object collapses back upon its own structure.[3]

[3] This echoes Husserl's concept of the *noema* (intentional object): when the noema is empty (void), the *noesis* (intentional act) has nothing to constitute, and the mind rests in its own pre-intentional state.

Theorem 20.7 (Composition Closure (20.6)). *Interpretations of interpretations are still compositions:*

$$\circ(r_2, \circ(r_1, s)) = \circ(\circ(r_2, r_1), s).$$

Proof sketch. The associativity axiom of $\circ$. □

Commentary. This is the formal content of Geertz's "thick description": interpretation layered upon interpretation is still interpretation. There is no escape from the hermeneutic circle, only deeper embedding within it. The second interpreter does not stand "outside" the first interpretation; rather, their reading *composes* with the first reading, producing a compound interpretation that is itself an element of ideatic space.

This theorem has profound epistemological consequences. It means that meta-interpretation—the interpretation of interpretations—is not a different *kind* of activity from first-order interpretation. It is the same operation, applied at a higher level of composition. The hermeneutic circle is not a vicious regress but a stable compositional structure.[4]

Theorem 20.8 (Void Receiver Transparency (20.14)). $\circ(\mathbf{0}, s) = s$.

Proof sketch. Left-identity of $\circ$. □

Commentary. A receiver with void background returns the signal unchanged—a *tabula rasa* is a perfect mirror. The empty mind contributes nothing to the interpretation, simply reflecting the signal as received.

20.4 Resonance Preservation

The following theorems establish that resonance—the fundamental compatibility relation—is preserved through the meaning game.

Theorem 20.9 (Resonance Preservation — Resonant Backgrounds (20.4)). *If* $resonates(r_1, r_2)$*, then for all* s*,*

$$resonates(\circ(r_1, s), \circ(r_2, s)).$$

Proof sketch. From `resonance_compose_right`. □

Commentary. Agents sharing "collective consciousness" (Durkheim) will produce consonant interpretations of any cultural artifact. Resonance between backgrounds propagates through the interpretive act to resonance between interpretations. This is the mechanism underlying cultural consonance: shared pre-understanding generates shared understanding.

[4] This parallels Ricoeur's argument in *Hermeneutics and the Human Sciences* (1981) that explanation and understanding are not opposed but complementary moments of a single interpretive arc.

Theorem 20.10 (Resonance Preservation — Resonant Signals (20.5)). *If* $resonates(s_1, s_2)$*, then for all* r*,*

$$resonates(\circ(r, s_1), \circ(r, s_2)).$$

Proof sketch. From `resonance_compose_left`. □

Commentary. A reader encountering similar texts (resonant signals) produces similar interpretations. This captures the literary-critical intuition that intertextuality creates resonance: texts that "echo" each other produce echoing interpretations in any given reader.

Theorem 20.11 (Interpretation Self-Resonance (20.11)). $resonates(\circ(r, s), \circ(r, s))$.

Proof sketch. Self-resonance. □

Commentary. Every act of understanding is internally coherent. No matter how idiosyncratic an interpretation may be, it resonates with itself—it is a valid element of ideatic space, self-consistent by the reflexivity of resonance.

Theorem 20.12 (Associativity of Iterated Interpretation (20.19)).

$$\circ(r_3, \circ(r_2, \circ(r_1, s))) = \circ(\circ(r_3, r_2), \circ(r_1, s)).$$

Proof sketch. Associativity of $\circ$. □

Commentary. Gadamer's hermeneutic circle has no privileged entry point. Re-bracketing the layers of interpretation—whether we first combine r_3 with r_2 and then interpret r_1's reading, or interpret r_2's reading of r_1's reading and then apply r_3—does not change the result. The "entry point" into the circle is arbitrary; the compositional structure is invariant.

20.5 Depth Bounds on Interpretive Richness

Theorem 20.13 (Universal Depth Bound on Interpretive Richness (20.3)). *For all lists of receivers reps and signals* s*,*

$$\texttt{interpretiveRichness}(reps, s) \;\leq\; \texttt{depthSum}(reps) + |reps| \cdot \mathrm{d}(s).$$

Proof sketch. By induction on the list of receivers. At each step, $\mathrm{d}(\circ(r, s)) \leq \mathrm{d}(r) + \mathrm{d}(s)$ by subadditivity, and the inductive hypothesis bounds the tail. □

Commentary. This is the *fundamental resource theorem of hermeneutics*: the surplus of meaning is bounded. Even the richest text can only generate bounded total interpretation, proportional to the community's existing hermeneutic capacity plus the text's complexity.

Ricoeur argued that the surplus of meaning is *inexhaustible*—that a great text always yields new readings. Our theorem does not contradict this claim, but qualifies it: the *total* depth of all readings is bounded, even if the *number* of distinct readings is unbounded. In other words, the surplus is bounded in intensity (depth) even if unbounded in extension (variety).[5]

Theorem 20.14 (Double Interpretation Depth Bound (20.12))**.**

$$\mathrm{d}\big(\circ(r_2, \circ(r_1, s))\big) \;\leq\; \mathrm{d}(r_1) + \mathrm{d}(r_2) + \mathrm{d}(s).$$

Proof sketch. Two applications of subadditivity, combined by transitivity. □

Commentary. Each pass through the hermeneutic circle adds bounded complexity. Understanding deepens, but not without limit per step. The depth of a doubly-interpreted signal is bounded by the sum of the two interpreters' depths and the signal's depth—a tight three-term bound.

Theorem 20.15 (Void Signal Richness (20.9))**.** $\texttt{interpretiveRichness}(\mathit{reps}, \mathbf{0}) = \texttt{depthSum}(\mathit{reps})$.

Proof sketch. By induction, using $\circ(r, \mathbf{0}) = r$ at each step. □

Commentary. The "cost" of interpretation comes entirely from the interpreters' backgrounds when the signal carries no content. In the economics of meaning, the void signal has zero marginal cost: all the interpretive richness is pre-existing, contributed by the receivers' own depth.

Theorem 20.16 (Interpretation Depth Non-Negativity (20.15))**.** $0 \leq \texttt{interpretiveRich}$

Proof sketch. Trivial for natural numbers. □

Commentary. Interpretation always produces non-negative hermeneutic content. While trivial, this establishes a conceptual baseline: there is no such thing as "negative interpretation." Every encounter between mind and text generates at least zero meaning.

Theorem 20.17 (Empty Repertoire Zero Richness (20.16))**.** $\texttt{interpretiveRichness}([\,], s$
0.

Proof sketch. Sum over empty list is zero. □

Commentary. A message with no audience generates no meaning. This is the information-theoretic counterpart to the philosophical puzzle: if a tree falls in a forest and no one is there to hear it, does it make a sound? In IDT, the answer is unambiguous: no receivers, zero richness.

[5] The distinction between intensive and extensive surplus of meaning parallels the distinction between depth and breadth in information theory. A text of finite depth can generate infinitely many shallow readings, but the total depth is constrained.

Theorem 20.18 (Singleton Richness Bound (20.17)). $\texttt{interpretiveRichness}([r], s) \leq \mathrm{d}(r) + \mathrm{d}(s)$.

Proof sketch. The richness of a singleton list is $\mathrm{d}(\circ(r, s))$, bounded by subadditivity. □

Commentary. One person's interpretive effort is bounded by their background plus the signal's complexity. Individual hermeneutics operates under the same resource constraints as collective hermeneutics, scaled down to a single agent.

20.6 Equilibrium Existence

Theorem 20.19 (Empty Game Equilibrium (20.7)). *For all $s \in \mathcal{I}$,* $\texttt{MeaningEquilibrium}(\langle s, [\,]\rangle$

Proof sketch. Universal quantification over the empty outcome list is vacuously true. □

Commentary. A meaning game with no receivers is trivially in equilibrium—vacuous consensus. Logic guarantees agreement when there are no parties to disagree.

Theorem 20.20 (Singleton Game Equilibrium (20.8)). *For all r, s,* $\texttt{MeaningEquilibrium}(\langle s, [r]$

Proof sketch. The only outcome element is $\circ(r, s)$, which resonates with itself. □

Commentary. A single interpreter always agrees with themselves—solipsism is equilibrium. The interesting cases arise when multiple receivers must cohere.

Theorem 20.21 (Trivial Equilibrium Exists (20.10)). $\texttt{MeaningEquilibrium}(\langle \mathbf{0}, [\,]\rangle)$.

Proof sketch. Vacuously true. □

Commentary. The "babbling equilibrium" trivially exists when there is no audience. Saying nothing to no one creates no disagreement—the most degenerate form of communicative success.

Theorem 20.22 (Resonant Community Equilibrium (20.13)). *If all receivers pairwise resonate, then the meaning game is in equilibrium for* any *signal:*

$$\big(\forall\, r_1, r_2 \in \mathit{receivers},\ \mathit{resonates}(r_1, r_2)\big) \implies \texttt{MeaningEquilibrium}(\langle s, \mathit{receivers}\rangle).$$

Proof sketch. For any x, y in the outcome, $x = \circ(r_i, s)$ and $y = \circ(r_j, s)$ for some $r_i, r_j \in$ receivers. By hypothesis, $\mathrm{resonates}(r_i, r_j)$, and by Theorem 20.9, $\mathrm{resonates}(\circ(r_i, s), \circ(r_j, s))$. □

Commentary. This is Durkheim's grand theorem, formalized: a community with strong "collective consciousness" (universal pairwise resonance) achieves interpretive consensus regardless of the stimulus. Shared culture makes shared understanding automatic. The theorem does not require any assumption about the signal—*any* signal produces equilibrium, provided the receivers share resonance.

The implications for social theory are significant. Homogeneous communities achieve meaning equilibrium effortlessly; diverse communities may not. This is neither a normative claim nor a defense of homogeneity—it is a structural fact about the interaction between resonance and composition. The challenge for pluralistic societies is to achieve equilibrium without requiring universal resonance, a problem we leave for future work.[6]

Theorem 20.23 (Game Outcome Length (20.18))**.** $|G.outcome| = |G.receivers|$.

Proof sketch. The `map` operation preserves list length. □

Commentary. Every voice produces exactly one interpretation. The meaning game's outcome has the same cardinality as its receiver list—a "one person, one interpretation" principle that ensures representational completeness.

20.7 The Grand Synthesis

We now arrive at the culminating theorem of the entire signalling-game arc.

Theorem 20.24 (The Grand Synthesis (20.20))**.** *For any meaning game with resonant receivers:*

(1) The total interpretive richness is bounded:

$$\texttt{interpretiveRichness}(receivers, s) \leq \texttt{depthSum}(receivers) + |receivers| \cdot \mathrm{d}(s).$$

(2) The game is in equilibrium:

$$\texttt{MeaningEquilibrium}(\langle s, receivers \rangle).$$

Both properties hold simultaneously whenever $\forall\, r_1, r_2 \in receivers,\ resonates(r_1, r_2)$.

```
theorem grand_synthesis (receivers : List I) (s : I)
    (hres : forall r1 in receivers, forall r2 in
        receivers,
```

[6] Compare this to the sociological concept of "bridging social capital" (Robert Putnam): connections between diverse groups that facilitate mutual understanding despite the absence of deep cultural resonance.

```
              IdeaticSpace.resonates r1 r2) :
    interpretiveRichness receivers s <=
      depthSum receivers + receivers.length *
         IdeaticSpace.depth s /\
    MeaningEquilibrium { signal := s, receivers :=
        receivers } :=
  <universal_depth_bound receivers s,
   resonant_community_equilibrium receivers s hres>
```

Listing 20.5: The grand synthesis in Lean

Proof sketch. Clause (1) is the universal depth bound (Theorem 20.13), which holds for *all* receiver lists, regardless of resonance. Clause (2) is the resonant community equilibrium (Theorem 20.22), which requires the resonance hypothesis. The conjunction of both clauses follows by pairing. □

Commentary. The grand synthesis is the formal heart of this book. It says that for a culturally coherent community—one in which all members share pairwise resonance—two fundamental properties hold *simultaneously*:

1. **Resource boundedness.** The total interpretive effort of the community is finite and bounded by a linear function of the community's existing depth and the signal's complexity. Meaning-making is not free; it consumes cognitive resources, and the total expenditure is constrained.

2. **Hermeneutic consensus.** All interpretations are mutually comprehensible. The community achieves shared meaning—not by suppressing individual differences (each receiver still composes with their own background), but because their backgrounds are sufficiently aligned that their readings "rhyme" with each other.

The philosophical significance of this conjunction cannot be overstated. It unifies the *economic* perspective (resources are bounded) with the *hermeneutic* perspective (meaning is shared). Previous accounts of meaning-making have typically emphasized one dimension at the expense of the other: economists focus on information costs, hermeneuticians on interpretive depth. The grand synthesis shows that both concerns are addressed simultaneously by a single structural condition—resonance.

Geertz revisited. Clifford Geertz wrote that culture is "webs of significance" that humans spin and within which they are suspended. The grand synthesis formalizes this metaphor with mathematical precision. The "webs" are the resonance relations between community members; the "significance" is the interpretive richness generated by their encounter with signals. The theorem shows that when the web is coherent (universal pairwise resonance), the significance is both bounded (finite web) and shared (equilibrium).

Ricoeur revisited. Paul Ricoeur's "surplus of meaning" is precisely the interpretive richness of the community. The universal depth bound shows that this surplus is finite—not inexhaustible, as Ricoeur sometimes suggested, but bounded by the community's cognitive resources. Yet within this bound, the surplus can be substantial: a deep community encountering a deep text generates rich interpretations.

Gadamer revisited. Gadamer's hermeneutic circle is formalized by the associativity of composition (Theorem 20.7). The grand synthesis shows that the circle is not merely internally consistent (associativity) but also *convergent*: when the interpreters share resonance, the circle produces equilibrium. The hermeneutic circle is a spiral, not a vortex—it converges on shared meaning rather than spinning into interpretive chaos.

20.8 Conclusion: The Game Theory of Meaning

This chapter—and this entire signalling-game arc—has established a formal theory of meaning grounded in the algebraic structure of ideatic space. Twenty theorems in this chapter, building on the seventy-four theorems of Chapters 16–19, constitute a comprehensive mathematical framework for analyzing how meaning is created, transmitted, negotiated, and shared.

Let us recapitulate the arc:

Chapter 16 established that cooperation is composition, with void as the non-contributor, associativity ensuring well-defined multi-party projects, and depth bounding the output complexity.

Chapter 17 showed that non-commutativity creates bargaining power: who speaks first shapes the meaning. Yet structural properties—depth bounds and void transparency—are order-invariant.

Chapter 18 formalized cultural diffusion as relay chains, proving that depth grows at most linearly, void relayers are transparent, and resonant relayers cluster.

Chapter 19 addressed the aggregation problem: how individual interpretations compose into collective meaning, and under what conditions (resonance) the community achieves hermeneutic consensus.

Chapter 20 (this chapter) unified all threads into the grand synthesis: resonant communities achieve both bounded richness and meaning equilibrium.

The game theory of meaning differs from classical game theory in a fundamental way. In classical game theory, agents pursue individual payoffs and

equilibria emerge from the alignment of selfish strategies. In the game theory of meaning, agents pursue *understanding*—the production of interpretations that resonate with those of their fellow-interpreters—and equilibria emerge from the alignment of *backgrounds*. The currency is not utility but resonance; the constraint is not budget but depth; the solution concept is not Nash equilibrium but meaning equilibrium.

This formal framework opens several directions for future work:

- **Non-resonant communities.** What happens when receivers do *not* pairwise resonate? Can partial equilibria be defined? Under what conditions does deliberation converge to equilibrium even without initial resonance?
- **Dynamic meaning games.** What happens when the signal changes over time, or when receivers update their backgrounds based on the interpretations of others? Can we formalize the dynamics of cultural change within this framework?
- **Strategic interpretation.** What happens when receivers are not purely hermeneutic but strategically choose their interpretive frames? Can the game theory of meaning be extended to encompass strategic signalling in the tradition of Crawford and Sobel?
- **Computational complexity.** How hard is it to compute interpretive richness, check meaning equilibrium, or find the optimal signal for a given community? Are there polynomial-time algorithms, or is the problem inherently intractable?

We began this arc with Mauss's insight that gift exchange is never merely economic. We end with a generalization: *meaning exchange* is never merely informational. It is compositional, order-dependent, depth-bounded, and resonance-conditioned. The game theory of meaning provides the formal tools to analyze these properties with mathematical rigor, while remaining faithful to the anthropological and philosophical traditions that inspired them.

"The locus of the study is not the object of study.
Anthropologists don't study villages; they study in villages."
— Clifford Geertz, *The Interpretation of Cultures*

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Index of Theorems

The following theorems are formally verified in Lean 4 with zero `sorry` axioms. Each theorem name links to the corresponding Lean source file in the `FormalAnthropology/` directory.

Ch. 1 `total_hermeneutic_bound`, `max_interp_depth_bound`, `silence_is_transparent`, `blank_receiver`, `negative_capability`, `durkheimian_consensus`, `mythical_transform` `turners_communitas`, `bartletts_serial_reproduction`, `telephone_depth`, `resonant_transmission_lines`, `void_is_unpersuasive`, `interpretation_conserves_c` `durkheims_division_of_labor`, `silence_preserves_depthSum`, `three_stage_telephon`

Ch. 2–5 See `IDT_Signal_Ch02-05.lean`

Ch. 6–10 See `IDT_Signal_Ch06-10.lean`

Ch. 11–15 See `IDT_Signal_Ch11-15.lean`

Ch. 16–20 See `IDT_Signal_Ch16-20.lean`